College of Engineering and Applied Sciences

Course Name: Lab. Intro. To Microprocessors

Course Number: ECE 2510

Instructors:
Christian Blanchard
Pravana Avuthu

Lab #3

Date: February 17rd 2026

Student:

De Leon Santos, Sebastian Manuel – WIN: 577866168

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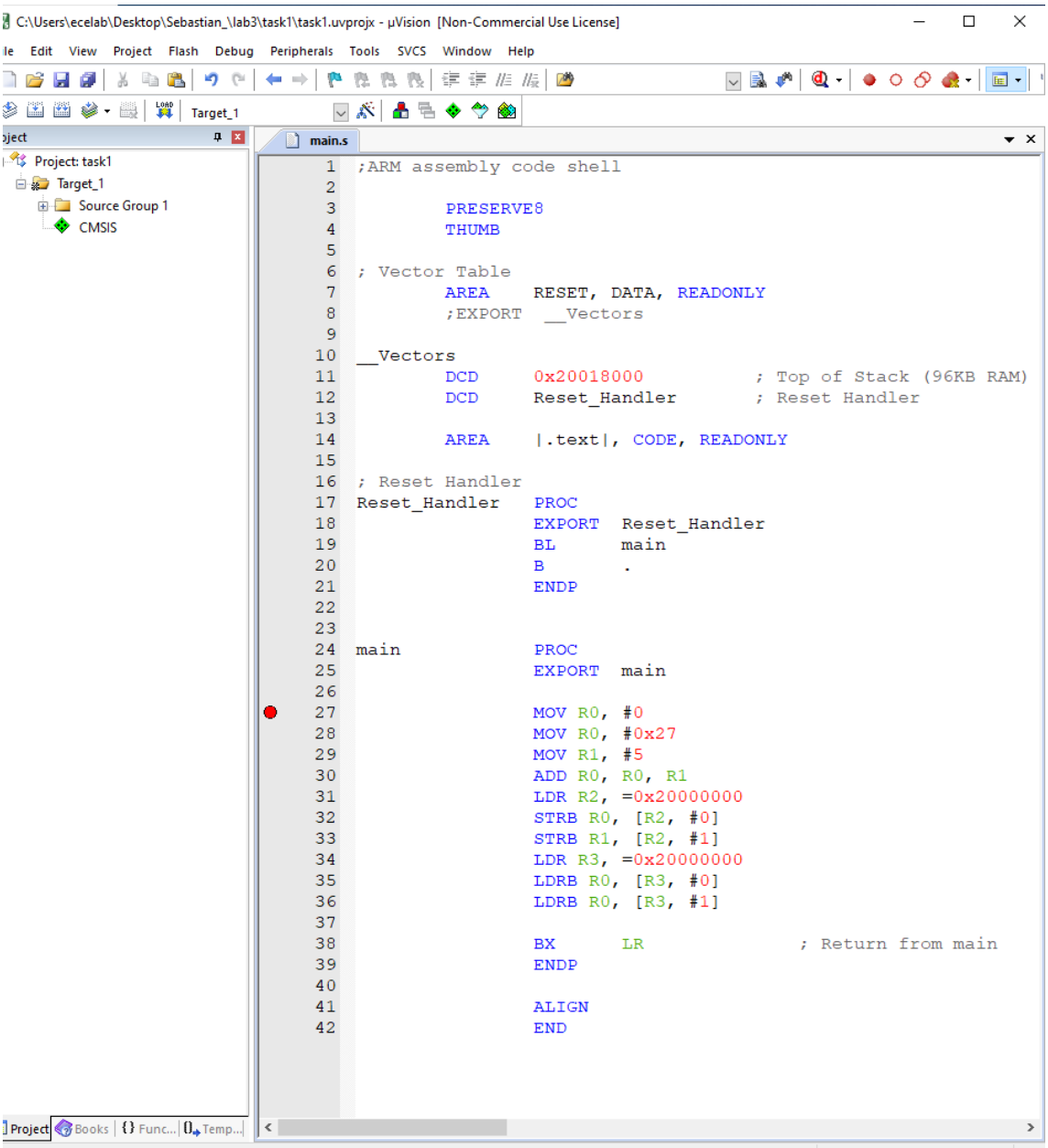
Task 1:

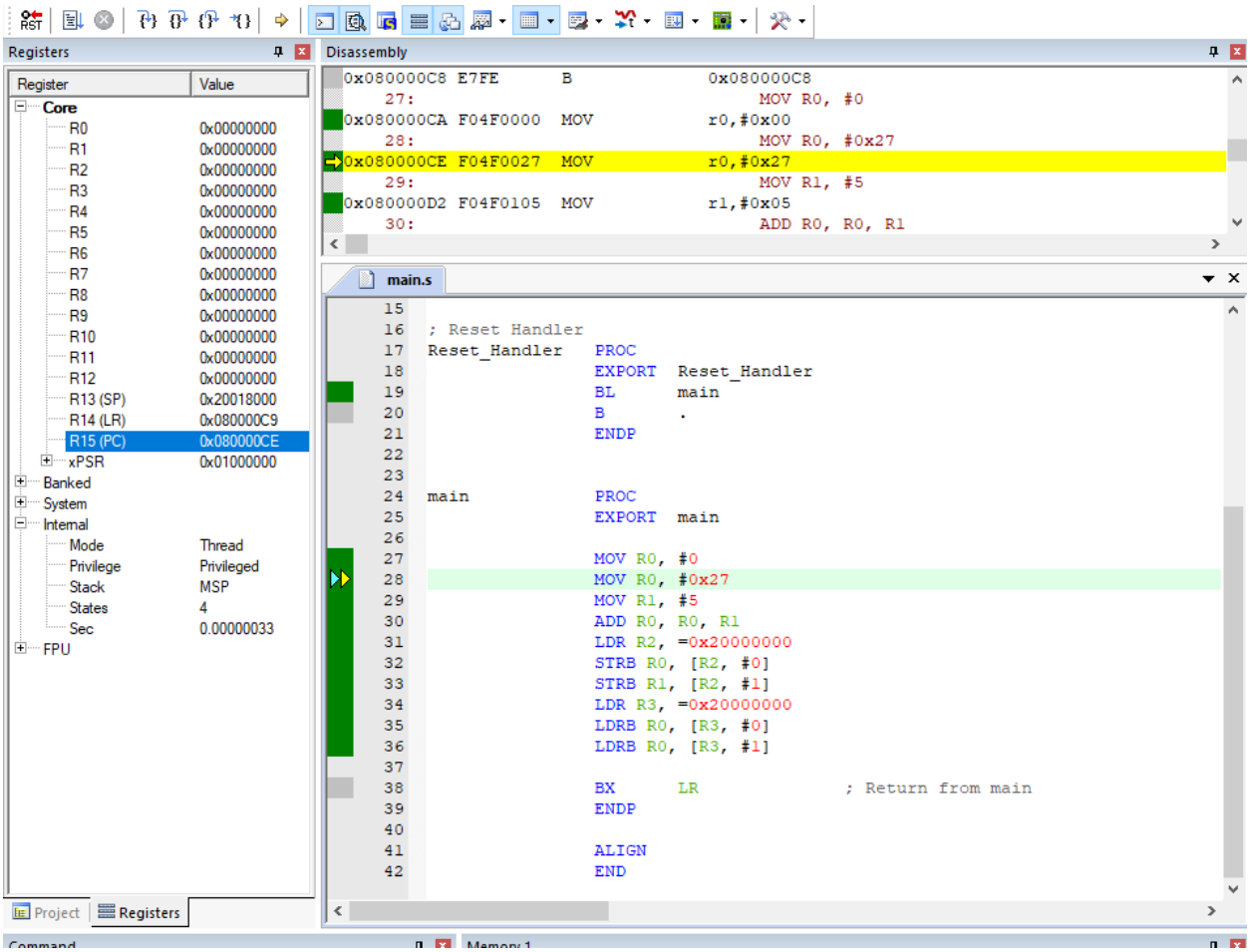
● ARM Assembly code with Register Transfer Level (RTL) and addressing modes for each instruction.

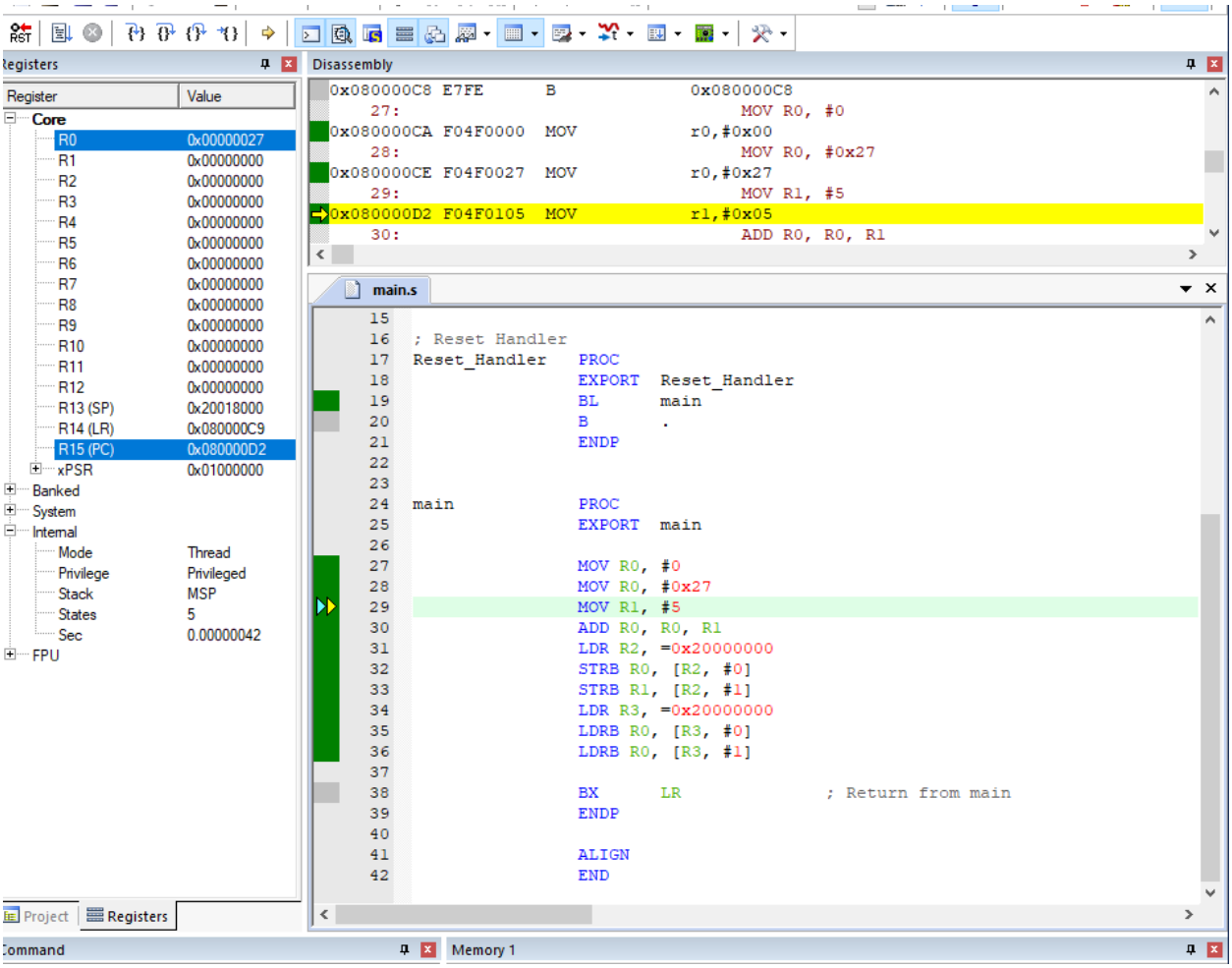
● Screenshots of each step of the program:

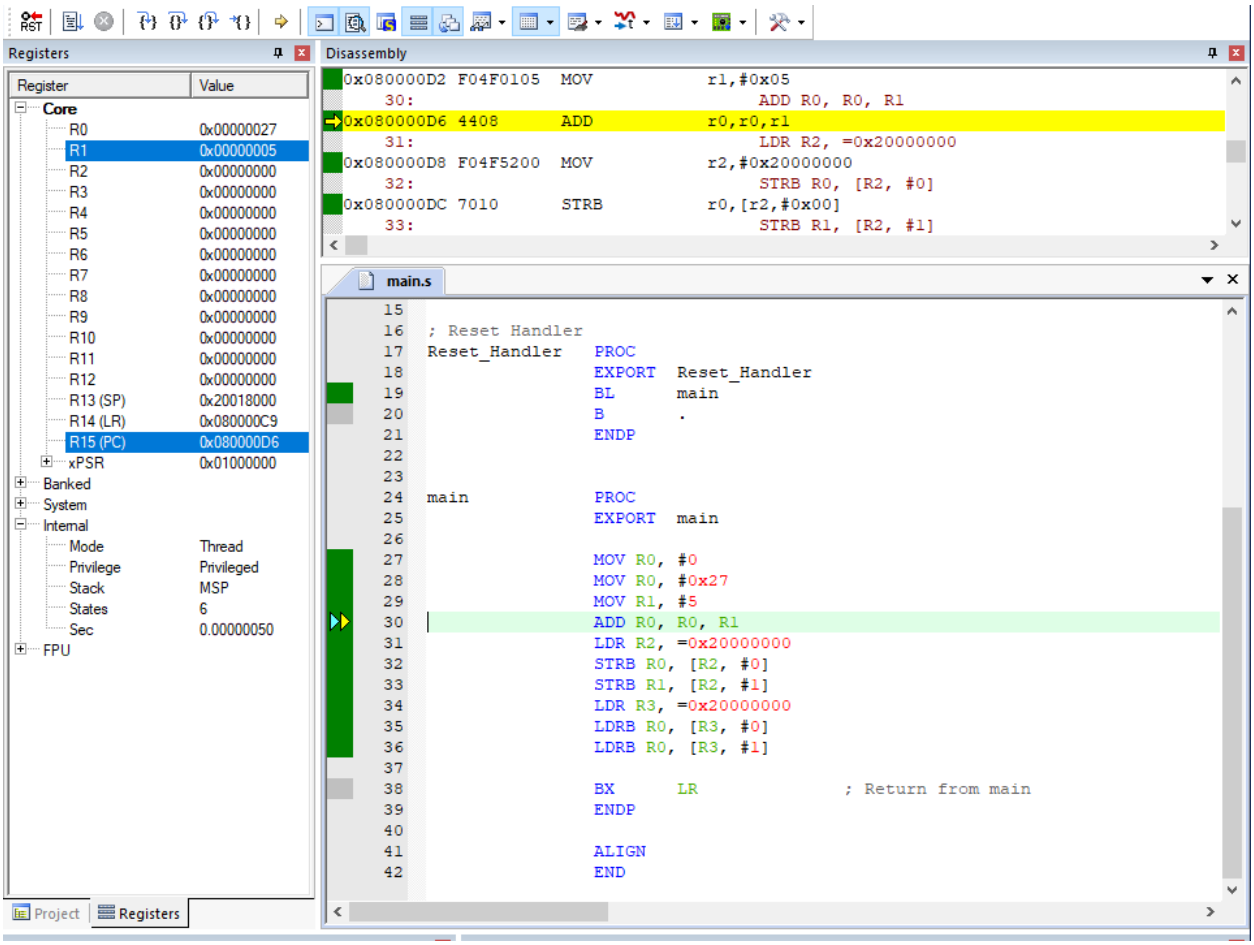
o If the instruction modifies the register, screenshot the Register Window.

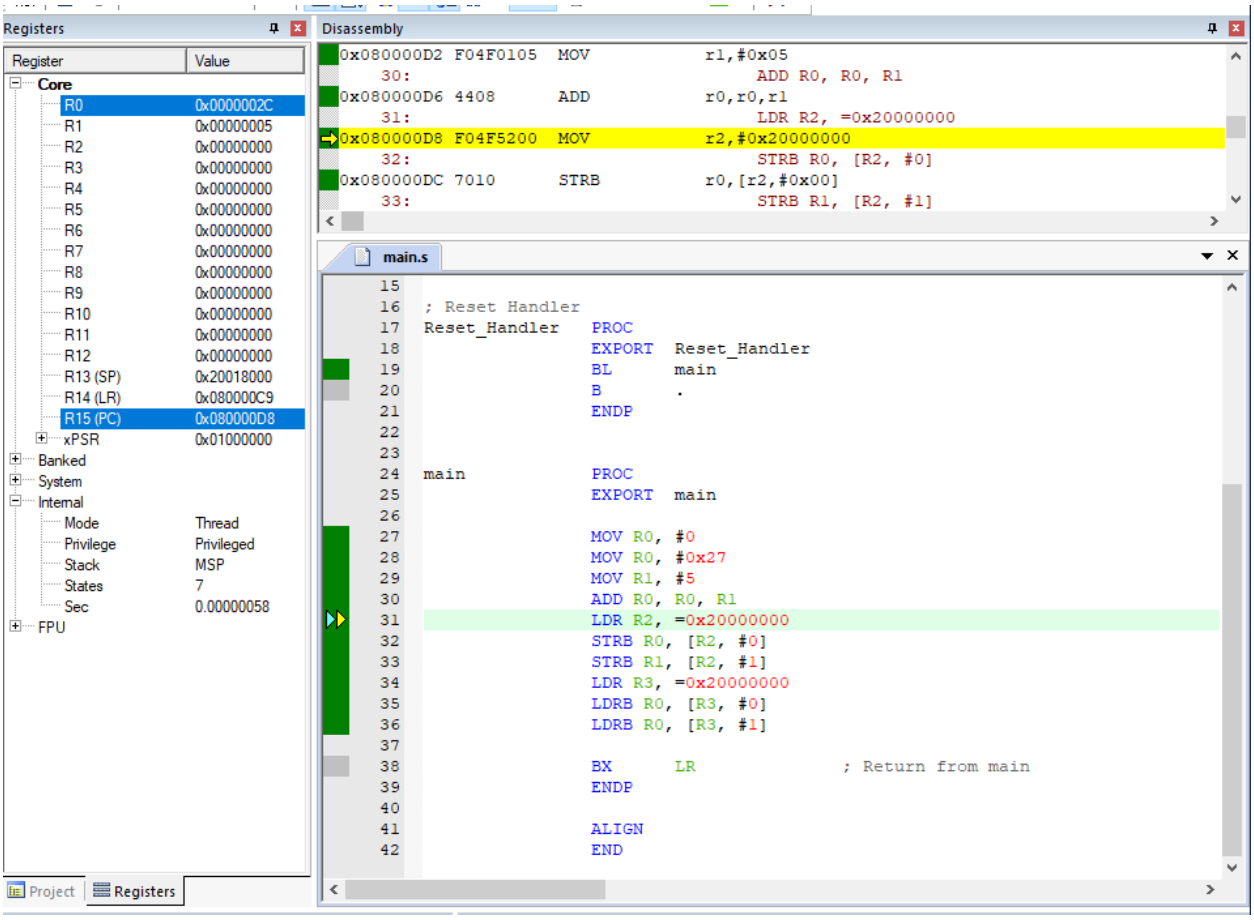
o If the instruction modifies memory, screenshot the Memory Window displaying the correct address.

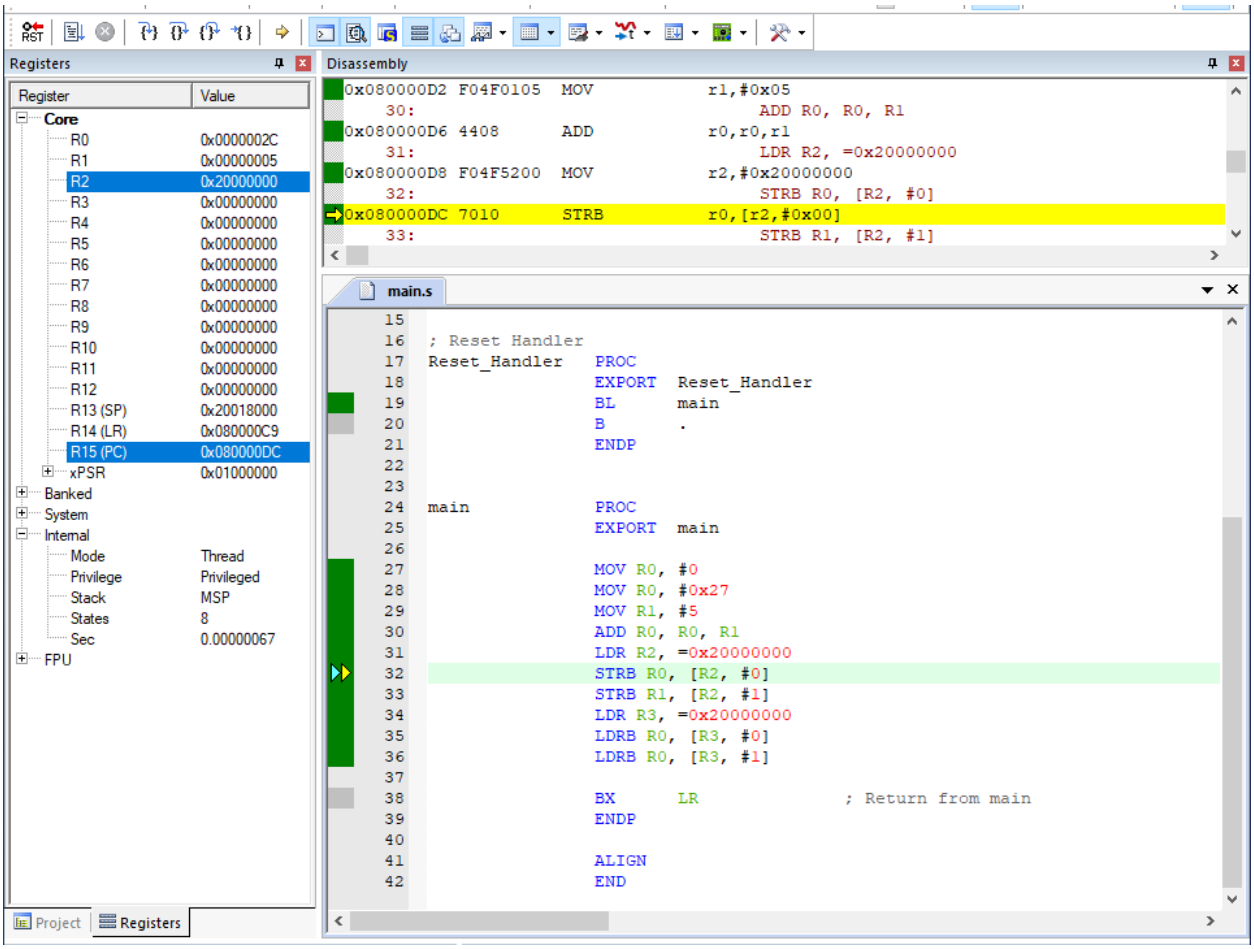


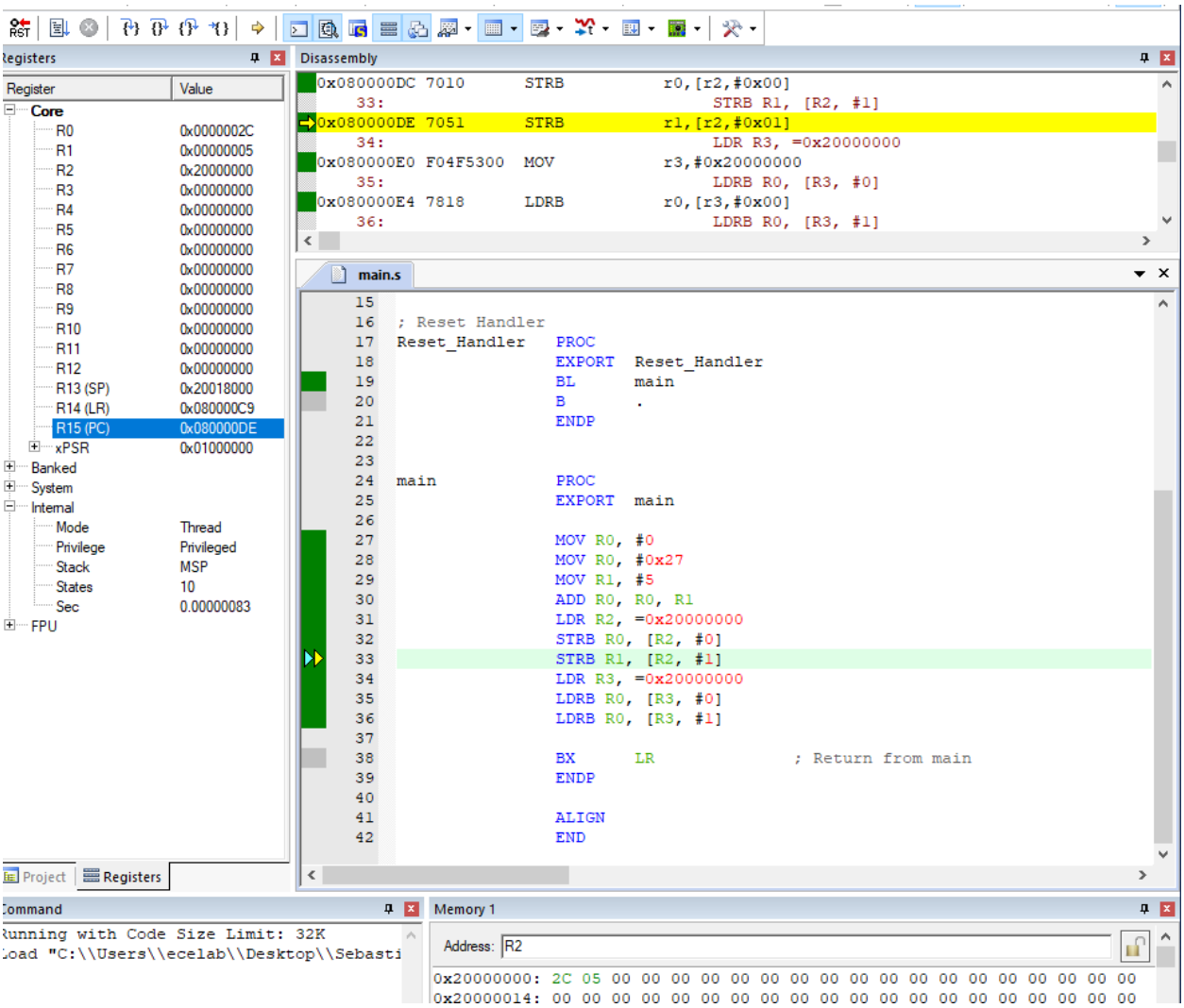

Registers
Register
Value
Core
R0 0x00000000
R1 0x00000000
R2 0x00000000
R3 0x00000000
R4 0x00000000
R5 0x00000000
R6 0x00000000
R7 0x00000000
R8 0x00000000
R9 0x00000000
R10 0x00000000
R11 0x00000000
R12 0x00000000
R13 (SP) 0x20018000
R14 (LR) 0x080000C9
R15 (PC) 0x080000CE
xPSR 0x01000000
Banked
System
Internal
Mode Thread
Privilege Privileged
Stack MSP
States 4
Sec 0.00000033
FPU
Project
Registers
Disassembly
0x080000C8 E7FE B 0x080000C8
27: MOV R0, #0
0x080000CA F04F0000 MOV r0,#0x00
28: MOV R0, #0x27
0x080000CE F04F0027 MOV r0,#0x27
29: MOV R1, #5
0x080000D2 F04F0105 MOV r1,#0x05
30: ADD R0, R0, R1
main.s
15
16 ; Reset Handler
17 Reset_Handler PROC
18 EXPORT Reset_Handler
19 BL main
20 B .
21 ENDP
22
23
24 main PROC
25 EXPORT main
26
27 MOV R0, #0
28 MOV R0, #0x27
29 MOV R1, #5
30 ADD R0, R0, R1
31 LDR R2, =0x20000000
32 STRB R0, [R2, #0]
33 STRB R1, [R2, #1]
34 LDR R3, =0x20000000
35 LDRB R0, [R3, #0]
36 LDRB R0, [R3, #1]
37
38 BX LR ; Return from main
39 ENDP
40
41 ALIGN
42 END
Command
Memory 1

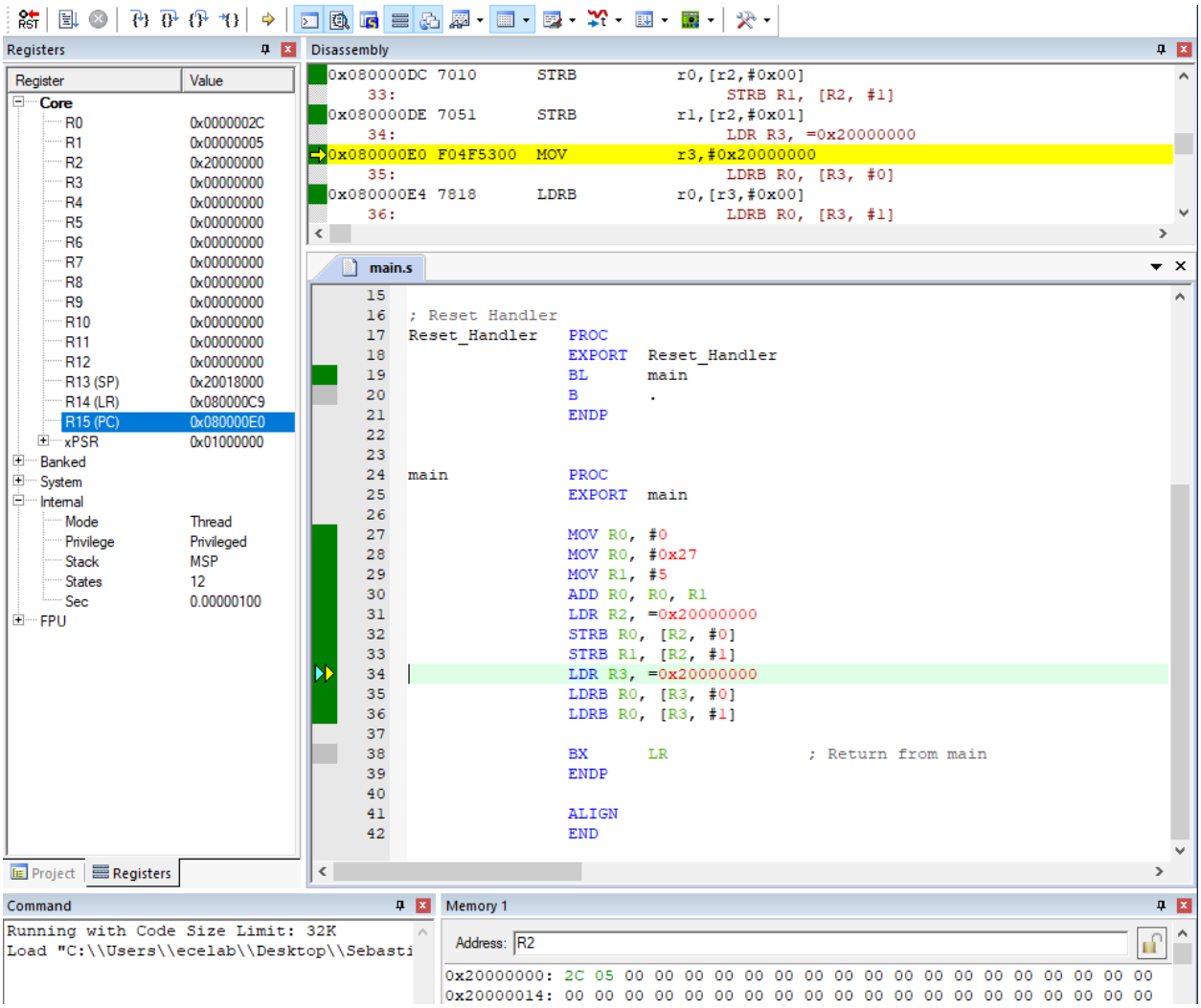

Registers
Register
Value
Core
R0 0x00000027
R1 0x00000000
R2 0x00000000
R3 0x00000000
R4 0x00000000
R5 0x00000000
R6 0x00000000
R7 0x00000000
R8 0x00000000
R9 0x00000000
R10 0x00000000
R11 0x00000000
R12 0x00000000
R13 (SP) 0x20018000
R14 (LR) 0x080000C9
R15 (PC) 0x080000D2
xPSR 0x01000000
Banked
System
Internal
Mode Thread
Privilege Privileged
Stack MSP
States 5
Sec 0.00000042
FPU
Project
Registers
Disassembly
0x080000C8 E7FE B 0x080000C8
27: MOV R0, #0
0x080000CA F04F0000 MOV r0,#0x00
28: MOV R0, #0x27
0x080000CE F04F0027 MOV r0,#0x27
29: MOV R1, #5
0x080000D2 F04F0105 MOV r1,#0x05
30: ADD R0, R0, R1
main.s
; Reset Handler
Reset_Handler PROC
EXPORT Reset_Handler
BL main
B .
ENDP
main PROC
EXPORT main
MOV R0, #0
MOV R0, #0x27
MOV R1, #5
ADD R0, R0, R1
LDR R2, =0x20000000
STRB R0, [R2, #0]
STRB R1, [R2, #1]
LDR R3, =0x20000000
LDRB R0, [R3, #0]
LDRB R0, [R3, #1]
BX LR ; Return from main
ENDP
ALIGN
END
Command
Memory 1

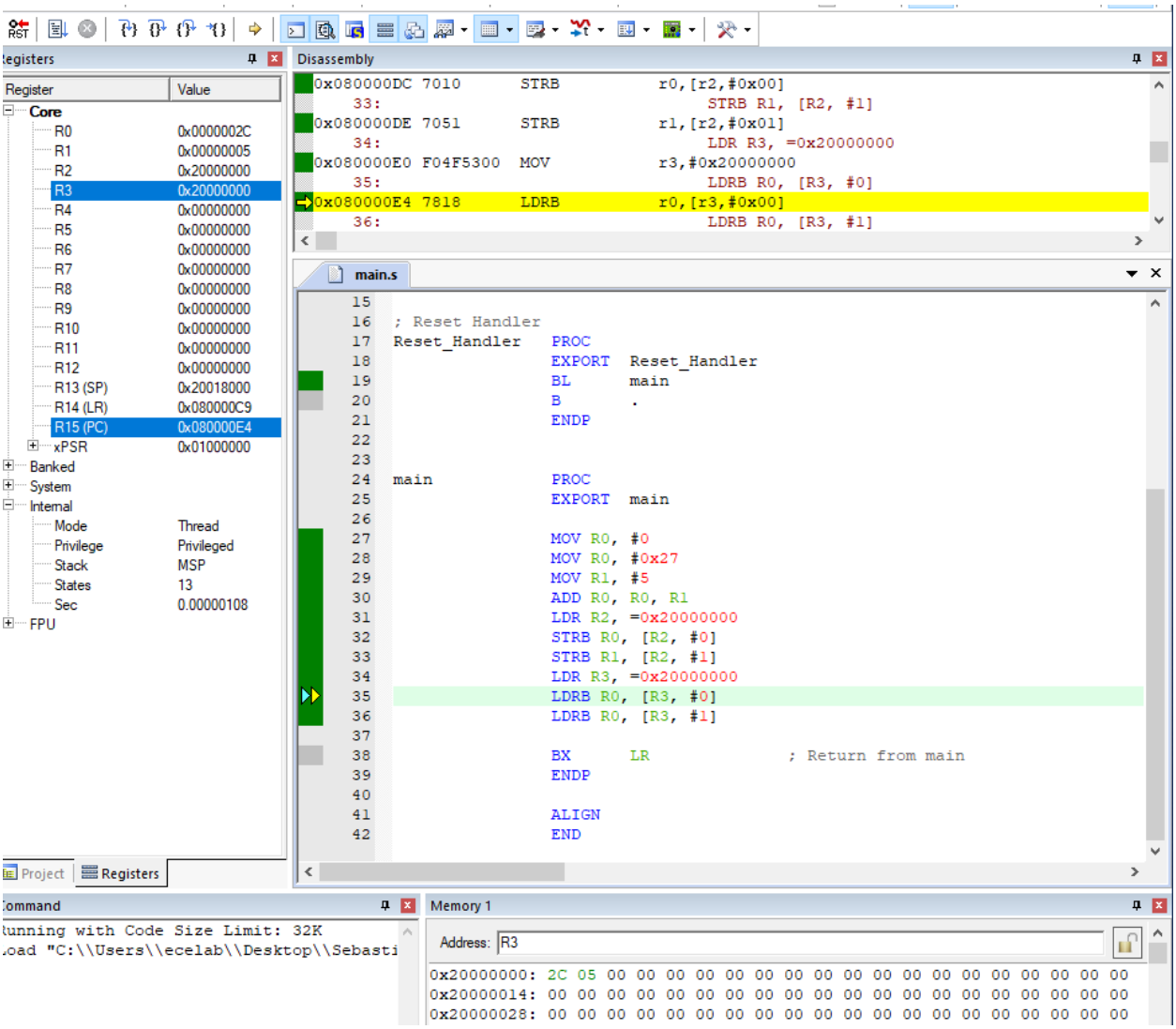

Registers
Register
Value
Core
R0 0x00000027
R1 0x00000005
R2 0x00000000
R3 0x00000000
R4 0x00000000
R5 0x00000000
R6 0x00000000
R7 0x00000000
R8 0x00000000
R9 0x00000000
R10 0x00000000
R11 0x00000000
R12 0x00000000
R13 (SP) 0x20018000
R14 (LR) 0x080000C9
R15 (PC) 0x080000D6
xPSR 0x01000000
Banked
System
Internal
Mode Thread
Privilege Privileged
Stack MSP
States 6
Sec 0.00000050
FPU
Project
Registers
Disassembly
0x080000D2 F04F0105 MOV r1,#0x05
30: ADD R0, R0, R1
0x080000D6 4408 ADD r0,r0,r1
31: LDR R2, =0x20000000
0x080000D8 F04F5200 MOV r2,#0x20000000
32: STRB R0, [R2, #0]
0x080000DC 7010 STRB r0,[r2,#0x00]
33: STRB R1, [R2, #1]
main.s
; Reset Handler
Reset_Handler PROC
EXPORT Reset_Handler
BL main
B .
ENDP
main PROC
EXPORT main
MOV R0, #0
MOV R0, #0x27
MOV R1, #5
ADD R0, R0, R1
LDR R2, =0x20000000
STRB R0, [R2, #0]
STRB R1, [R2, #1]
LDR R3, =0x20000000
LDRB R0, [R3, #0]
LDRB R0, [R3, #1]
BX LR ; Return from main
ENDP
ALIGN
END

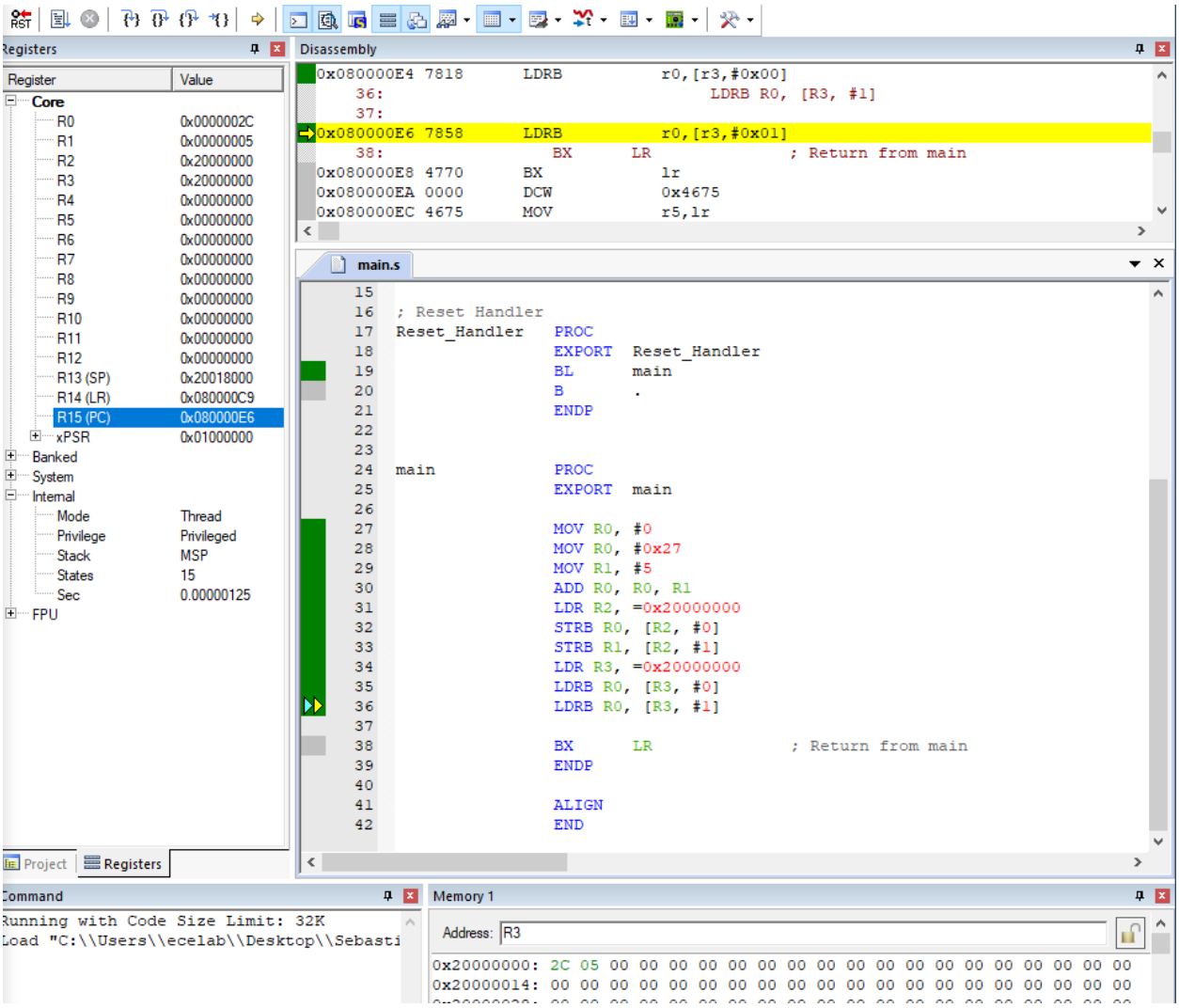

Registers
Register
Value
Core
R0 0x0000002C
R1 0x00000005
R2 0x00000000
R3 0x00000000
R4 0x00000000
R5 0x00000000
R6 0x00000000
R7 0x00000000
R8 0x00000000
R9 0x00000000
R10 0x00000000
R11 0x00000000
R12 0x00000000
R13 (SP) 0x20018000
R14 (LR) 0x080000C9
R15 (PC) 0x080000D8
xPSR 0x01000000
Banked
System
Internal
Mode Thread
Privilege Privileged
Stack MSP
States 7
Sec 0.00000058
FPU
Project
Registers
Disassembly
0x080000D2 F04F0105 MOV r1,#0x05
30: ADD R0, R0, R1
0x080000D6 4408 ADD r0,r0,r1
31: LDR R2, =0x20000000
0x080000D8 F04F5200 MOV r2,#0x20000000
32: STRB R0, [R2, #0]
0x080000DC 7010 STRB r0,[r2,#0x00]
33: STRB R1, [R2, #1]
main.s
15
16 ; Reset Handler
17 Reset_Handler PROC
18 EXPORT Reset_Handler
19 BL main
20 B .
21 ENDP
22
23
24 main PROC
25 EXPORT main
26
27 MOV R0, #0
28 MOV R0, #0x27
29 MOV R1, #5
30 ADD R0, R0, R1
31 LDR R2, =0x20000000
32 STRB R0, [R2, #0]
33 STRB R1, [R2, #1]
34 LDR R3, =0x20000000
35 LDRB R0, [R3, #0]
36 LDRB R0, [R3, #1]
37
38 BX LR ; Return from main
39 ENDP
40
41 ALIGN
42 END

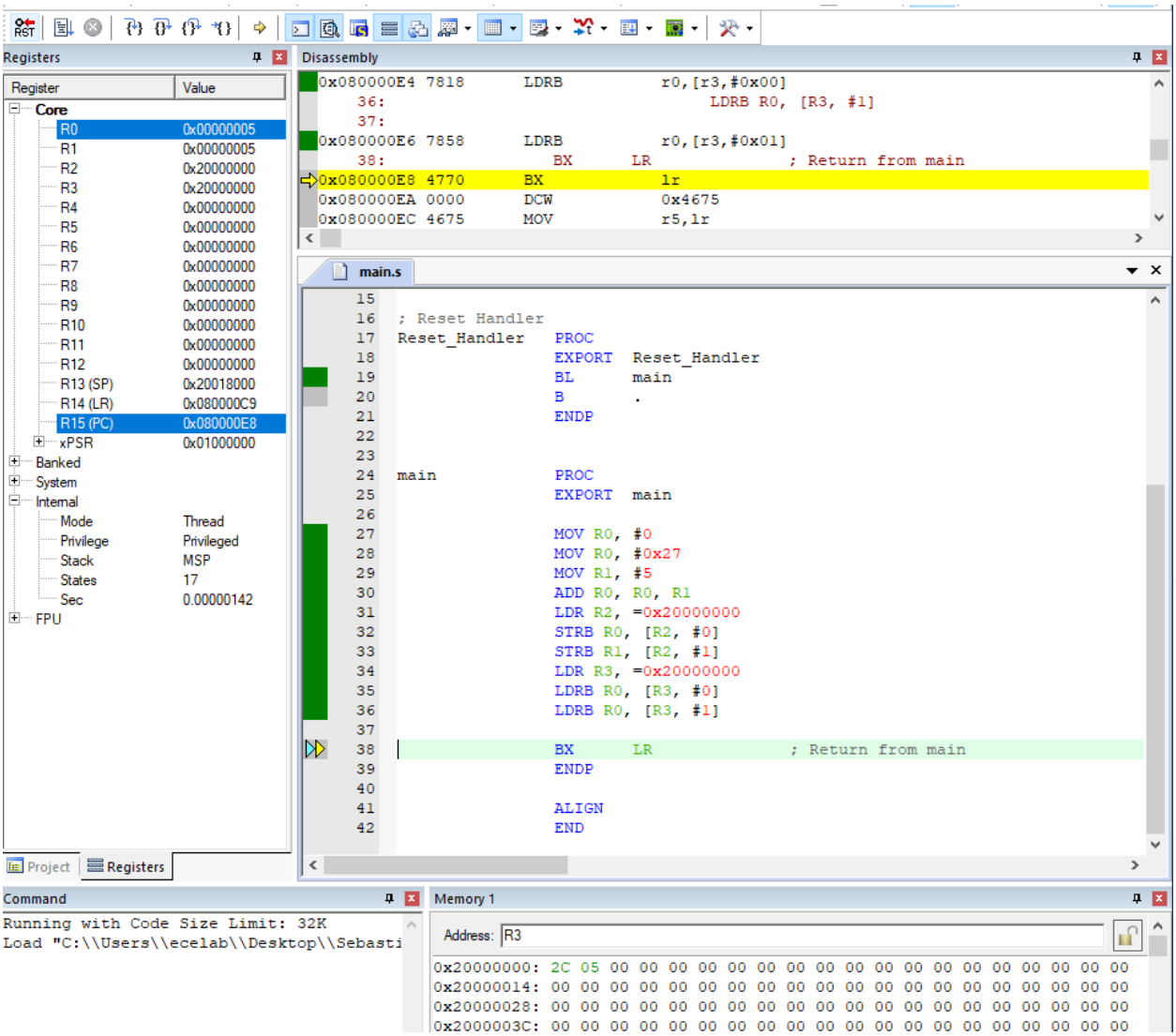

Registers
Register
Value
Core
R0 0x0000002C
R1 0x00000005
R2 0x20000000
R3 0x00000000
R4 0x00000000
R5 0x00000000
R6 0x00000000
R7 0x00000000
R8 0x00000000
R9 0x00000000
R10 0x00000000
R11 0x00000000
R12 0x00000000
R13 (SP) 0x20018000
R14 (LR) 0x080000C9
R15 (PC) 0x080000DC
xPSR 0x01000000
Banked
System
Internal
Mode Thread
Privilege Privileged
Stack MSP
States 8
Sec 0.00000067
FPU
Project
Registers
Disassembly
0x080000D2 F04F0105 MOV r1,#0x05
30: ADD R0, R0, R1
0x080000D6 4408 ADD r0,r0,r1
31: LDR R2, =0x20000000
0x080000D8 F04F5200 MOV r2,#0x20000000
32: STRB R0, [R2, #0]
0x080000DC 7010 STRB r0,[r2,#0x00]
33: STRB R1, [R2, #1]
main.s
; Reset Handler
Reset_Handler PROC
EXPORT Reset_Handler
BL main
B .
ENDP
main PROC
EXPORT main
MOV R0, #0
MOV R0, #0x27
MOV R1, #5
ADD R0, R0, R1
LDR R2, =0x20000000
STRB R0, [R2, #0]
STRB R1, [R2, #1]
LDR R3, =0x20000000
LDRB R0, [R3, #0]
LDRB R0, [R3, #1]
BX LR ; Return from main
ENDP
ALIGN
END


Registers
Register
Value
Core
R0 0x0000002C
R1 0x00000005
R2 0x20000000
R3 0x00000000
R4 0x00000000
R5 0x00000000
R6 0x00000000
R7 0x00000000
R8 0x00000000
R9 0x00000000
R10 0x00000000
R11 0x00000000
R12 0x00000000
R13 (SP) 0x20018000
R14 (LR) 0x080000C9
R15 (PC) 0x080000DE
xPSR 0x01000000
Banked
System
Internal
Mode Thread
Privilege Privileged
Stack MSP
States 10
Sec 0.00000083
FPU
Project
Registers
Disassembly
0x080000DC 7010 STRB r0,[r2,#0x00]
33: STRB R1, [R2, #1]
0x080000DE 7051 STRB r1,[r2,#0x01]
34: LDR R3, =0x20000000
0x080000E0 F04F5300 MOV r3,#0x20000000
35: LDRB R0, [R3, #0]
0x080000E4 7818 LDRB r0,[r3,#0x00]
36: LDRB R0, [R3, #1]
main.s
; Reset Handler
Reset_Handler PROC
EXPORT Reset_Handler
BL main
B .
ENDP
main PROC
EXPORT main
MOV R0, #0
MOV R0, #0x27
MOV R1, #5
ADD R0, R0, R1
LDR R2, =0x20000000
STRB R0, [R2, #0]
STRB R1, [R2, #1]
LDR R3, =0x20000000
LDRB R0, [R3, #0]
LDRB R0, [R3, #1]
BX LR ; Return from main
ENDP
ALIGN
END
Command
Running with Code Size Limit: 32K
Load "C:\\Users\\ecelab\\Desktop\\Sebasti
Memory 1
Address: R2
0x20000000: 2C 05 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0x20000014: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00


Registers
Register
Value
Core
R0 0x0000002C
R1 0x00000005
R2 0x20000000
R3 0x00000000
R4 0x00000000
R5 0x00000000
R6 0x00000000
R7 0x00000000
R8 0x00000000
R9 0x00000000
R10 0x00000000
R11 0x00000000
R12 0x00000000
R13 (SP) 0x20018000
R14 (LR) 0x080000C9
R15 (PC) 0x080000E0
xPSR 0x01000000
Banked
System
Internal
Mode Thread
Privilege Privileged
Stack MSP
States 12
Sec 0.00000100
FPU
Project
Registers
Disassembly
0x080000DC 7010 STRB r0,[r2,#0x00]
33: STRB R1, [R2, #1]
0x080000DE 7051 STRB r1,[r2,#0x01]
34: LDR R3, =0x20000000
0x080000E0 F04F5300 MOV r3,#0x20000000
35: LDRB R0, [R3, #0]
0x080000E4 7818 LDRB r0,[r3,#0x00]
36: LDRB R0, [R3, #1]
main.s
; Reset Handler
Reset_Handler PROC
EXPORT Reset_Handler
BL main
B .
ENDP
main PROC
EXPORT main
MOV R0, #0
MOV R0, #0x27
MOV R1, #5
ADD R0, R0, R1
LDR R2, =0x20000000
STRB R0, [R2, #0]
STRB R1, [R2, #1]
LDR R3, =0x20000000
LDRB R0, [R3, #0]
LDRB R0, [R3, #1]
BX LR ; Return from main
ENDP
ALIGN
END
Command
Running with Code Size Limit: 32K
Load "C:\\Users\\ecelab\\Desktop\\Sebasti
Memory 1
Address: R2
0x20000000: 2C 05 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0x20000014: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00


Registers
Register
Value
Core
R0 0x0000002C
R1 0x00000005
R2 0x20000000
R3 0x20000000
R4 0x00000000
R5 0x00000000
R6 0x00000000
R7 0x00000000
R8 0x00000000
R9 0x00000000
R10 0x00000000
R11 0x00000000
R12 0x00000000
R13 (SP) 0x20018000
R14 (LR) 0x080000C9
R15 (PC) 0x080000E4
xPSR 0x01000000
Banked
System
Internal
Mode Thread
Privilege Privileged
Stack MSP
States 13
Sec 0.00000108
FPU
Project
Registers
Disassembly
0x080000DC 7010 STRB r0,[r2,#0x00]
33: STRB R1, [R2, #1]
0x080000DE 7051 STRB r1,[r2,#0x01]
34: LDR R3, =0x20000000
0x080000E0 F04F5300 MOV r3,#0x20000000
35: LDRB R0, [R3, #0]
0x080000E4 7818 LDRB r0,[r3,#0x00]
36: LDRB R0, [R3, #1]
main.s
15
16 ; Reset Handler
17 Reset_Handler PROC
18 EXPORT Reset_Handler
19 BL main
20 B .
21 ENDP
22
23
24 main PROC
25 EXPORT main
26
27 MOV R0, #0
28 MOV R0, #0x27
29 MOV R1, #5
30 ADD R0, R0, R1
31 LDR R2, =0x20000000
32 STRB R0, [R2, #0]
33 STRB R1, [R2, #1]
34 LDR R3, =0x20000000
35 LDRB R0, [R3, #0]
36 LDRB R0, [R3, #1]
37
38 BX LR ; Return from main
39 ENDP
40
41 ALIGN
42 END
Command
Running with Code Size Limit: 32K
Load "C:\\Users\\ecelab\\Desktop\\Sebasti
Memory 1
Address: R3
0x20000000: 2C 05 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0x20000014: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0x20000028: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00


Registers
Register
Value
Core
R0 0x0000002C
R1 0x00000005
R2 0x20000000
R3 0x20000000
R4 0x00000000
R5 0x00000000
R6 0x00000000
R7 0x00000000
R8 0x00000000
R9 0x00000000
R10 0x00000000
R11 0x00000000
R12 0x00000000
R13 (SP) 0x20018000
R14 (LR) 0x080000C9
R15 (PC) 0x080000E6
xPSR 0x01000000
Banked
System
Internal
Mode Thread
Privilege Privileged
Stack MSP
States 15
Sec 0.00000125
FPU
Project
Registers
Disassembly
0x080000E4 7818 LDRB r0,[r3,#0x00]
36: LDRB R0, [R3, #1]
37:
0x080000E6 7858 LDRB r0,[r3,#0x01]
38: BX LR ; Return from main
0x080000E8 4770 BX lr
0x080000EA 0000 DCW 0x4675
0x080000EC 4675 MOV r5,lr
main.s
; Reset Handler
Reset_Handler PROC
EXPORT Reset_Handler
BL main
B .
ENDP
main PROC
EXPORT main
MOV R0, #0
MOV R0, #0x27
MOV R1, #5
ADD R0, R0, R1
LDR R2, =0x20000000
STRB R0, [R2, #0]
STRB R1, [R2, #1]
LDR R3, =0x20000000
LDRB R0, [R3, #0]
LDRB R0, [R3, #1]
BX LR ; Return from main
ENDP
ALIGN
END
Command
Running with Code Size Limit: 32K
Load "C:\\Users\\ecelab\\Desktop\\Sebasti
Memory 1
Address: R3
0x20000000: 2C 05 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0x20000014: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00

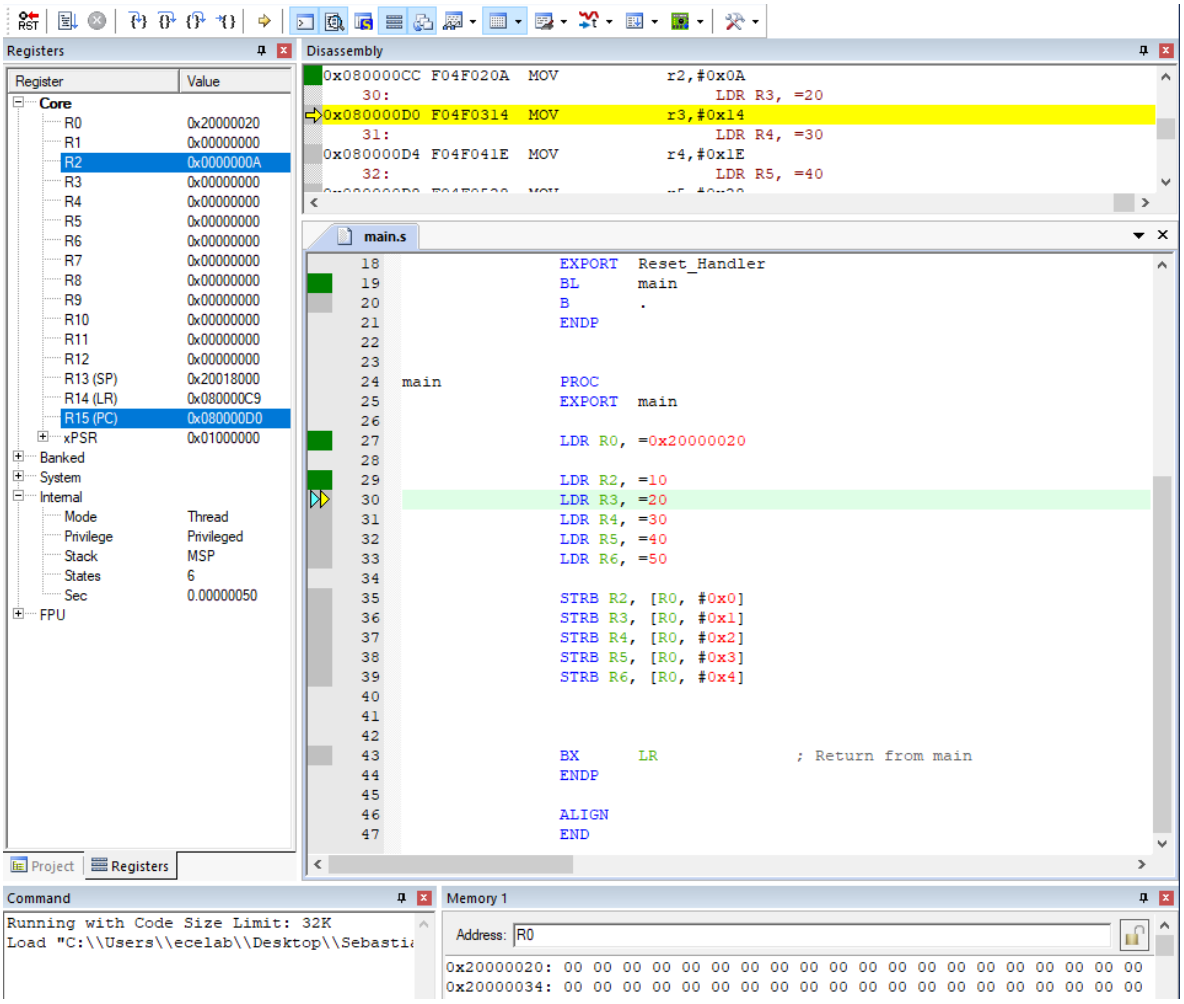

Registers
Register
Value
Core
R0 0x00000005
R1 0x00000005
R2 0x20000000
R3 0x20000000
R4 0x00000000
R5 0x00000000
R6 0x00000000
R7 0x00000000
R8 0x00000000
R9 0x00000000
R10 0x00000000
R11 0x00000000
R12 0x00000000
R13 (SP) 0x20018000
R14 (LR) 0x080000C9
R15 (PC) 0x080000E8
xPSR 0x01000000
Banked
System
Internal
Mode Thread
Privilege Privileged
Stack MSP
States 17
Sec 0.00000142
FPU
Project
Registers
Disassembly
0x080000E4 7818 LDRB r0,[r3,#0x00]
36: LDRB R0, [R3, #1]
37:
0x080000E6 7858 LDRB r0,[r3,#0x01]
38: BX LR ; Return from main
0x080000E8 4770 BX lr
0x080000EA 0000 DCW 0x4675
0x080000EC 4675 MOV r5,lr
main.s
; Reset Handler
Reset_Handler PROC
EXPORT Reset_Handler
BL main
B .
ENDP
main PROC
EXPORT main
MOV R0, #0
MOV R0, #0x27
MOV R1, #5
ADD R0, R0, R1
LDR R2, =0x20000000
STRB R0, [R2, #0]
STRB R1, [R2, #1]
LDR R3, =0x20000000
LDRB R0, [R3, #0]
LDRB R0, [R3, #1]
BX LR ; Return from main
ENDP
ALIGN
END
Command
Running with Code Size Limit: 32K
Load "C:\\Users\\ecelab\\Desktop\\Sebasti
Memory 1
Address: R3
0x20000000: 2C 05 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0x20000014: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0x20000028: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0x2000003C: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00

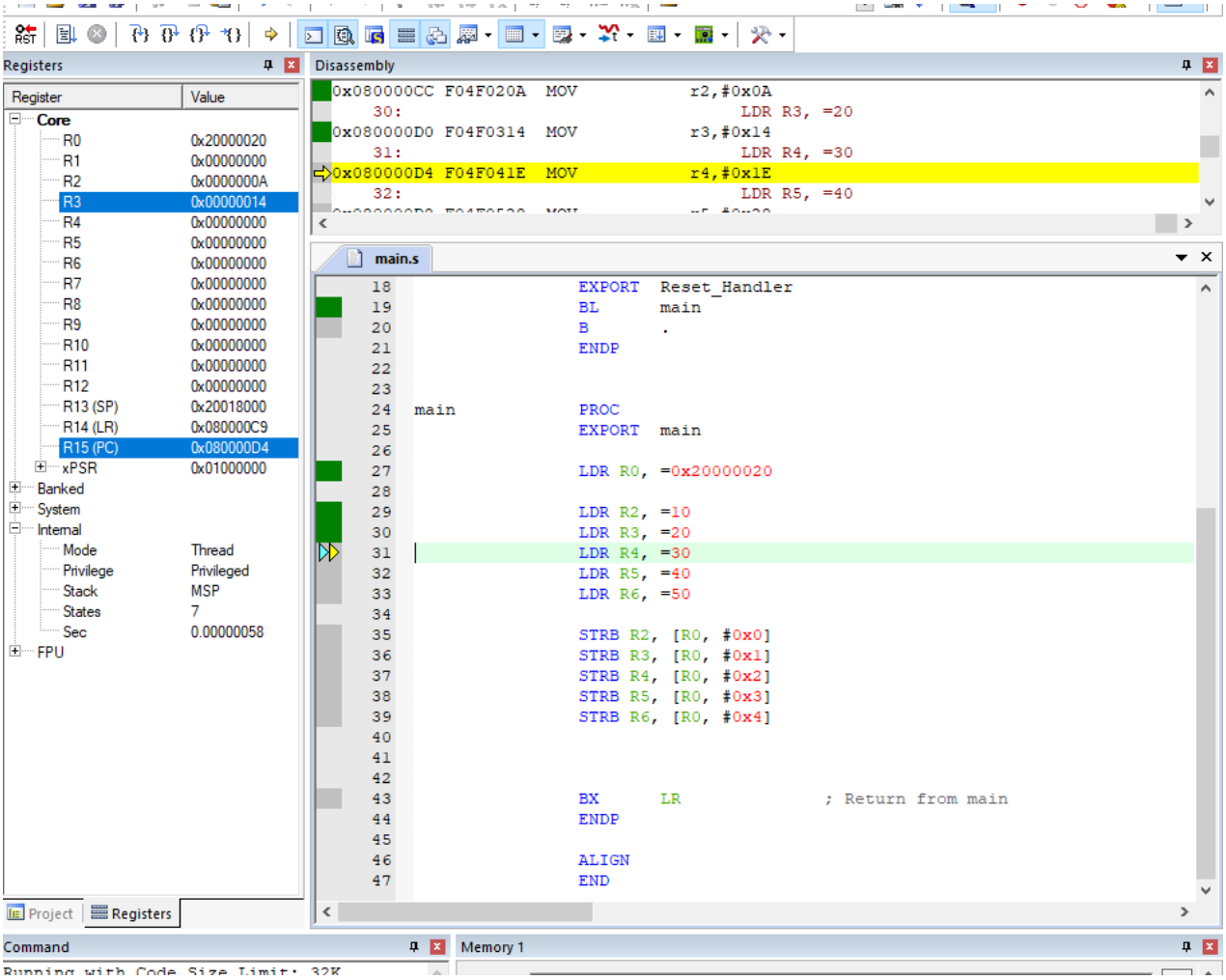
Task 2:

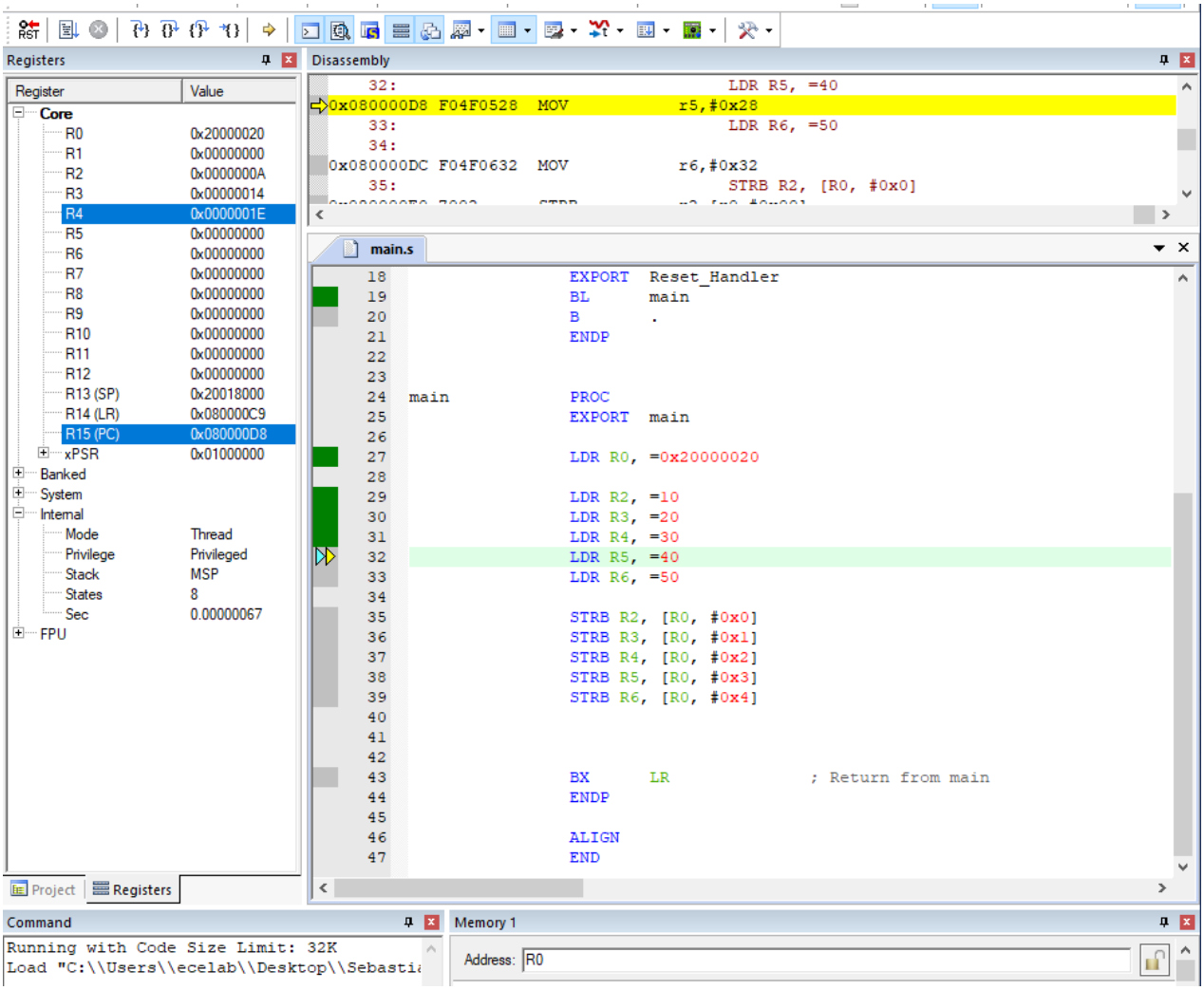
- ARM Assembly code.
- Screenshot of each step of the program.

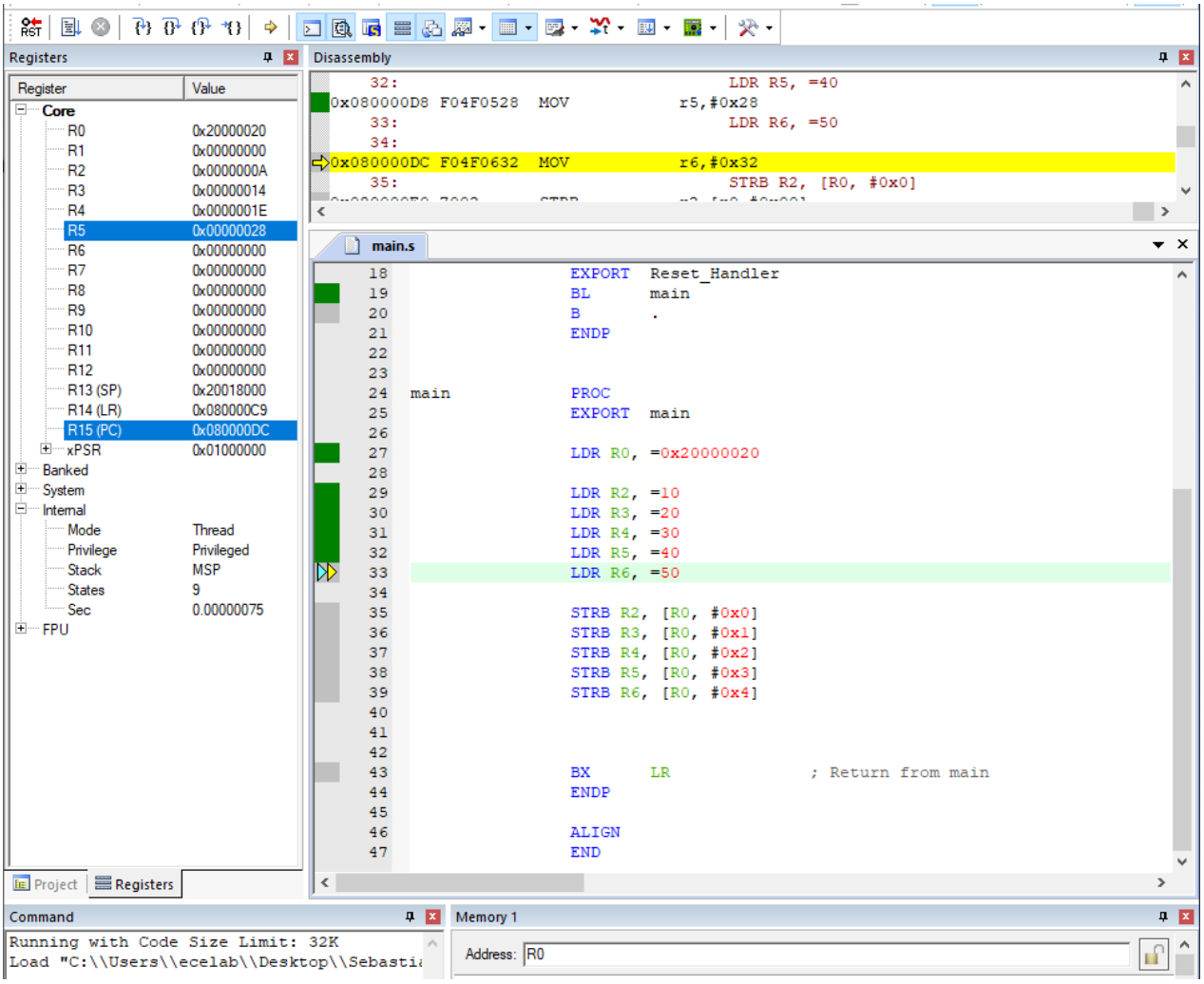
Task 3:

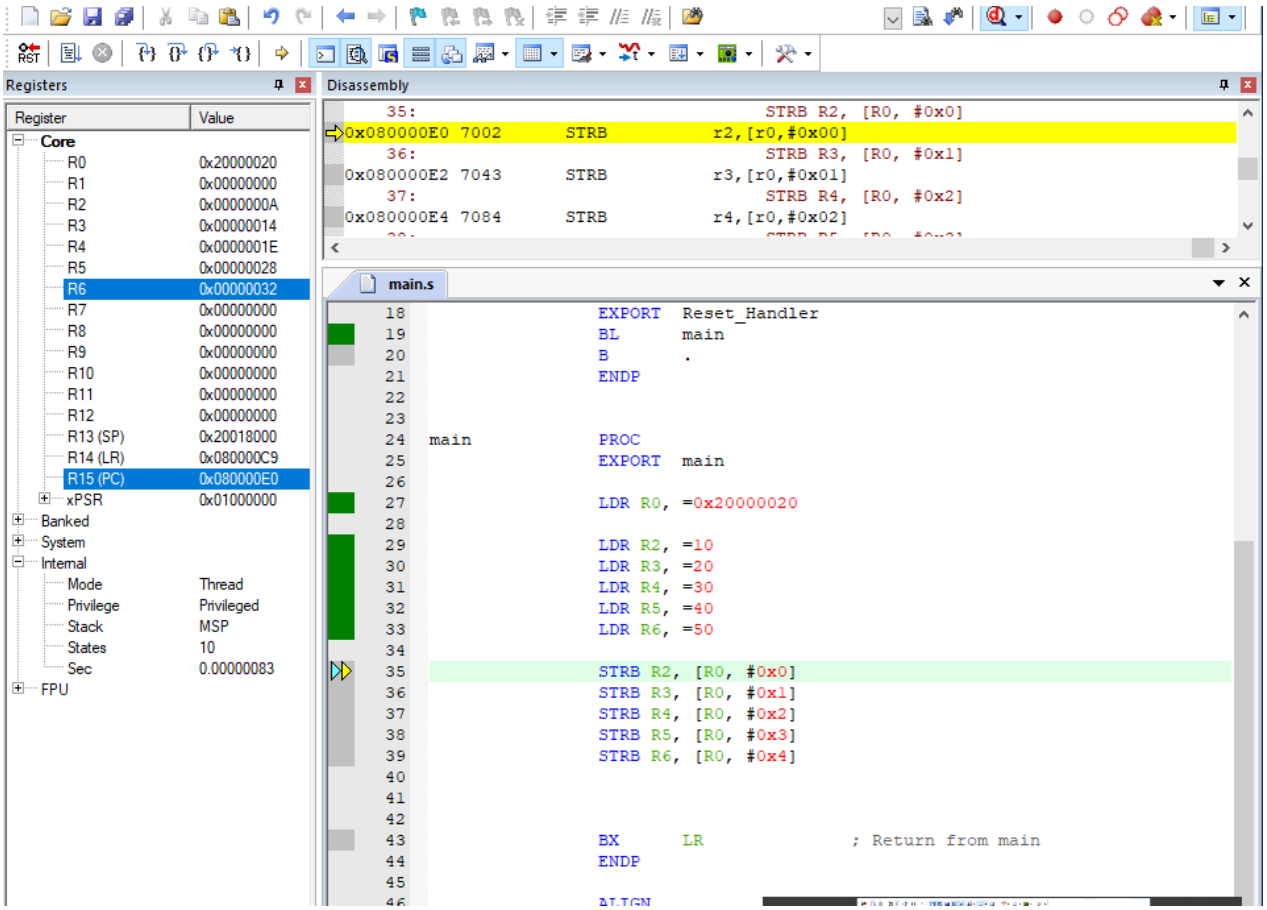
- ASM code version 1 (Immediate Offset).


Registers
Register
Value
Core
R0 0x20000020
R1 0x00000000
R2 0x0000000A
R3 0x00000000
R4 0x00000000
R5 0x00000000
R6 0x00000000
R7 0x00000000
R8 0x00000000
R9 0x00000000
R10 0x00000000
R11 0x00000000
R12 0x00000000
R13 (SP) 0x20018000
R14 (LR) 0x080000C9
R15 (PC) 0x080000D0
xPSR 0x01000000
Banked
System
Internal
Mode Thread
Privilege Privileged
Stack MSP
States 6
Sec 0.00000050
FPU
Project
Registers
Disassembly
0x080000CC F04F020A MOV r2,#0x0A
30: LDR R3, =20
0x080000D0 F04F0314 MOV r3,#0x14
31: LDR R4, =30
0x080000D4 F04F041E MOV r4,#0x1E
32: LDR R5, =40
main.s
EXPORT Reset_Handler
BL main
B .
ENDP
main PROC
EXPORT main
LDR R0, =0x20000020
LDR R2, =10
LDR R3, =20
LDR R4, =30
LDR R5, =40
LDR R6, =50
STRB R2, [R0, #0x0]
STRB R3, [R0, #0x1]
STRB R4, [R0, #0x2]
STRB R5, [R0, #0x3]
STRB R6, [R0, #0x4]
BX LR ; Return from main
ENDP
ALIGN
END
Command
Running with Code Size Limit: 32K
Load "C:\\Users\\ecelab\\Desktop\\Sebasti
Memory 1
Address: R0
0x20000020: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0x20000034: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00

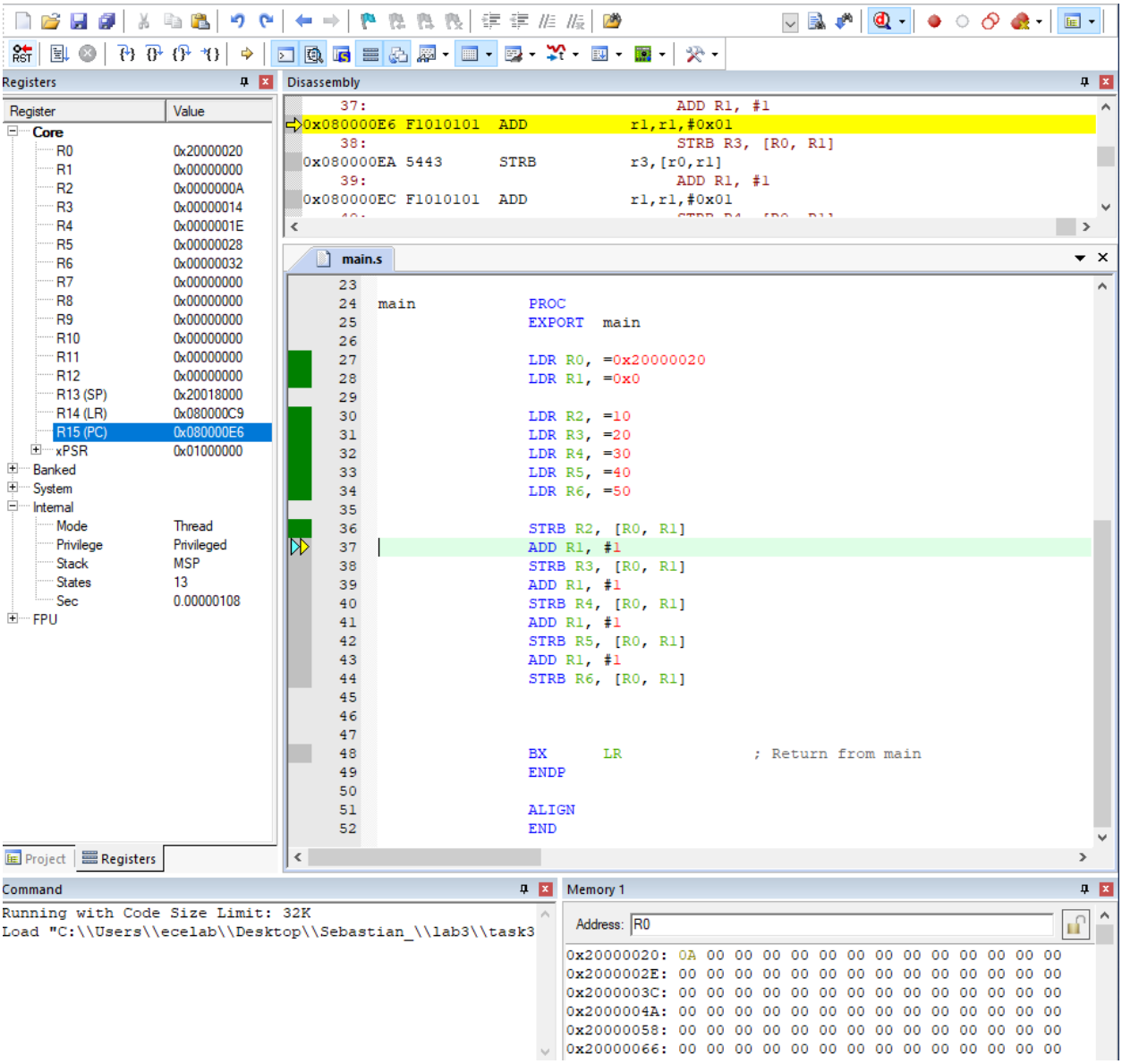

Registers
Register
Value
Core
R0 0x20000020
R1 0x00000000
R2 0x0000000A
R3 0x00000014
R4 0x00000000
R5 0x00000000
R6 0x00000000
R7 0x00000000
R8 0x00000000
R9 0x00000000
R10 0x00000000
R11 0x00000000
R12 0x00000000
R13 (SP) 0x20018000
R14 (LR) 0x080000C9
R15 (PC) 0x080000D4
xPSR 0x01000000
Banked
System
Internal
Mode Thread
Privilege Privileged
Stack MSP
States 7
Sec 0.00000058
FPU
Project
Registers
Disassembly
0x080000CC F04F020A MOV r2,#0x0A
30: LDR R3, =20
0x080000D0 F04F0314 MOV r3,#0x14
31: LDR R4, =30
0x080000D4 F04F041E MOV r4,#0x1E
32: LDR R5, =40
main.s
18 EXPORT Reset_Handler
19 BL main
20 B .
21 ENDP
22
23
24 main PROC
25 EXPORT main
26
27 LDR R0, =0x20000020
28
29 LDR R2, =10
30 LDR R3, =20
31 LDR R4, =30
32 LDR R5, =40
33 LDR R6, =50
34
35 STRB R2, [R0, #0x0]
36 STRB R3, [R0, #0x1]
37 STRB R4, [R0, #0x2]
38 STRB R5, [R0, #0x3]
39 STRB R6, [R0, #0x4]
40
41
42
43 BX LR ; Return from main
44 ENDP
45
46 ALIGN
47 END
Command
Memory 1

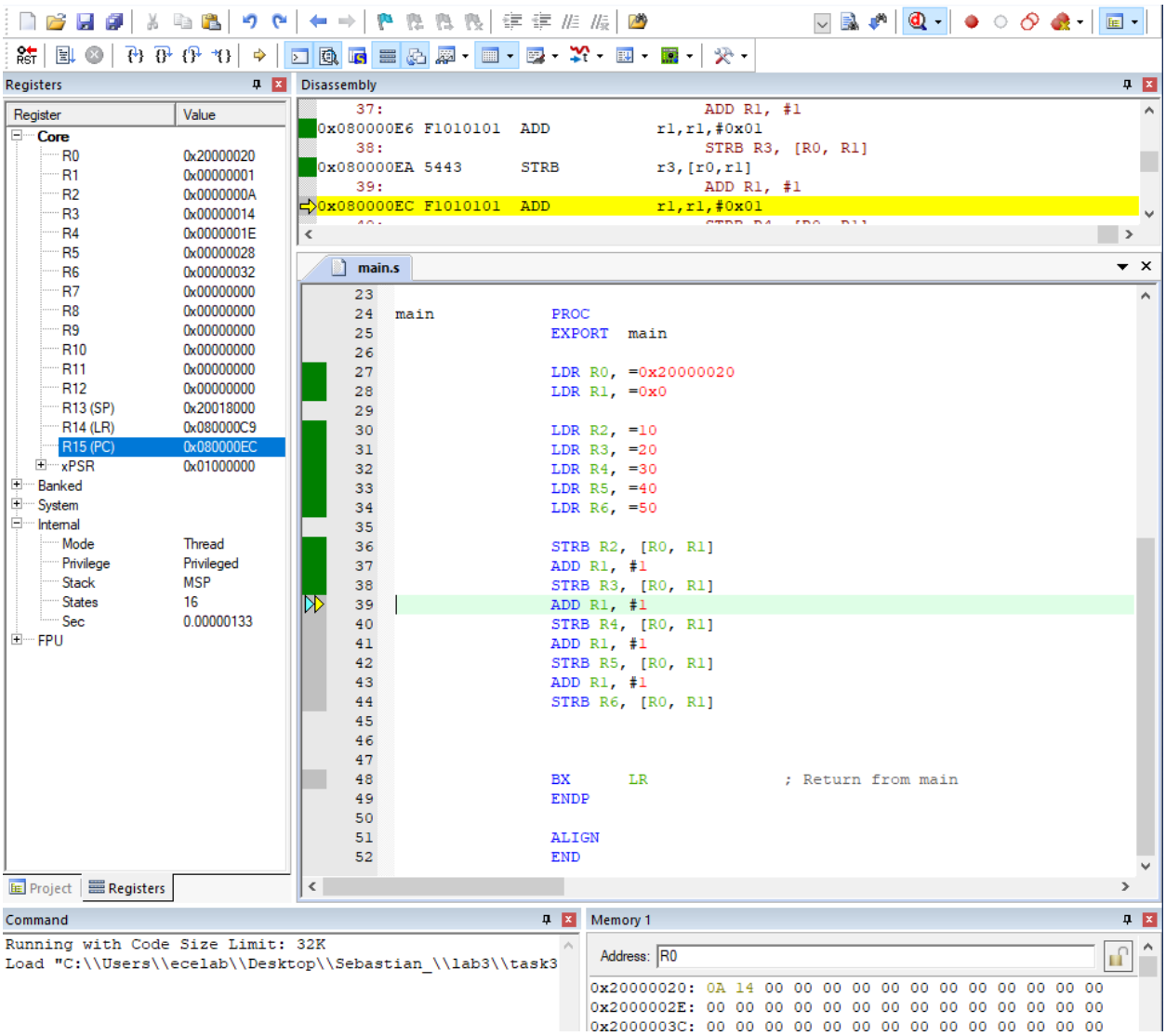

Registers
Register
Value
Core
R0 0x20000020
R1 0x00000000
R2 0x0000000A
R3 0x00000014
R4 0x0000001E
R5 0x00000000
R6 0x00000000
R7 0x00000000
R8 0x00000000
R9 0x00000000
R10 0x00000000
R11 0x00000000
R12 0x00000000
R13 (SP) 0x20018000
R14 (LR) 0x080000C9
R15 (PC) 0x080000D8
xPSR 0x01000000
Banked
System
Internal
Mode Thread
Privilege Privileged
Stack MSP
States 8
Sec 0.00000067
FPU
Project
Registers
Disassembly
32: LDR R5, =40
0x080000D8 F04F0528 MOV r5,#0x28
33: LDR R6, =50
34:
0x080000DC F04F0632 MOV r6,#0x32
35: STRB R2, [R0, #0x0]
main.s
18 EXPORT Reset_Handler
19 BL main
20 B .
21 ENDP
22
23
24 main PROC
25 EXPORT main
26
27 LDR R0, =0x20000020
28
29 LDR R2, =10
30 LDR R3, =20
31 LDR R4, =30
32 LDR R5, =40
33 LDR R6, =50
34
35 STRB R2, [R0, #0x0]
36 STRB R3, [R0, #0x1]
37 STRB R4, [R0, #0x2]
38 STRB R5, [R0, #0x3]
39 STRB R6, [R0, #0x4]
40
41
42
43 BX LR ; Return from main
44 ENDP
45
46 ALIGN
47 END
Command
Running with Code Size Limit: 32K
Load "C:\\Users\\ecelab\\Desktop\\Sebasti
Memory 1
Address: R0

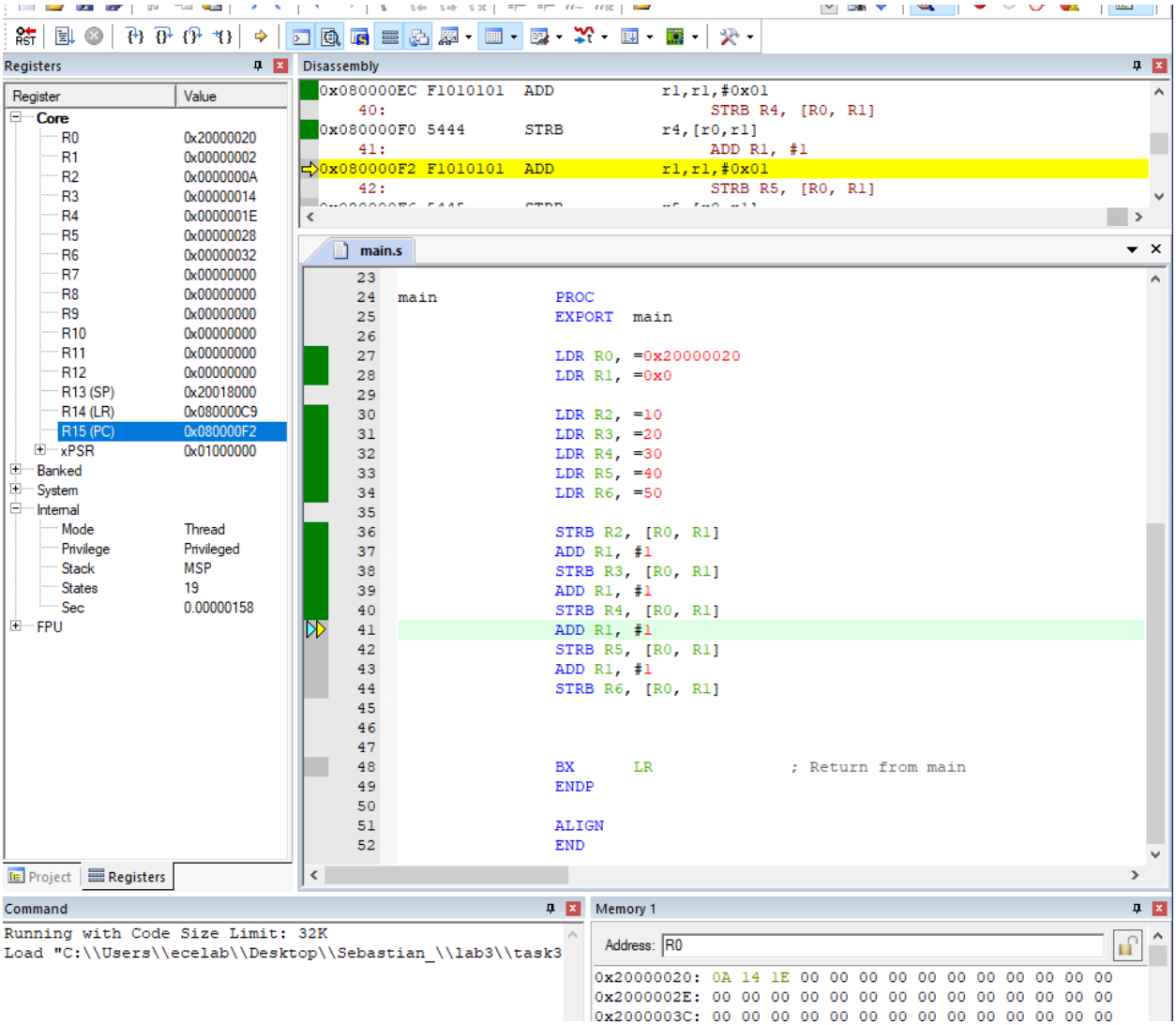

Registers
Register
Value
Core
R0 0x20000020
R1 0x00000000
R2 0x0000000A
R3 0x00000014
R4 0x0000001E
R5 0x00000028
R6 0x00000000
R7 0x00000000
R8 0x00000000
R9 0x00000000
R10 0x00000000
R11 0x00000000
R12 0x00000000
R13 (SP) 0x20018000
R14 (LR) 0x080000C9
R15 (PC) 0x080000DC
xPSR 0x01000000
Banked
System
Internal
Mode Thread
Privilege Privileged
Stack MSP
States 9
Sec 0.00000075
FPU
Project
Registers
Disassembly
32: LDR R5, =40
0x080000D8 F04F0528 MOV r5,#0x28
33: LDR R6, =50
34:
0x080000DC F04F0632 MOV r6,#0x32
35: STRB R2, [R0, #0x0]
main.s
18 EXPORT Reset_Handler
19 BL main
20 B .
21 ENDP
22
23
24 main PROC
25 EXPORT main
26
27 LDR R0, =0x20000020
28
29 LDR R2, =10
30 LDR R3, =20
31 LDR R4, =30
32 LDR R5, =40
33 LDR R6, =50
34
35 STRB R2, [R0, #0x0]
36 STRB R3, [R0, #0x1]
37 STRB R4, [R0, #0x2]
38 STRB R5, [R0, #0x3]
39 STRB R6, [R0, #0x4]
40
41
42
43 BX LR ; Return from main
44 ENDP
45
46 ALIGN
47 END
Command
Running with Code Size Limit: 32K
Load "C:\\Users\\ecelab\\Desktop\\Sebasti
Memory 1
Address: R0

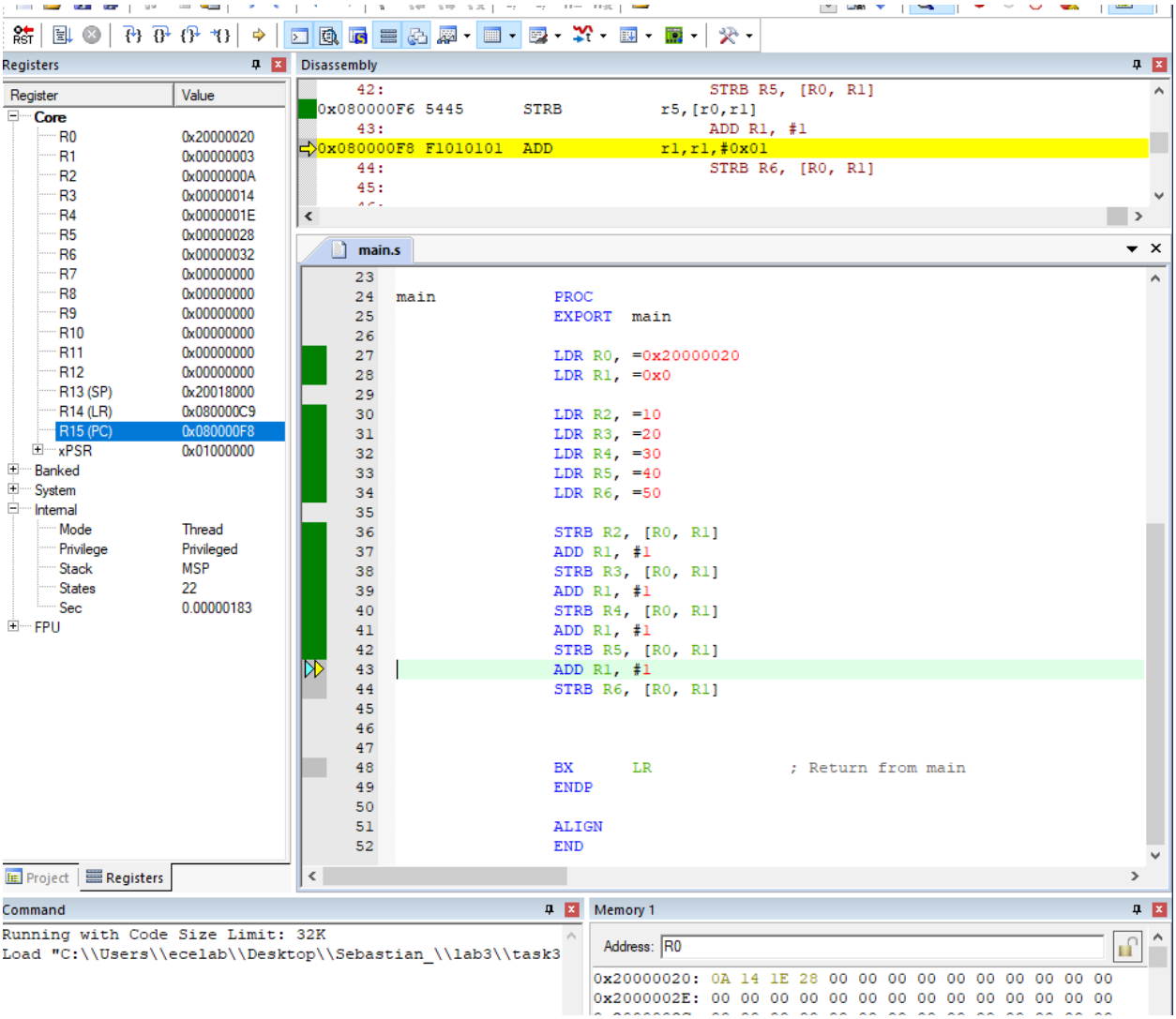

Registers
Register
Value
Core
R0 0x20000020
R1 0x00000000
R2 0x0000000A
R3 0x00000014
R4 0x0000001E
R5 0x00000028
R6 0x00000032
R7 0x00000000
R8 0x00000000
R9 0x00000000
R10 0x00000000
R11 0x00000000
R12 0x00000000
R13 (SP) 0x20018000
R14 (LR) 0x080000C9
R15 (PC) 0x080000E0
xPSR 0x01000000
Banked
System
Internal
Mode Thread
Privilege Privileged
Stack MSP
States 10
Sec 0.00000083
FPU
Disassembly
35: STRB R2, [R0, #0x0]
0x080000E0 7002 STRB r2,[r0,#0x00]
36: STRB R3, [R0, #0x1]
0x080000E2 7043 STRB r3,[r0,#0x01]
37: STRB R4, [R0, #0x2]
0x080000E4 7084 STRB r4,[r0,#0x02]
main.s
EXPORT Reset_Handler
BL main
B .
ENDP
main PROC
EXPORT main
LDR R0, =0x20000020
LDR R2, =10
LDR R3, =20
LDR R4, =30
LDR R5, =40
LDR R6, =50
STRB R2, [R0, #0x0]
STRB R3, [R0, #0x1]
STRB R4, [R0, #0x2]
STRB R5, [R0, #0x3]
STRB R6, [R0, #0x4]
BX LR ; Return from main
ENDP

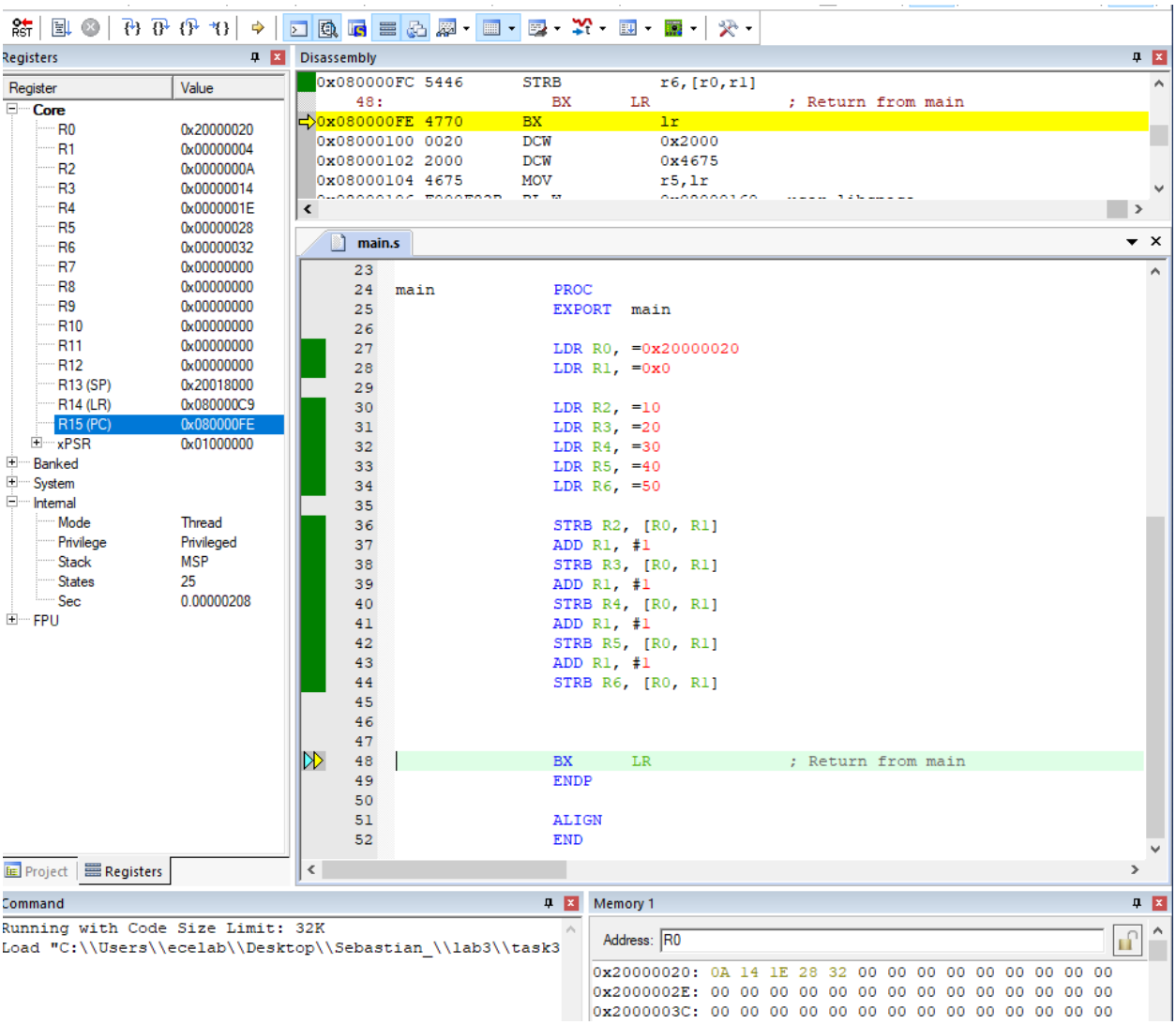
● ASM code version 2 (Register Offset).




Registers
Register
Value
Core
R0 0x20000020
R1 0x00000001
R2 0x0000000A
R3 0x00000014
R4 0x0000001E
R5 0x00000028
R6 0x00000032
R7 0x00000000
R8 0x00000000
R9 0x00000000
R10 0x00000000
R11 0x00000000
R12 0x00000000
R13 (SP) 0x20018000
R14 (LR) 0x080000C9
R15 (PC) 0x080000EC
xPSR 0x01000000
Banked
System
Internal
Mode Thread
Privilege Privileged
Stack MSP
States 16
Sec 0.00000133
FPU
Project
Registers
Disassembly
37: ADD R1, #1
0x080000E6 F1010101 ADD r1,r1,#0x01
38: STRB R3, [R0, R1]
0x080000EA 5443 STRB r3,[r0,r1]
39: ADD R1, #1
0x080000EC F1010101 ADD r1,r1,#0x01
main.s
main PROC
EXPORT main
LDR R0, =0x20000020
LDR R1, =0x0
LDR R2, =10
LDR R3, =20
LDR R4, =30
LDR R5, =40
LDR R6, =50
STRB R2, [R0, R1]
ADD R1, #1
STRB R3, [R0, R1]
ADD R1, #1
STRB R4, [R0, R1]
ADD R1, #1
STRB R5, [R0, R1]
ADD R1, #1
STRB R6, [R0, R1]
BX LR ; Return from main
ENDP
ALIGN
END
Command
Running with Code Size Limit: 32K
Load "C:\\Users\\ecelab\\Desktop\\Sebastian_\\lab3\\task3
Memory 1
Address: R0
0x20000020: 0A 14 00 00 00 00 00 00 00 00 00 00 00 00
0x2000002E: 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0x2000003C: 00 00 00 00 00 00 00 00 00 00 00 00 00 00

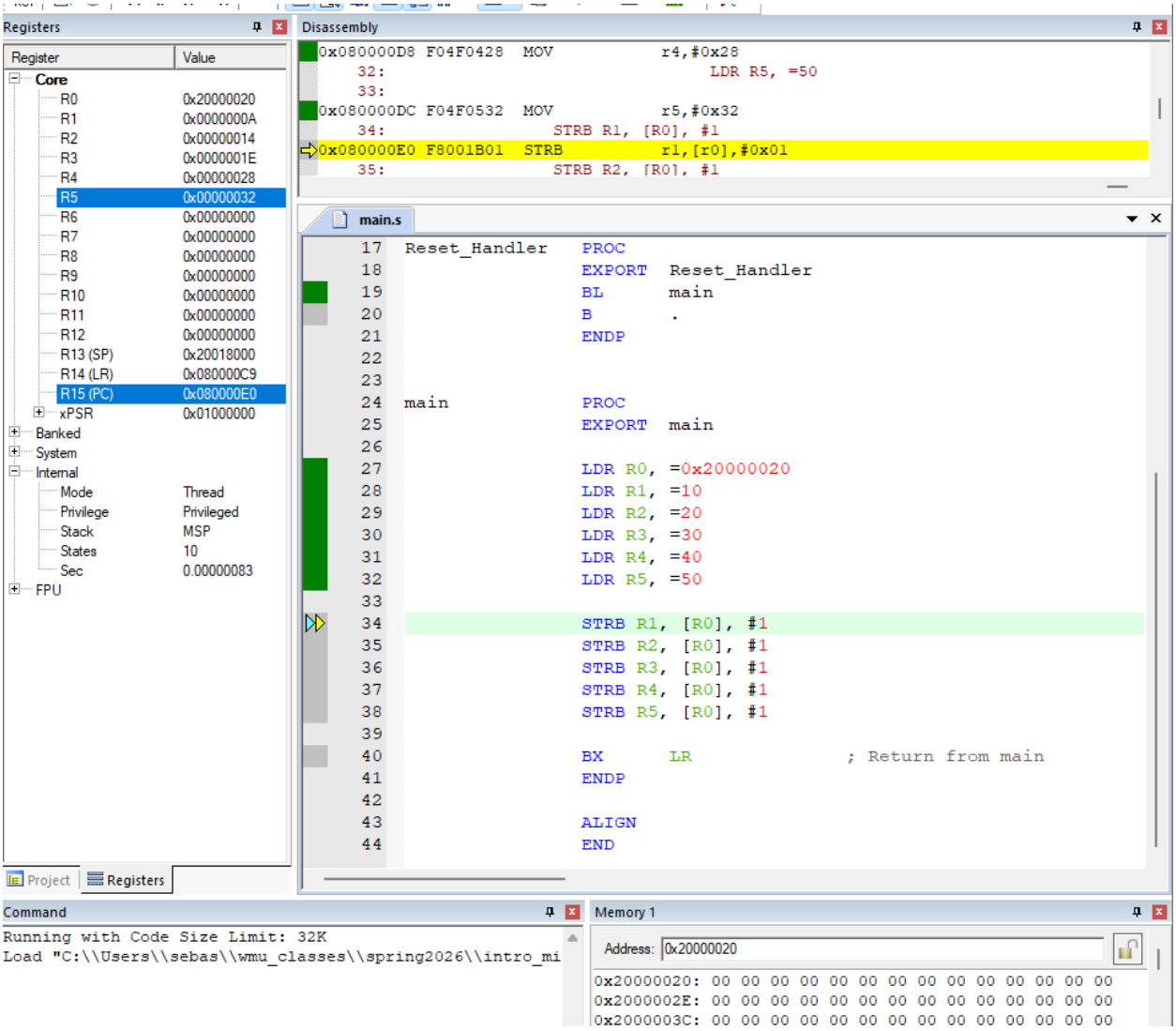

Registers
Register
Value
Core
R0 0x20000020
R1 0x00000002
R2 0x0000000A
R3 0x00000014
R4 0x0000001E
R5 0x00000028
R6 0x00000032
R7 0x00000000
R8 0x00000000
R9 0x00000000
R10 0x00000000
R11 0x00000000
R12 0x00000000
R13 (SP) 0x20018000
R14 (LR) 0x080000C9
R15 (PC) 0x080000F2
xPSR 0x01000000
Banked
System
Internal
Mode Thread
Privilege Privileged
Stack MSP
States 19
Sec 0.00000158
FPU
Project
Registers
Disassembly
0x080000EC F1010101 ADD r1,r1,#0x01
40: STRB R4, [R0, R1]
0x080000F0 5444 STRB r4,[r0,r1]
41: ADD R1, #1
0x080000F2 F1010101 ADD r1,r1,#0x01
42: STRB R5, [R0, R1]
main.s
23
24 main PROC
25 EXPORT main
26
27 LDR R0, =0x20000020
28 LDR R1, =0x0
29
30 LDR R2, =10
31 LDR R3, =20
32 LDR R4, =30
33 LDR R5, =40
34 LDR R6, =50
35
36 STRB R2, [R0, R1]
37 ADD R1, #1
38 STRB R3, [R0, R1]
39 ADD R1, #1
40 STRB R4, [R0, R1]
41 ADD R1, #1
42 STRB R5, [R0, R1]
43 ADD R1, #1
44 STRB R6, [R0, R1]
45
46
47
48 BX LR ; Return from main
49 ENDP
50
51 ALIGN
52 END
Command
Running with Code Size Limit: 32K
Load "C:\\Users\\ecelab\\Desktop\\Sebastian_\\lab3\\task3
Memory 1
Address: R0
0x20000020: 0A 14 1E 00 00 00 00 00 00 00 00 00 00 00
0x2000002E: 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0x2000003C: 00 00 00 00 00 00 00 00 00 00 00 00 00 00

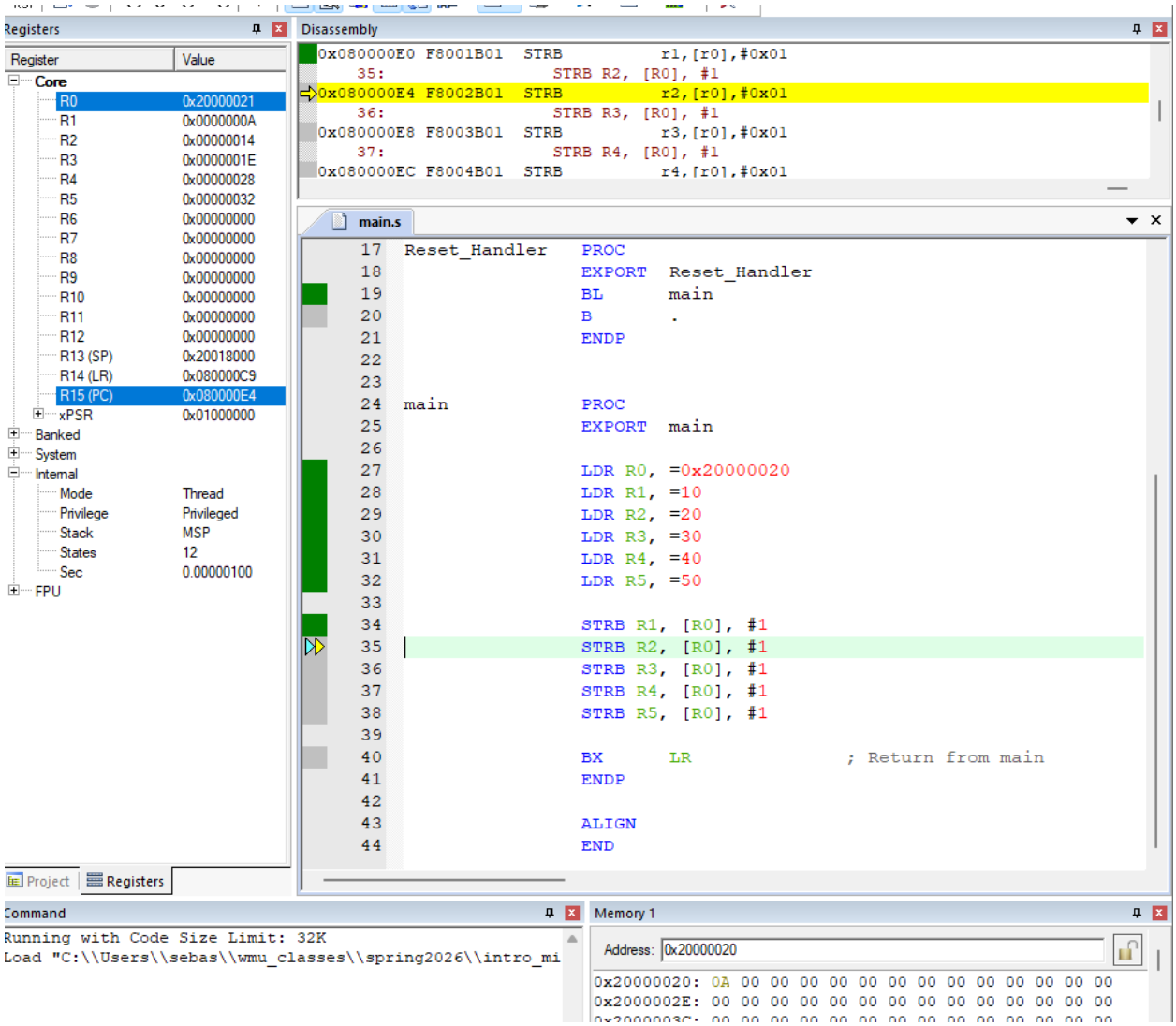

Registers
Register
Value
Core
R0 0x20000020
R1 0x00000003
R2 0x0000000A
R3 0x00000014
R4 0x0000001E
R5 0x00000028
R6 0x00000032
R7 0x00000000
R8 0x00000000
R9 0x00000000
R10 0x00000000
R11 0x00000000
R12 0x00000000
R13 (SP) 0x20018000
R14 (LR) 0x080000C9
R15 (PC) 0x080000F8
xPSR 0x01000000
Banked
System
Internal
Mode Thread
Privilege Privileged
Stack MSP
States 22
Sec 0.00000183
FPU
Project
Registers
Disassembly
42: STRB R5, [R0, R1]
0x080000F6 5445 STRB r5,[r0,r1]
43: ADD R1, #1
0x080000F8 F1010101 ADD r1,r1,#0x01
44: STRB R6, [R0, R1]
45:
main.s
23
24 main PROC
25 EXPORT main
26
27 LDR R0, =0x20000020
28 LDR R1, =0x0
29
30 LDR R2, =10
31 LDR R3, =20
32 LDR R4, =30
33 LDR R5, =40
34 LDR R6, =50
35
36 STRB R2, [R0, R1]
37 ADD R1, #1
38 STRB R3, [R0, R1]
39 ADD R1, #1
40 STRB R4, [R0, R1]
41 ADD R1, #1
42 STRB R5, [R0, R1]
43 ADD R1, #1
44 STRB R6, [R0, R1]
45
46
47
48 BX LR ; Return from main
49 ENDP
50
51 ALIGN
52 END
Command
Running with Code Size Limit: 32K
Load "C:\\Users\\ecelab\\Desktop\\Sebastian_\\lab3\\task3
Memory 1
Address: R0
0x20000020: 0A 14 1E 28 00 00 00 00 00 00 00 00 00 00
0x2000002E: 00 00 00 00 00 00 00 00 00 00 00 00 00 00

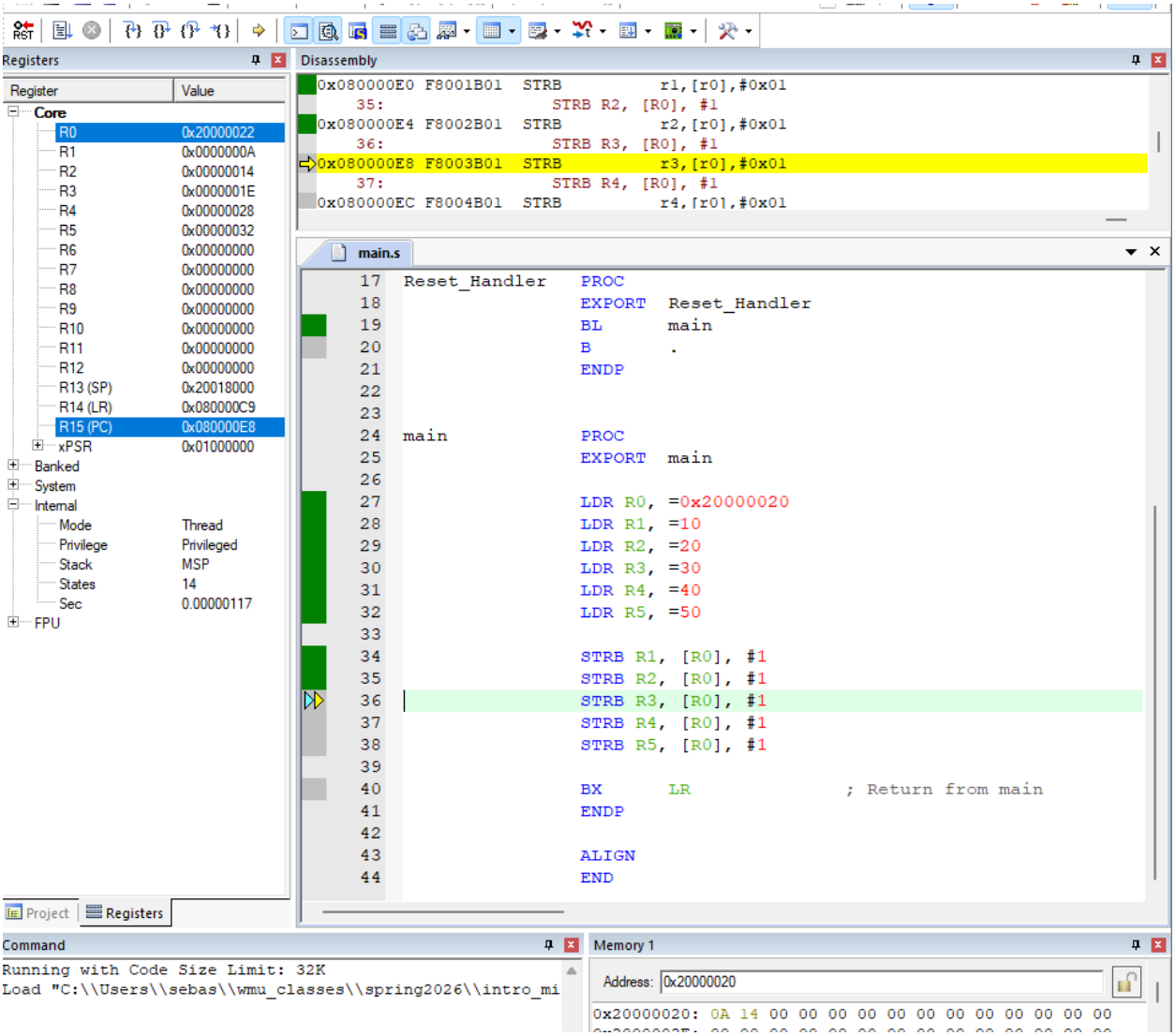

Registers
Register
Value
Core
R0 0x20000020
R1 0x00000004
R2 0x0000000A
R3 0x00000014
R4 0x0000001E
R5 0x00000028
R6 0x00000032
R7 0x00000000
R8 0x00000000
R9 0x00000000
R10 0x00000000
R11 0x00000000
R12 0x00000000
R13 (SP) 0x20018000
R14 (LR) 0x080000C9
R15 (PC) 0x080000FE
xPSR 0x01000000
Banked
System
Internal
Mode Thread
Privilege Privileged
Stack MSP
States 25
Sec 0.00000208
FPU
Project
Registers
Disassembly
0x080000FC 5446 STRB r6,[r0,r1]
48: BX LR ; Return from main
0x080000FE 4770 BX lr
0x08000100 0020 DCW 0x2000
0x08000102 2000 DCW 0x4675
0x08000104 4675 MOV r5,lr
main.s
23
24 main PROC
25 EXPORT main
26
27 LDR R0, =0x20000020
28 LDR R1, =0x0
29
30 LDR R2, =10
31 LDR R3, =20
32 LDR R4, =30
33 LDR R5, =40
34 LDR R6, =50
35
36 STRB R2, [R0, R1]
37 ADD R1, #1
38 STRB R3, [R0, R1]
39 ADD R1, #1
40 STRB R4, [R0, R1]
41 ADD R1, #1
42 STRB R5, [R0, R1]
43 ADD R1, #1
44 STRB R6, [R0, R1]
45
46
47
48 BX LR ; Return from main
49 ENDP
50
51 ALIGN
52 END
Command
Running with Code Size Limit: 32K
Load "C:\\Users\\ecelab\\Desktop\\Sebastian_\\lab3\\task3
Memory 1
Address: R0
0x20000020: 0A 14 1E 28 32 00 00 00 00 00 00 00 00 00
0x2000002E: 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0x2000003C: 00 00 00 00 00 00 00 00 00 00 00 00 00 00

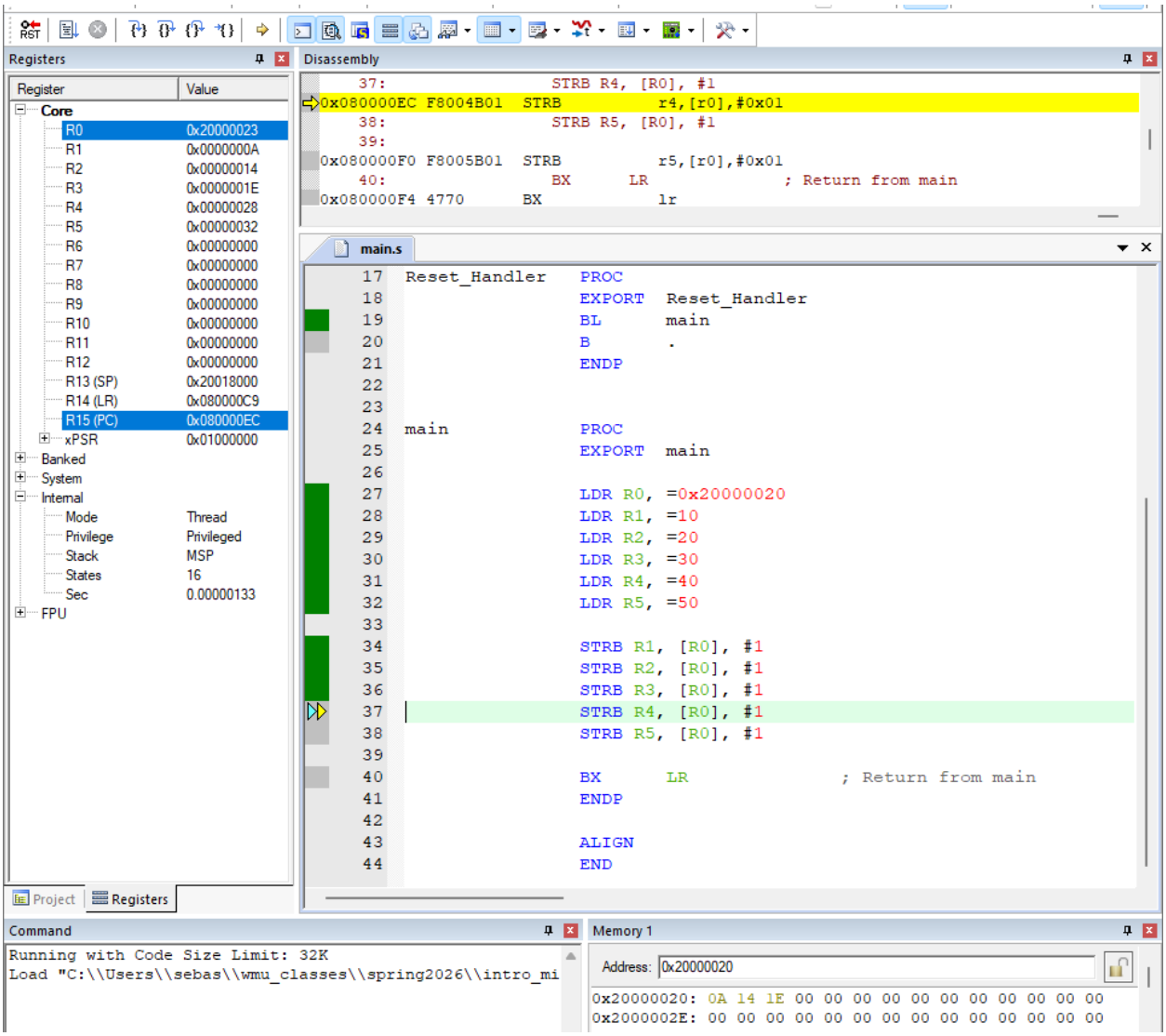
- ASM code version 3 (Post-Index).
- Screenshot showing the values in memory at the correct address.

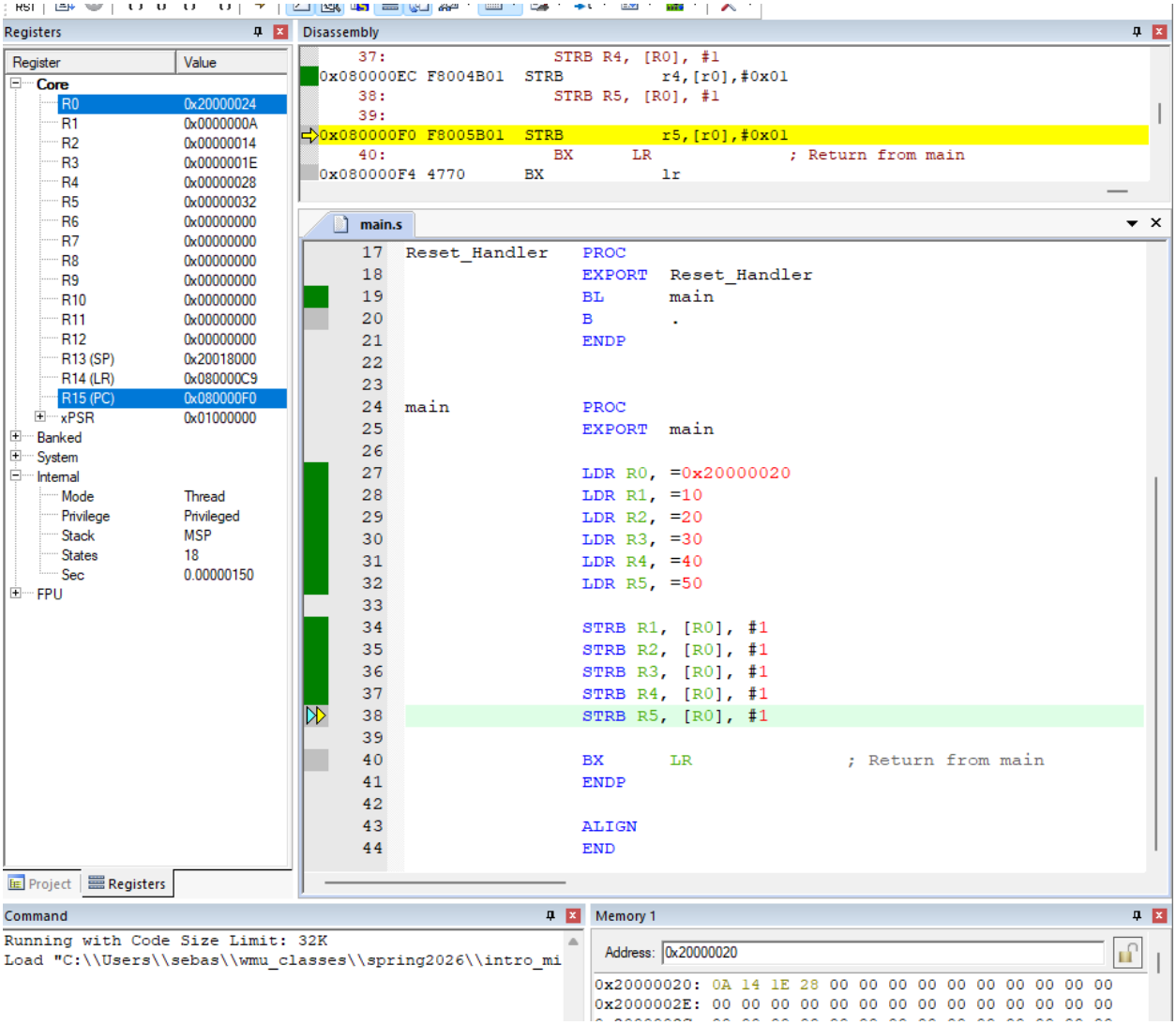
```
        THUMB

; Vector Table
        AREA    RESET, DATA, READONLY
        EXPORT  __Vectors

__Vectors
        DCD     0x20018000          ; Top of Stack (96KB RAM)
        DCD     Reset_Handler       ; Reset Handler

        AREA    |.text|, CODE, READONLY

; Reset Handler
Reset_Handler   PROC
                EXPORT  Reset_Handler
                BL      main
                B       .
                ENDP


main            PROC
                EXPORT  main

                LDR R0, =0x20000020
                LDR R1, #10             ; First value
                LDR R2, #20             ; Second value
                LDR R3, #30
                LDR R4, #40
                LDR R5, #50

                STRB R1, [R0], #1
                STRB R2, [R0], #1
                STRB R3, [R0], #1
                STRB R4, [R0], #1
                STRB R5, [R0], #1

                BX      LR              ; Return from main
                ENDP

                ALIGN
                END
```

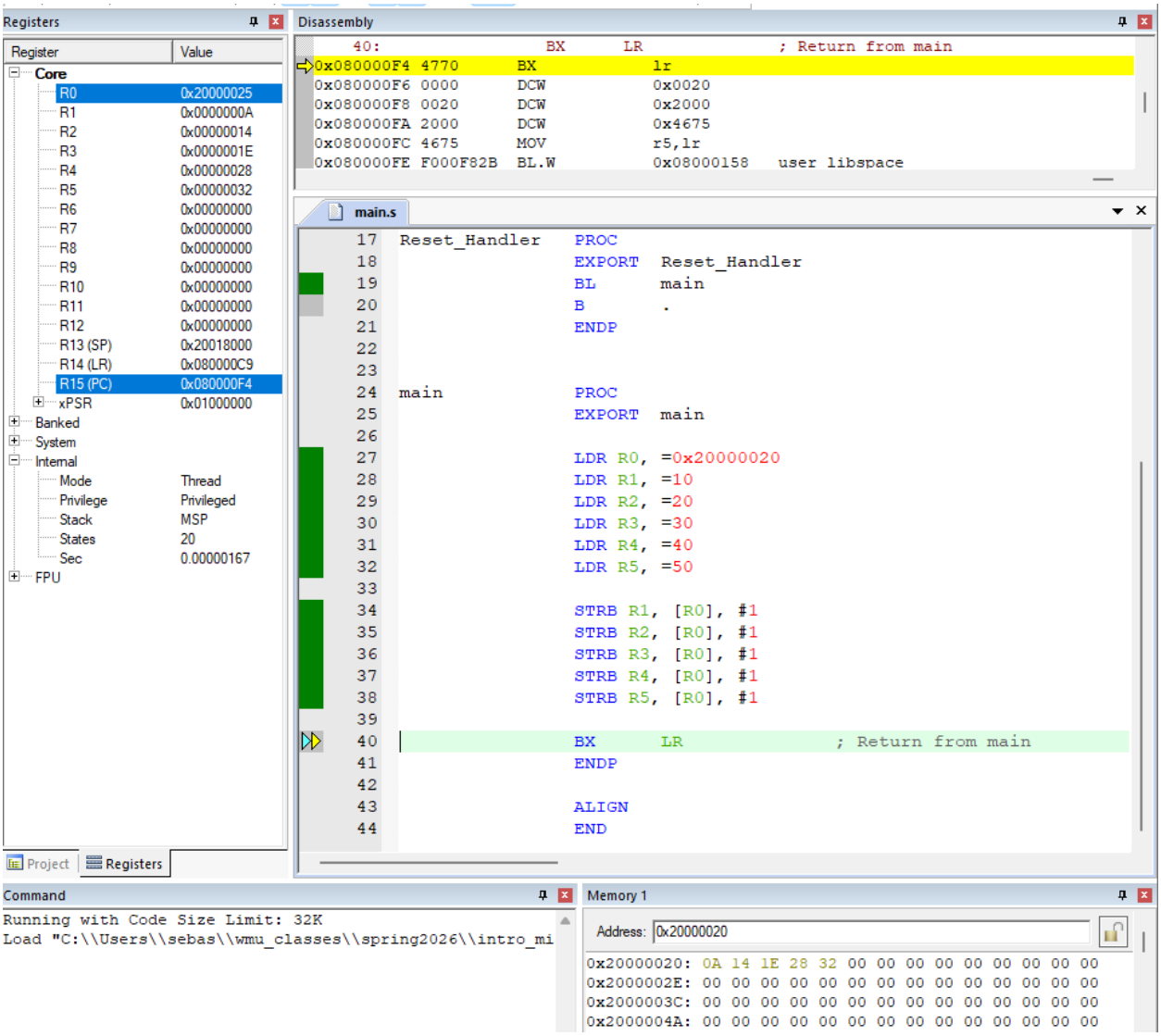

Registers
Register
Value
Core
R0 0x20000020
R1 0x0000000A
R2 0x00000014
R3 0x0000001E
R4 0x00000028
R5 0x00000032
R6 0x00000000
R7 0x00000000
R8 0x00000000
R9 0x00000000
R10 0x00000000
R11 0x00000000
R12 0x00000000
R13 (SP) 0x20018000
R14 (LR) 0x080000C9
R15 (PC) 0x080000E0
xPSR 0x01000000
Banked
System
Internal
Mode Thread
Privilege Privileged
Stack MSP
States 10
Sec 0.00000083
FPU
Project
Registers
Disassembly
0x080000D8 F04F0428 MOV r4,#0x28
32: LDR R5, =50
33:
0x080000DC F04F0532 MOV r5,#0x32
34: STRB R1, [R0], #1
0x080000E0 F8001B01 STRB r1,[r0],#0x01
35: STRB R2, [R0], #1
main.s
17 Reset_Handler PROC
18 EXPORT Reset_Handler
19 BL main
20 B .
21 ENDP
22
23
24 main PROC
25 EXPORT main
26
27 LDR R0, =0x20000020
28 LDR R1, =10
29 LDR R2, =20
30 LDR R3, =30
31 LDR R4, =40
32 LDR R5, =50
33
34 STRB R1, [R0], #1
35 STRB R2, [R0], #1
36 STRB R3, [R0], #1
37 STRB R4, [R0], #1
38 STRB R5, [R0], #1
39
40 BX LR ; Return from main
41 ENDP
42
43 ALIGN
44 END
Command
Running with Code Size Limit: 32K
Load "C:\\Users\\sebas\\wmu_classes\\spring2026\\intro_mi
Memory 1
Address: 0x20000020
0x20000020: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0x2000002E: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0x2000003C: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00


Registers
Register
Value
Core
R0 0x20000021
R1 0x0000000A
R2 0x00000014
R3 0x0000001E
R4 0x00000028
R5 0x00000032
R6 0x00000000
R7 0x00000000
R8 0x00000000
R9 0x00000000
R10 0x00000000
R11 0x00000000
R12 0x00000000
R13 (SP) 0x20018000
R14 (LR) 0x080000C9
R15 (PC) 0x080000E4
xPSR 0x01000000
Banked
System
Internal
Mode Thread
Privilege Privileged
Stack MSP
States 12
Sec 0.00000100
FPU
Project
Registers
Disassembly
0x080000E0 F8001B01 STRB r1,[r0],#0x01
35: STRB R2, [R0], #1
0x080000E4 F8002B01 STRB r2,[r0],#0x01
36: STRB R3, [R0], #1
0x080000E8 F8003B01 STRB r3,[r0],#0x01
37: STRB R4, [R0], #1
0x080000EC F8004B01 STRB r4,[r0],#0x01
main.s
17 Reset_Handler PROC
18 EXPORT Reset_Handler
19 BL main
20 B .
21 ENDP
22
23
24 main PROC
25 EXPORT main
26
27 LDR R0, =0x20000020
28 LDR R1, =10
29 LDR R2, =20
30 LDR R3, =30
31 LDR R4, =40
32 LDR R5, =50
33
34 STRB R1, [R0], #1
35 STRB R2, [R0], #1
36 STRB R3, [R0], #1
37 STRB R4, [R0], #1
38 STRB R5, [R0], #1
39
40 BX LR ; Return from main
41 ENDP
42
43 ALIGN
44 END
Command
Running with Code Size Limit: 32K
Load "C:\\Users\\sebas\\wmu_classes\\spring2026\\intro_mi
Memory 1
Address: 0x20000020
0x20000020: 0A 00 00 00 00 00 00 00 00 00 00 00 00 00
0x2000002E: 00 00 00 00 00 00 00 00 00 00 00 00 00 00


Registers
Register
Value
Core
R0 0x20000022
R1 0x0000000A
R2 0x00000014
R3 0x0000001E
R4 0x00000028
R5 0x00000032
R6 0x00000000
R7 0x00000000
R8 0x00000000
R9 0x00000000
R10 0x00000000
R11 0x00000000
R12 0x00000000
R13 (SP) 0x20018000
R14 (LR) 0x080000C9
R15 (PC) 0x080000E8
xPSR 0x01000000
Banked
System
Internal
Mode Thread
Privilege Privileged
Stack MSP
States 14
Sec 0.00000117
FPU
Project
Registers
Disassembly
0x080000E0 F8001B01 STRB r1,[r0],#0x01
35: STRB R2, [R0], #1
0x080000E4 F8002B01 STRB r2,[r0],#0x01
36: STRB R3, [R0], #1
0x080000E8 F8003B01 STRB r3,[r0],#0x01
37: STRB R4, [R0], #1
0x080000EC F8004B01 STRB r4,[r0],#0x01
main.s
17 Reset_Handler PROC
18 EXPORT Reset_Handler
19 BL main
20 B .
21 ENDP
22
23
24 main PROC
25 EXPORT main
26
27 LDR R0, =0x20000020
28 LDR R1, =10
29 LDR R2, =20
30 LDR R3, =30
31 LDR R4, =40
32 LDR R5, =50
33
34 STRB R1, [R0], #1
35 STRB R2, [R0], #1
36 STRB R3, [R0], #1
37 STRB R4, [R0], #1
38 STRB R5, [R0], #1
39
40 BX LR ; Return from main
41 ENDP
42
43 ALIGN
44 END
Command
Running with Code Size Limit: 32K
Load "C:\\Users\\sebas\\wmu_classes\\spring2026\\intro_mi
Memory 1
Address: 0x20000020
0x20000020: 0A 14 00 00 00 00 00 00 00 00 00 00 00 00

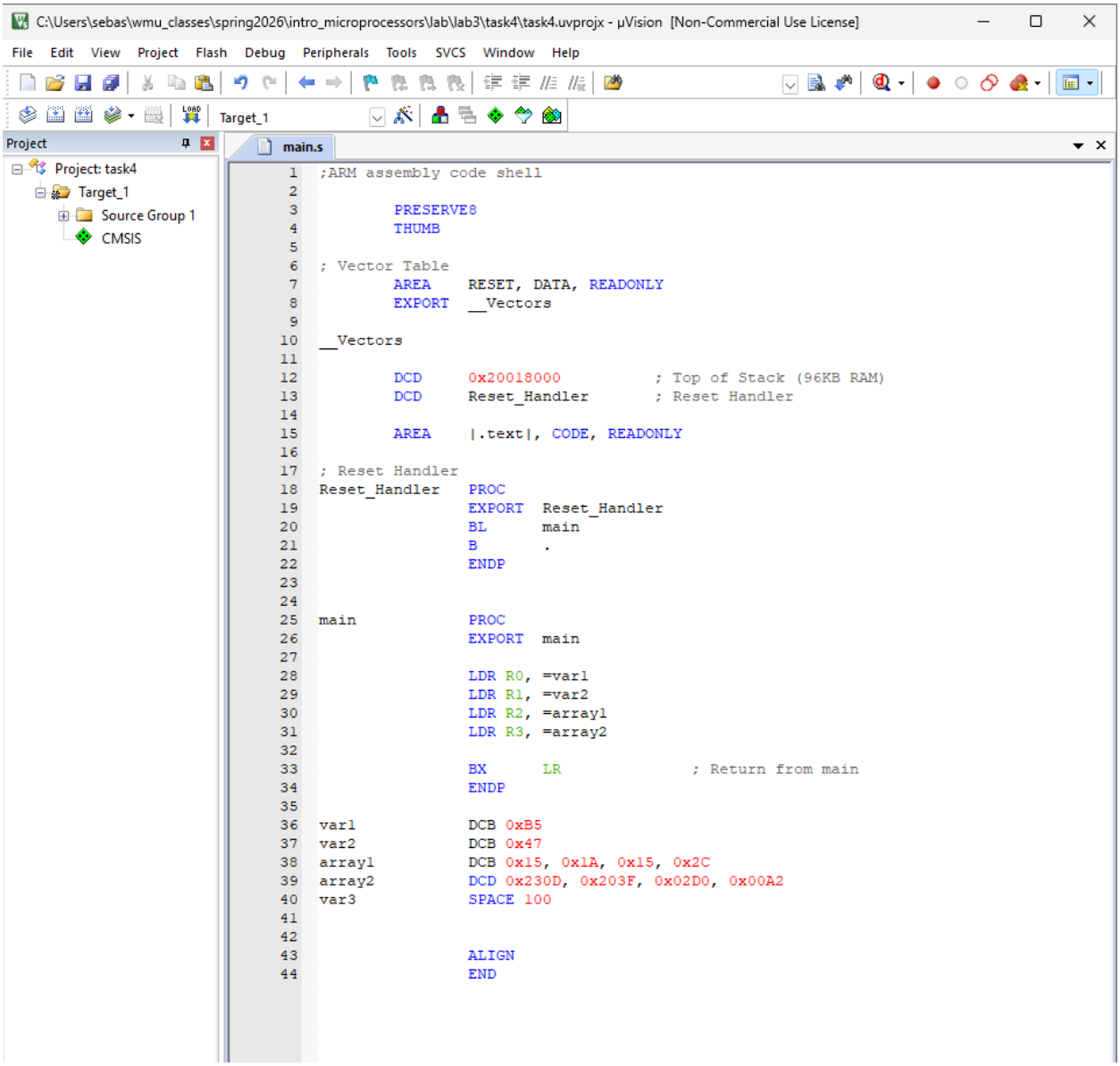

Registers
Register
Value
Core
R0 0x20000023
R1 0x0000000A
R2 0x00000014
R3 0x0000001E
R4 0x00000028
R5 0x00000032
R6 0x00000000
R7 0x00000000
R8 0x00000000
R9 0x00000000
R10 0x00000000
R11 0x00000000
R12 0x00000000
R13 (SP) 0x20018000
R14 (LR) 0x080000C9
R15 (PC) 0x080000EC
xPSR 0x01000000
Banked
System
Internal
Mode Thread
Privilege Privileged
Stack MSP
States 16
Sec 0.00000133
FPU
Project
Registers
Disassembly
37: STRB R4, [R0], #1
0x080000EC F8004B01 STRB r4,[r0],#0x01
38: STRB R5, [R0], #1
39:
0x080000F0 F8005B01 STRB r5,[r0],#0x01
40: BX LR ; Return from main
0x080000F4 4770 BX lr
main.s
17 Reset_Handler PROC
18 EXPORT Reset_Handler
19 BL main
20 B .
21 ENDP
22
23
24 main PROC
25 EXPORT main
26
27 LDR R0, =0x20000020
28 LDR R1, =10
29 LDR R2, =20
30 LDR R3, =30
31 LDR R4, =40
32 LDR R5, =50
33
34 STRB R1, [R0], #1
35 STRB R2, [R0], #1
36 STRB R3, [R0], #1
37 STRB R4, [R0], #1
38 STRB R5, [R0], #1
39
40 BX LR ; Return from main
41 ENDP
42
43 ALIGN
44 END
Command
Running with Code Size Limit: 32K
Load "C:\\Users\\sebas\\wmu_classes\\spring2026\\intro_mi
Memory 1
Address: 0x20000020
0x20000020: 0A 14 1E 00 00 00 00 00 00 00 00 00 00 00
0x2000002E: 00 00 00 00 00 00 00 00 00 00 00 00 00 00


Registers
Register
Value
Core
R0 0x20000024
R1 0x0000000A
R2 0x00000014
R3 0x0000001E
R4 0x00000028
R5 0x00000032
R6 0x00000000
R7 0x00000000
R8 0x00000000
R9 0x00000000
R10 0x00000000
R11 0x00000000
R12 0x00000000
R13 (SP) 0x20018000
R14 (LR) 0x080000C9
R15 (PC) 0x080000F0
xPSR 0x01000000
Banked
System
Internal
Mode Thread
Privilege Privileged
Stack MSP
States 18
Sec 0.00000150
FPU
Project
Registers
Disassembly
37: STRB R4, [R0], #1
0x080000EC F8004B01 STRB r4,[r0],#0x01
38: STRB R5, [R0], #1
39:
0x080000F0 F8005B01 STRB r5,[r0],#0x01
40: BX LR ; Return from main
0x080000F4 4770 BX lr
main.s
17 Reset_Handler PROC
18 EXPORT Reset_Handler
19 BL main
20 B .
21 ENDP
22
23
24 main PROC
25 EXPORT main
26
27 LDR R0, =0x20000020
28 LDR R1, =10
29 LDR R2, =20
30 LDR R3, =30
31 LDR R4, =40
32 LDR R5, =50
33
34 STRB R1, [R0], #1
35 STRB R2, [R0], #1
36 STRB R3, [R0], #1
37 STRB R4, [R0], #1
38 STRB R5, [R0], #1
39
40 BX LR ; Return from main
41 ENDP
42
43 ALIGN
44 END
Command
Running with Code Size Limit: 32K
Load "C:\\Users\\sebas\\wmu_classes\\spring2026\\intro_mi
Memory 1
Address: 0x20000020
0x20000020: 0A 14 1E 28 00 00 00 00 00 00 00 00 00 00
0x2000002E: 00 00 00 00 00 00 00 00 00 00 00 00 00 00


Registers
Register
Value
Core
R0 0x20000025
R1 0x0000000A
R2 0x00000014
R3 0x0000001E
R4 0x00000028
R5 0x00000032
R6 0x00000000
R7 0x00000000
R8 0x00000000
R9 0x00000000
R10 0x00000000
R11 0x00000000
R12 0x00000000
R13 (SP) 0x20018000
R14 (LR) 0x080000C9
R15 (PC) 0x080000F4
xPSR 0x01000000
Banked
System
Internal
Mode Thread
Privilege Privileged
Stack MSP
States 20
Sec 0.00000167
FPU
Project
Registers
Disassembly
40: BX LR ; Return from main
0x080000F4 4770 BX lr
0x080000F6 0000 DCW 0x0020
0x080000F8 0020 DCW 0x2000
0x080000FA 2000 DCW 0x4675
0x080000FC 4675 MOV r5,lr
0x080000FE F000F82B BL.W 0x08000158 user libspace
main.s
17 Reset_Handler PROC
18 EXPORT Reset_Handler
19 BL main
20 B .
21 ENDP
22
23
24 main PROC
25 EXPORT main
26
27 LDR R0, =0x20000020
28 LDR R1, =10
29 LDR R2, =20
30 LDR R3, =30
31 LDR R4, =40
32 LDR R5, =50
33
34 STRB R1, [R0], #1
35 STRB R2, [R0], #1
36 STRB R3, [R0], #1
37 STRB R4, [R0], #1
38 STRB R5, [R0], #1
39
40 BX LR ; Return from main
41 ENDP
42
43 ALIGN
44 END
Command
Running with Code Size Limit: 32K
Load "C:\\Users\\sebas\\wmu_classes\\spring2026\\intro_mi
Memory 1
Address: 0x20000020
0x20000020: 0A 14 1E 28 32 00 00 00 00 00 00 00 00 00
0x2000002E: 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0x2000003C: 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0x2000004A: 00 00 00 00 00 00 00 00 00 00 00 00 00 00

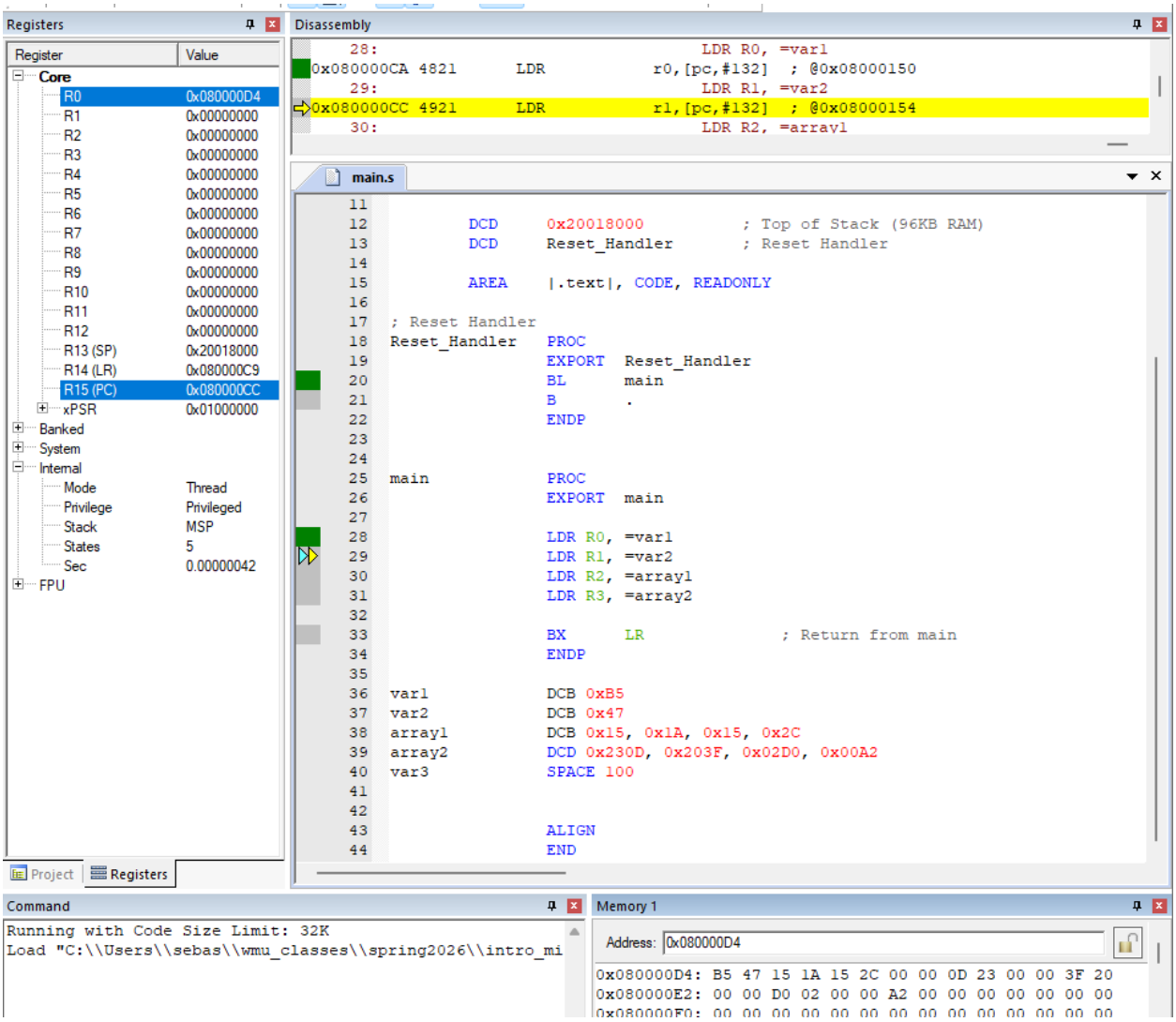
Task 4:

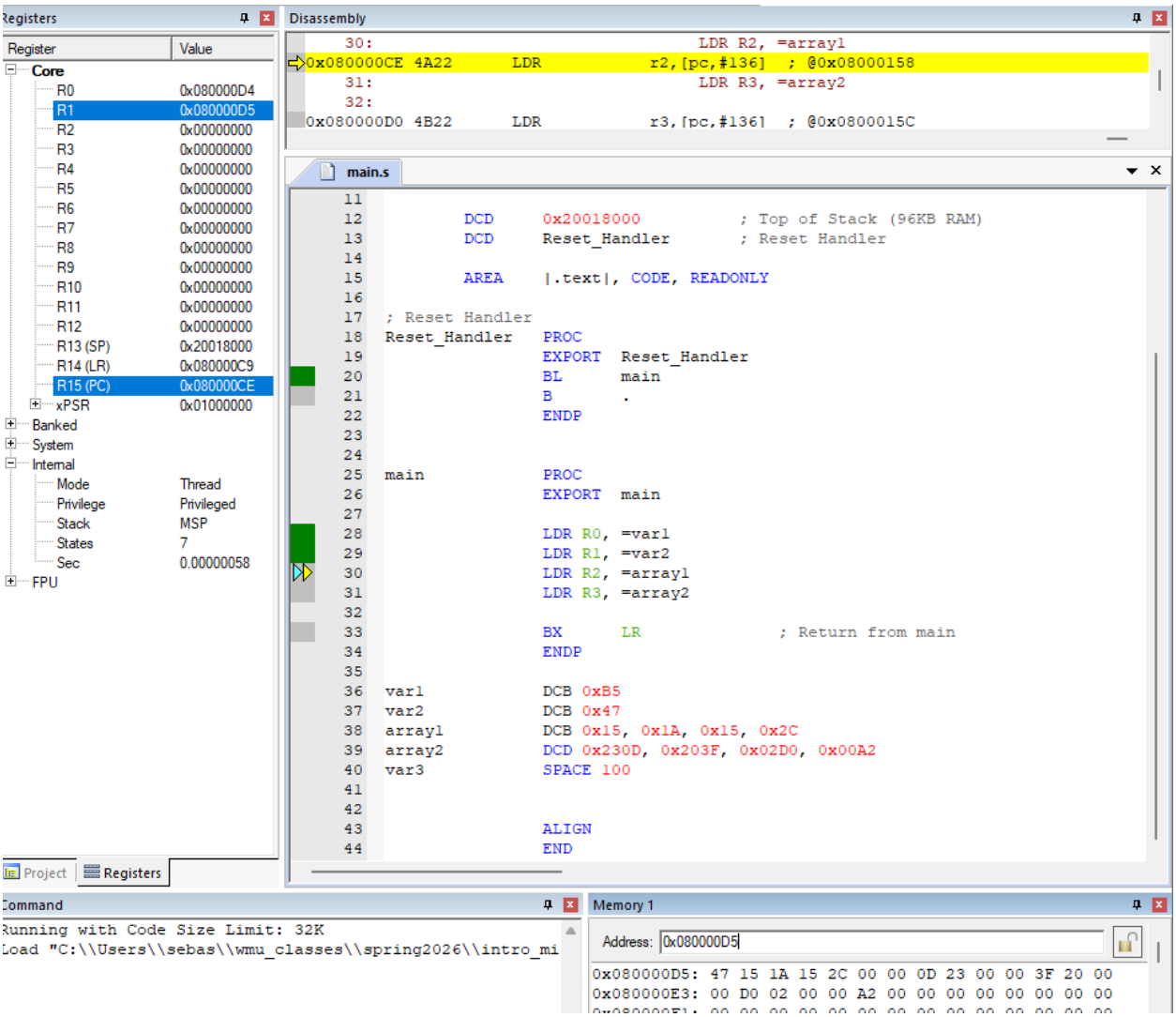
- Code.

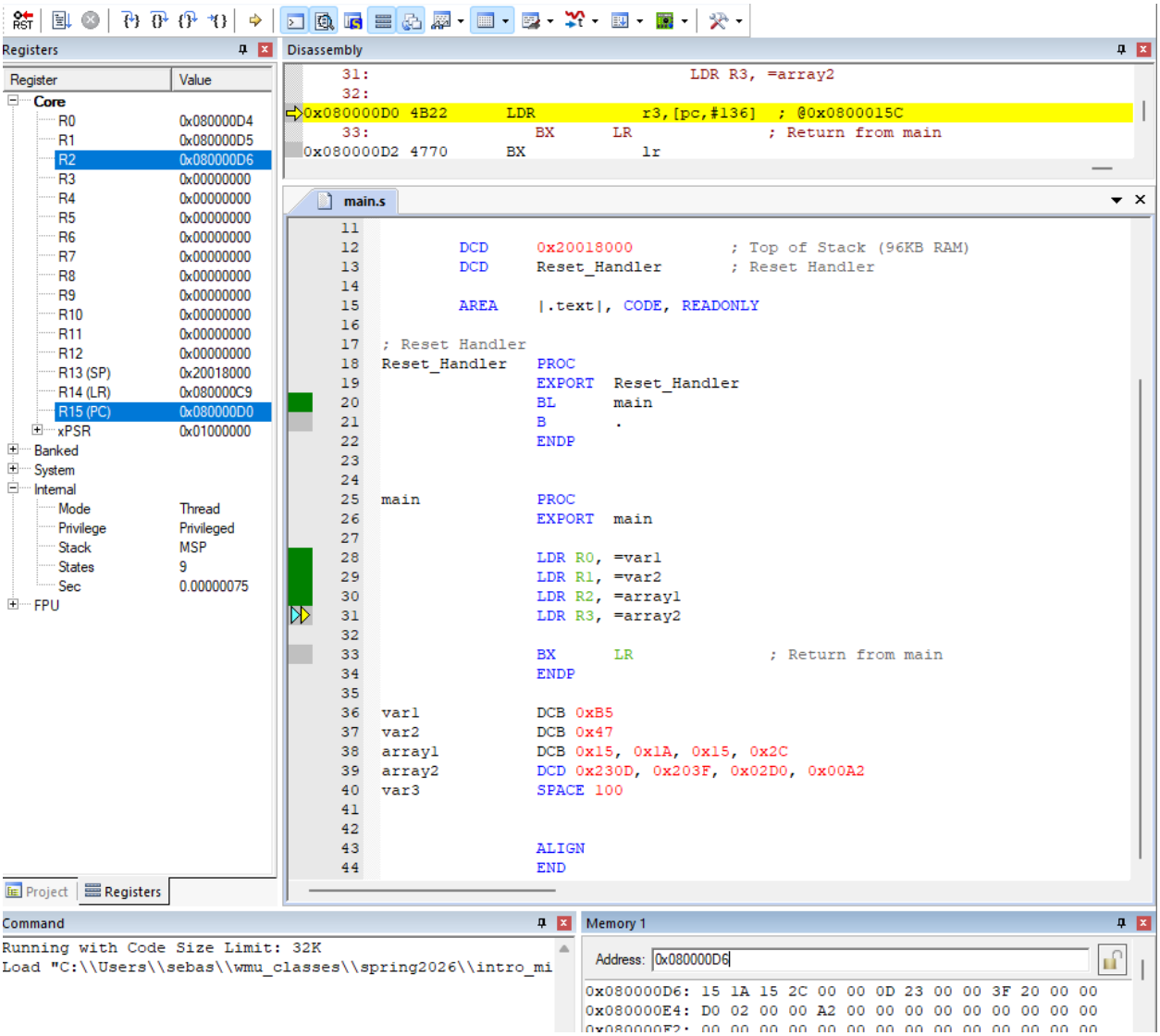


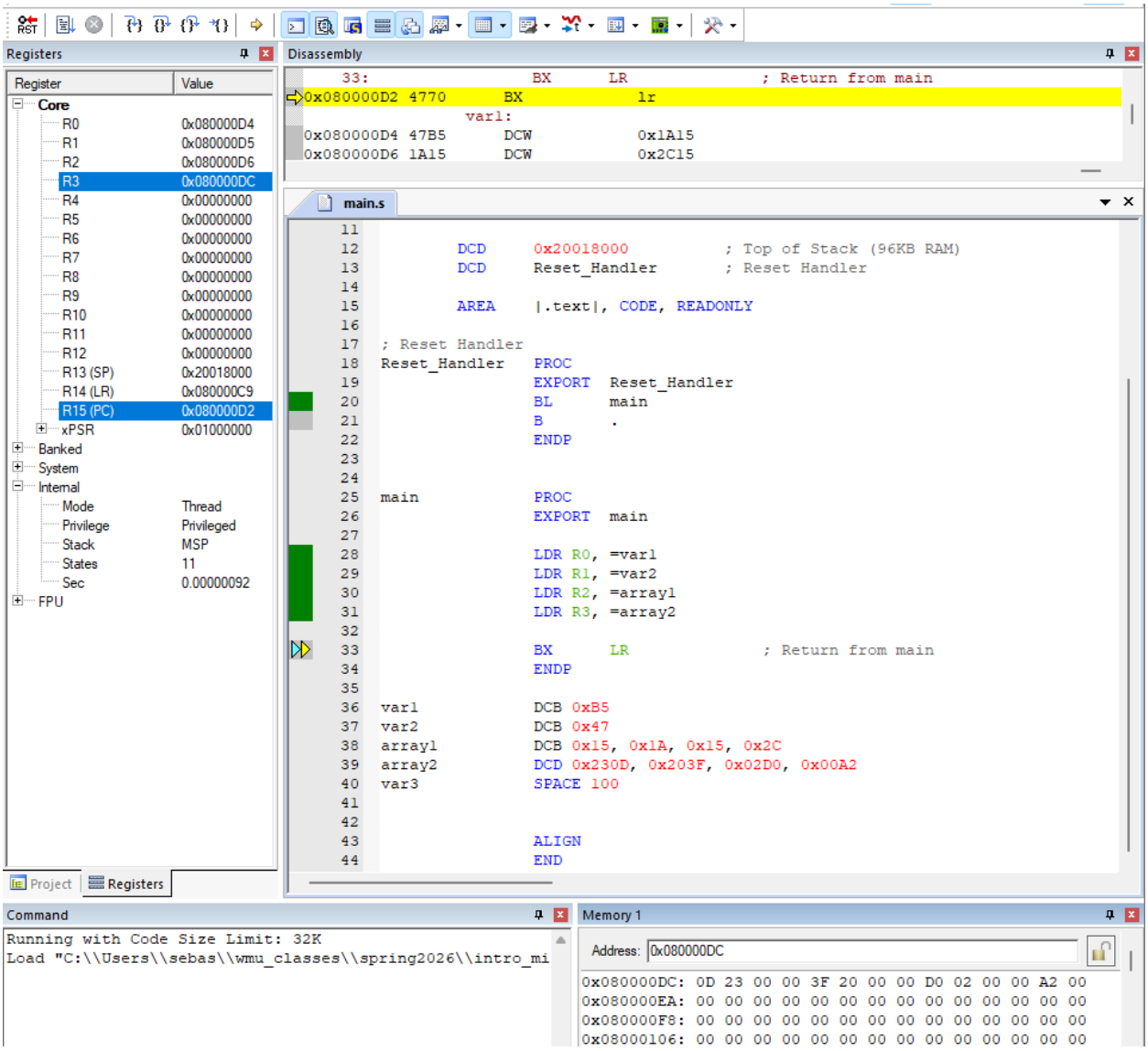

```
;ARM assembly code shell

        PRESERVE8
        THUMB

; Vector Table
        AREA    RESET, DATA, READONLY
        EXPORT  __Vectors

__Vectors

        DCD     0x20018000          ; Top of Stack (96KB RAM)
        DCD     Reset_Handler       ; Reset Handler

        AREA    |.text|, CODE, READONLY

; Reset Handler
Reset_Handler   PROC
                EXPORT  Reset_Handler
                BL      main
                B       .
                ENDP


main            PROC
                EXPORT  main

                LDR R0, =var1
                LDR R1, =var2
                LDR R2, =array1
                LDR R3, =array2

                BX      LR              ; Return from main
                ENDP

var1            DCB 0xB5
var2            DCB 0x47
array1          DCB 0x15, 0x1A, 0x15, 0x2C
array2          DCD 0x230D, 0x203F, 0x02D0, 0x00A2
var3            SPACE 100


                ALIGN
                END
```

● Memory snapshots.




Registers
Register
Value
Core
R0 0x080000D4
R1 0x080000D5
R2 0x00000000
R3 0x00000000
R4 0x00000000
R5 0x00000000
R6 0x00000000
R7 0x00000000
R8 0x00000000
R9 0x00000000
R10 0x00000000
R11 0x00000000
R12 0x00000000
R13 (SP) 0x20018000
R14 (LR) 0x080000C9
R15 (PC) 0x080000CE
xPSR 0x01000000
Banked
System
Internal
Mode Thread
Privilege Privileged
Stack MSP
States 7
Sec 0.00000058
FPU
Project
Registers
Disassembly
30: LDR R2, =array1
0x080000CE 4A22 LDR r2,[pc,#136] ; @0x08000158
31: LDR R3, =array2
32:
0x080000D0 4B22 LDR r3,[pc,#136] ; @0x0800015C
main.s
DCD 0x20018000 ; Top of Stack (96KB RAM)
DCD Reset_Handler ; Reset Handler
AREA |.text|, CODE, READONLY
; Reset Handler
Reset_Handler PROC
EXPORT Reset_Handler
BL main
B .
ENDP
main PROC
EXPORT main
LDR R0, =var1
LDR R1, =var2
LDR R2, =array1
LDR R3, =array2
BX LR ; Return from main
ENDP
var1 DCB 0xB5
var2 DCB 0x47
array1 DCB 0x15, 0x1A, 0x15, 0x2C
array2 DCD 0x230D, 0x203F, 0x02D0, 0x00A2
var3 SPACE 100
ALIGN
END
Command
Running with Code Size Limit: 32K
Load "C:\\Users\\sebas\\wmu_classes\\spring2026\\intro_mi
Memory 1
Address: 0x080000D5
0x080000D5: 47 15 1A 15 2C 00 00 0D 23 00 00 3F 20 00
0x080000E3: 00 D0 02 00 00 A2 00 00 00 00 00 00 00 00


Registers
Register
Value
Core
R0 0x080000D4
R1 0x080000D5
R2 0x080000D6
R3 0x00000000
R4 0x00000000
R5 0x00000000
R6 0x00000000
R7 0x00000000
R8 0x00000000
R9 0x00000000
R10 0x00000000
R11 0x00000000
R12 0x00000000
R13 (SP) 0x20018000
R14 (LR) 0x080000C9
R15 (PC) 0x080000D0
xPSR 0x01000000
Banked
System
Internal
Mode Thread
Privilege Privileged
Stack MSP
States 9
Sec 0.00000075
FPU
Project
Registers
Disassembly
31: LDR R3, =array2
32:
0x080000D0 4B22 LDR r3,[pc,#136] ; @0x0800015C
33: BX LR ; Return from main
0x080000D2 4770 BX lr
main.s
11
12 DCD 0x20018000 ; Top of Stack (96KB RAM)
13 DCD Reset_Handler ; Reset Handler
14
15 AREA |.text|, CODE, READONLY
16
17 ; Reset Handler
18 Reset_Handler PROC
19 EXPORT Reset_Handler
20 BL main
21 B .
22 ENDP
23
24
25 main PROC
26 EXPORT main
27
28 LDR R0, =var1
29 LDR R1, =var2
30 LDR R2, =array1
31 LDR R3, =array2
32
33 BX LR ; Return from main
34 ENDP
35
36 var1 DCB 0xB5
37 var2 DCB 0x47
38 array1 DCB 0x15, 0x1A, 0x15, 0x2C
39 array2 DCB 0x230D, 0x203F, 0x02D0, 0x00A2
40 var3 SPACE 100
41
42
43 ALIGN
44 END
Command
Running with Code Size Limit: 32K
Load "C:\\Users\\sebas\\wmu_classes\\spring2026\\intro_mi
Memory 1
Address: 0x080000D6
0x080000D6: 15 1A 15 2C 00 00 0D 23 00 00 3F 20 00 00
0x080000E4: D0 02 00 00 A2 00 00 00 00 00 00 00 00 00



● Comment on the difference between the assembler directives used (DCB, DCD, SPACE).

DCB, Defines Constant Bytes is used to declare bytes. To Allocate one or more bytes (8 bits) of data

DCD is used to declare words. To Allocate one or more words (32 bits) of data

SPACE is used to reserve a number of bytes in memory initialized to 0

Task 5:

- Code.

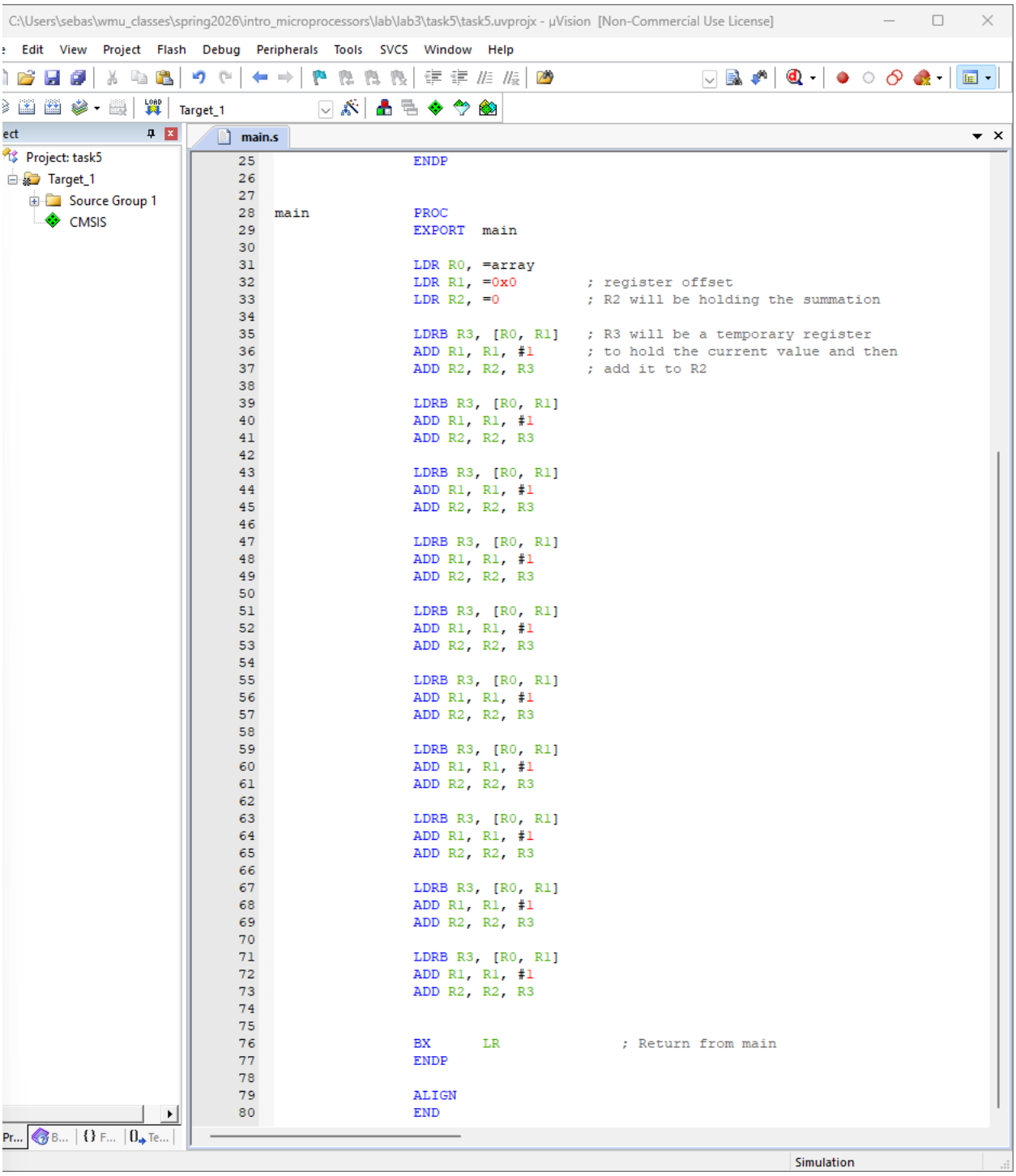
C:\Users\sebas\wmu_classes\spring2026\intro_microprocessors\lab\lab3\task5\task5.uvprojx - µVision [Non-Commercial Use License]

Project: task5
Target_1
Source Group 1
CMSIS

main.s

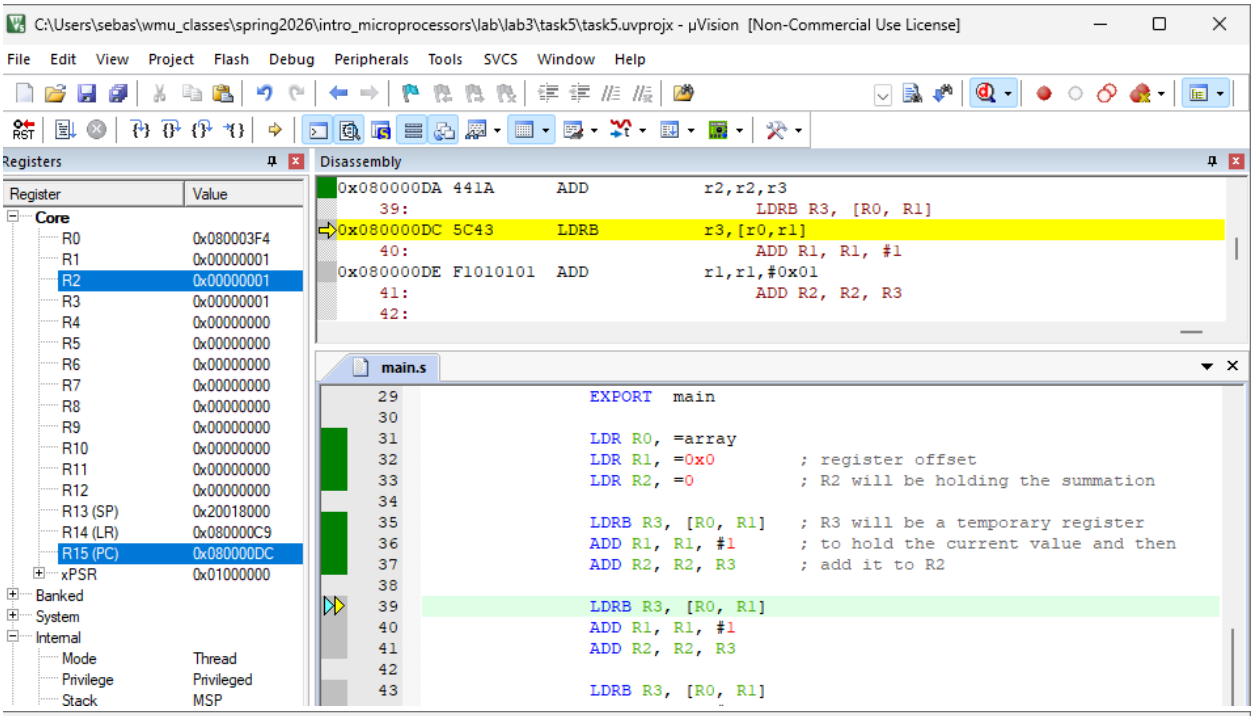
```
;ARM assembly code shell

        PRESERVE8
        THUMB

; Vector Table
        AREA    RESET, DATA, READONLY
        EXPORT  __Vectors

__Vectors
        DCD     0x20018000          ; Top of Stack (96KB RAM)
        DCD     Reset_Handler       ; Reset Handler


        AREA myData, DATA, READONLY
array       DCB 1, 2, 3, 4, 5, 6, 7, 8, 9, 10

        AREA    |.text|, CODE, READONLY

; Reset Handler
Reset_Handler   PROC
                EXPORT  Reset_Handler
                BL      main
                B       .
                ENDP


main            PROC
                EXPORT  main

                LDR R0, =array
                LDR R1, =0x0        ; register offset
                LDR R2, =0          ; R2 will be holding the summation

                LDRB R3, [R0, R1]   ; R3 will be a temporary register
                ADD R1, R1, #1      ; to hold the current value and then
                ADD R2, R2, R3      ; add it to R2

                LDRB R3, [R0, R1]
                ADD R1, R1, #1
                ADD R2, R2, R3

                LDRB R3, [R0, R1]
                ADD R1, R1, #1
                ADD R2, R2, R3

                LDRB R3, [R0, R1]
                ADD R1, R1, #1
                ADD R2, R2, R3

                LDRB R3, [R0, R1]
                ADD R1, R1, #1
                ADD R2, R2, R3

                LDRB R3, [R0, R1]
                ADD R1, R1, #1
```

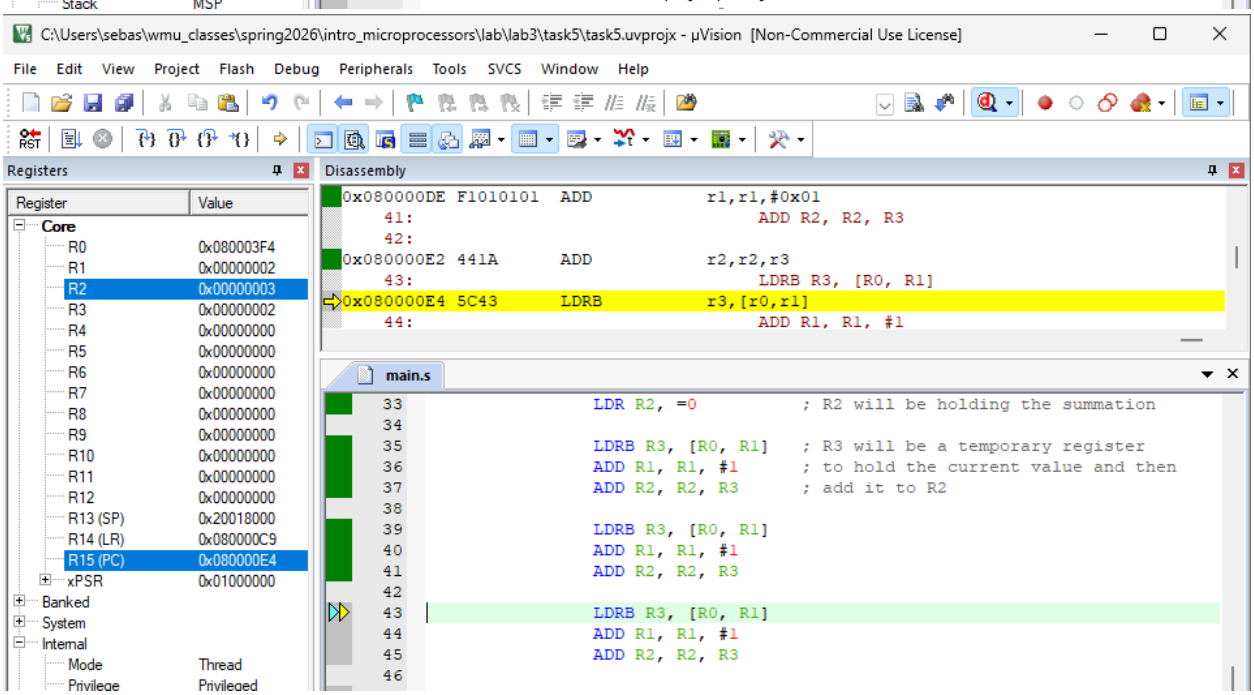
Simulation

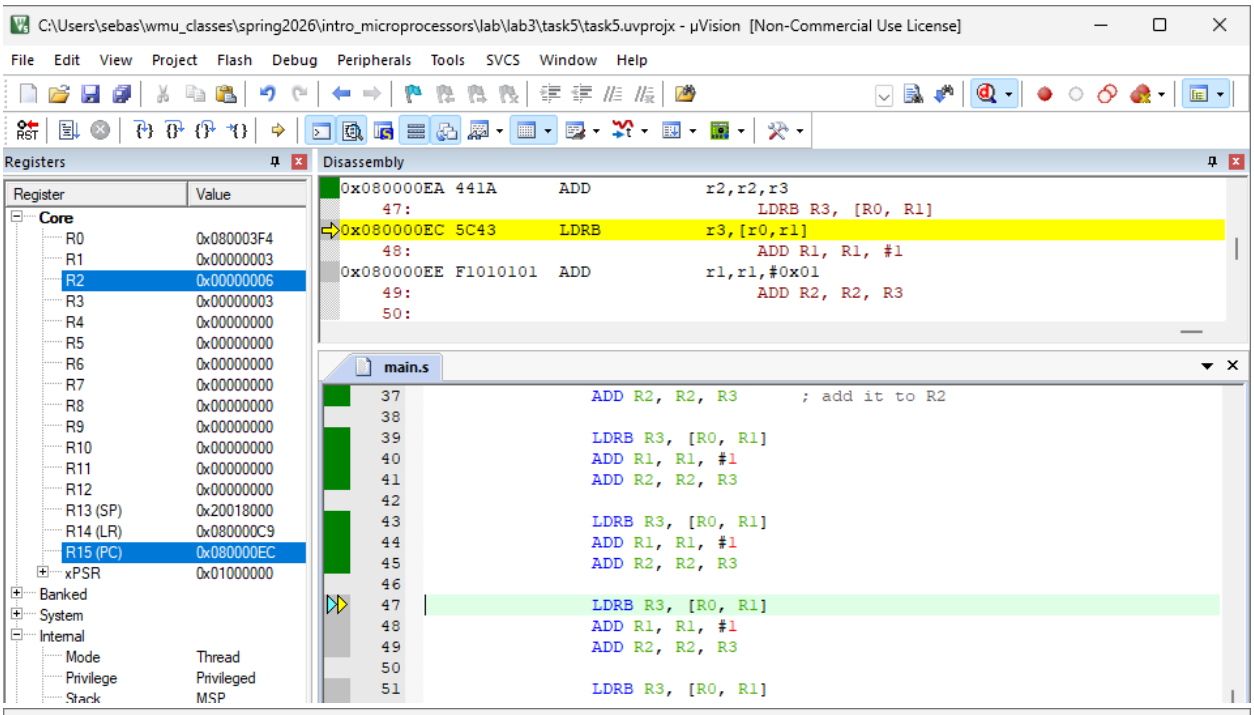


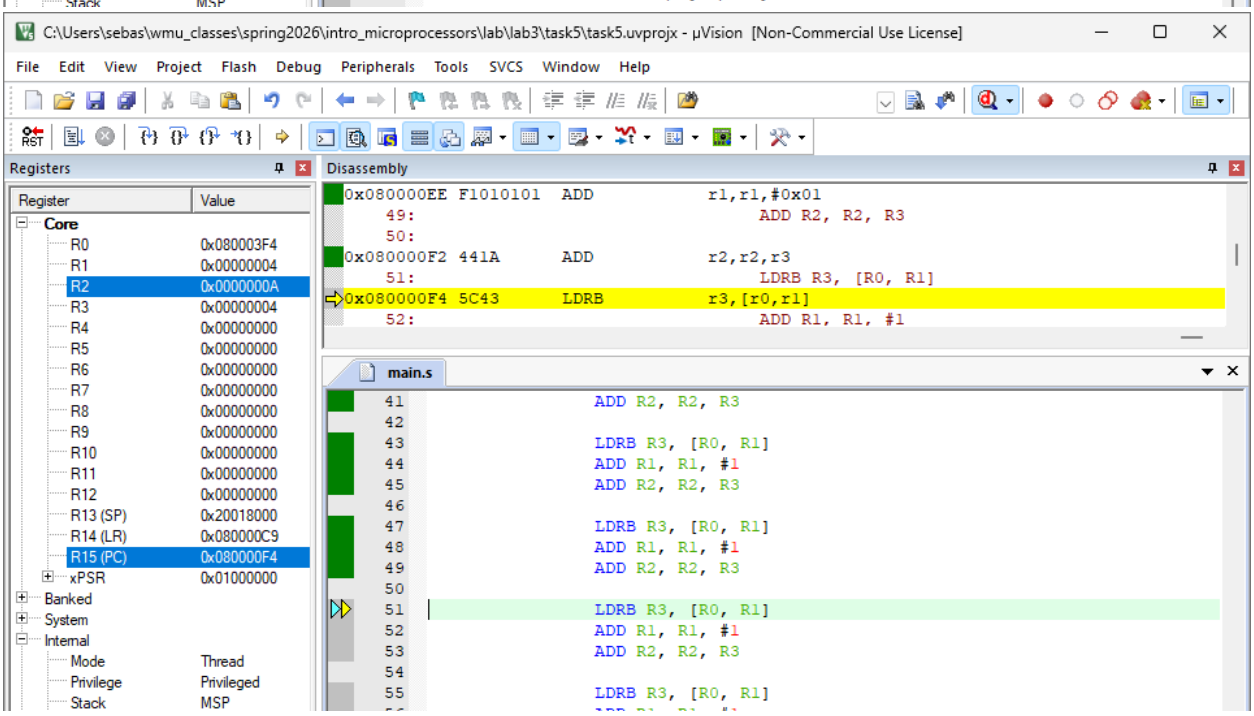
- Memory snapshots.

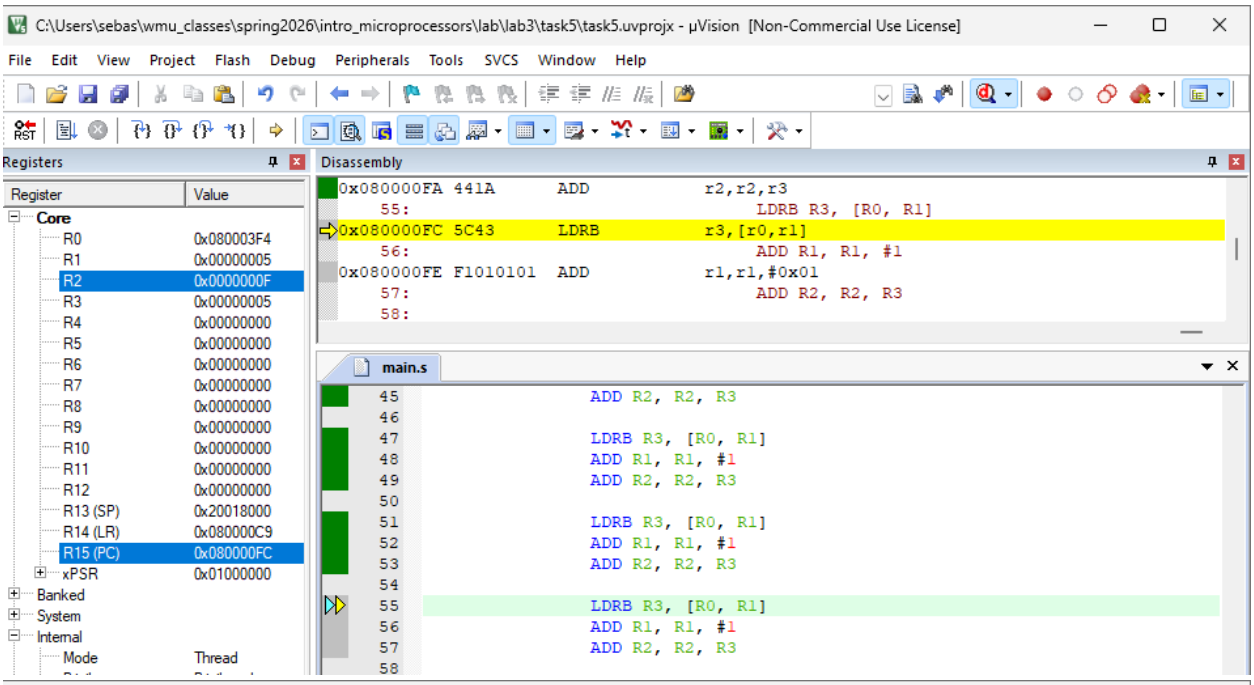
For Register offset:

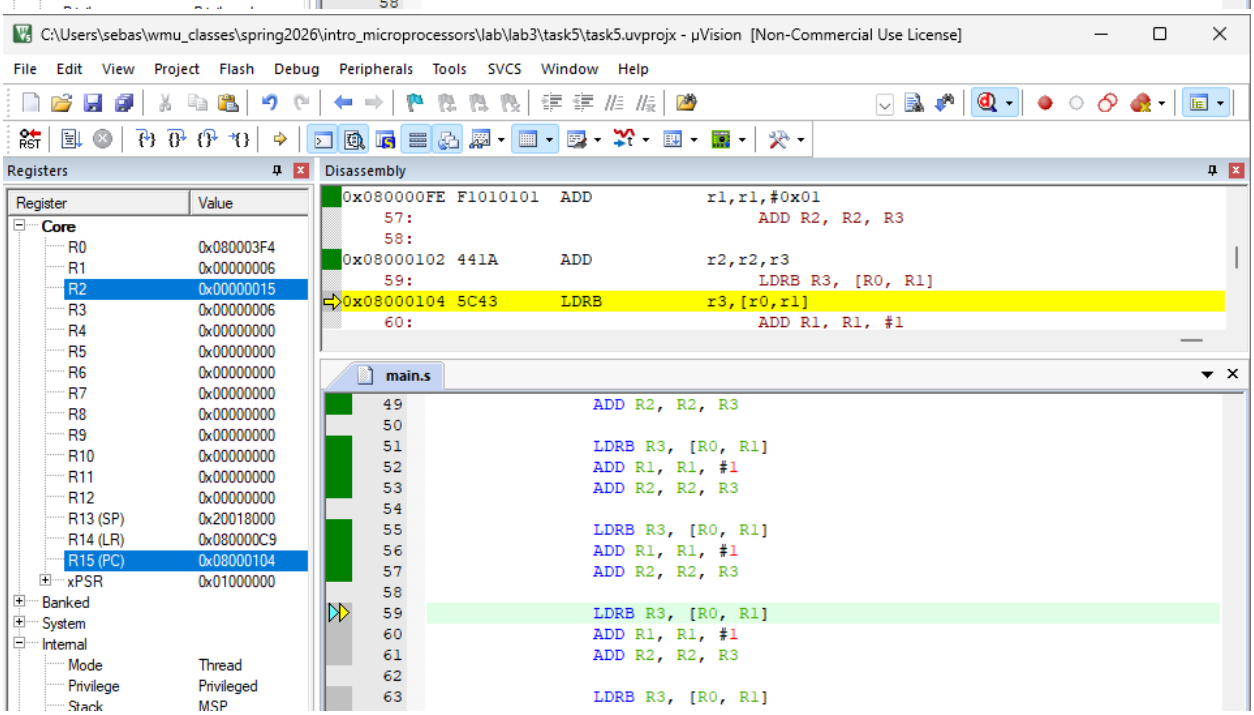





C:\Users\sebas\wmu_classes\spring2026\intro_microprocessors\lab\lab3\task5\task5.uvprojx - µVision [Non-Commercial Use License]
File Edit View Project Flash Debug Peripherals Tools SVCS Window Help
Registers
Register Value
Core
R0 0x080003F4
R1 0x00000003
R2 0x00000006
R3 0x00000003
R4 0x00000000
R5 0x00000000
R6 0x00000000
R7 0x00000000
R8 0x00000000
R9 0x00000000
R10 0x00000000
R11 0x00000000
R12 0x00000000
R13 (SP) 0x20018000
R14 (LR) 0x080000C9
R15 (PC) 0x080000EC
xPSR 0x01000000
Banked
System
Internal
Mode Thread
Privilege Privileged
Stack MSP
Disassembly
0x080000EA 441A ADD r2,r2,r3
47: LDRB R3, [R0, R1]
0x080000EC 5C43 LDRB r3,[r0,r1]
48: ADD R1, R1, #1
0x080000EE F1010101 ADD r1,r1,#0x01
49: ADD R2, R2, R3
50:
main.s
37 ADD R2, R2, R3 ; add it to R2
38
39 LDRB R3, [R0, R1]
40 ADD R1, R1, #1
41 ADD R2, R2, R3
42
43 LDRB R3, [R0, R1]
44 ADD R1, R1, #1
45 ADD R2, R2, R3
46
47 LDRB R3, [R0, R1]
48 ADD R1, R1, #1
49 ADD R2, R2, R3
50
51 LDRB R3, [R0, R1]


C:\Users\sebas\wmu_classes\spring2026\intro_microprocessors\lab\lab3\task5\task5.uvprojx - µVision [Non-Commercial Use License]
File Edit View Project Flash Debug Peripherals Tools SVCS Window Help
Registers
Register Value
Core
R0 0x080003F4
R1 0x00000004
R2 0x0000000A
R3 0x00000004
R4 0x00000000
R5 0x00000000
R6 0x00000000
R7 0x00000000
R8 0x00000000
R9 0x00000000
R10 0x00000000
R11 0x00000000
R12 0x00000000
R13 (SP) 0x20018000
R14 (LR) 0x080000C9
R15 (PC) 0x080000F4
xPSR 0x01000000
Banked
System
Internal
Mode Thread
Privilege Privileged
Stack MSP
Disassembly
0x080000EE F1010101 ADD r1,r1,#0x01
49: ADD R2, R2, R3
50:
0x080000F2 441A ADD r2,r2,r3
51: LDRB R3, [R0, R1]
0x080000F4 5C43 LDRB r3,[r0,r1]
52: ADD R1, R1, #1
main.s
41 ADD R2, R2, R3
42
43 LDRB R3, [R0, R1]
44 ADD R1, R1, #1
45 ADD R2, R2, R3
46
47 LDRB R3, [R0, R1]
48 ADD R1, R1, #1
49 ADD R2, R2, R3
50
51 LDRB R3, [R0, R1]
52 ADD R1, R1, #1
53 ADD R2, R2, R3
54
55 LDRB R3, [R0, R1]


C:\Users\sebas\wmu_classes\spring2026\intro_microprocessors\lab\lab3\task5\task5.uvprojx - µVision [Non-Commercial Use License]
File Edit View Project Flash Debug Peripherals Tools SVCS Window Help
Registers
Register Value
Core
R0 0x080003F4
R1 0x00000005
R2 0x0000000F
R3 0x00000005
R4 0x00000000
R5 0x00000000
R6 0x00000000
R7 0x00000000
R8 0x00000000
R9 0x00000000
R10 0x00000000
R11 0x00000000
R12 0x00000000
R13 (SP) 0x20018000
R14 (LR) 0x080000C9
R15 (PC) 0x080000FC
xPSR 0x01000000
Banked
System
Internal
Mode Thread
Disassembly
0x080000FA 441A ADD r2,r2,r3
55: LDRB R3, [R0, R1]
0x080000FC 5C43 LDRB r3,[r0,r1]
56: ADD R1, R1, #1
0x080000FE F1010101 ADD r1,r1,#0x01
57: ADD R2, R2, R3
58:
main.s
45 ADD R2, R2, R3
46
47 LDRB R3, [R0, R1]
48 ADD R1, R1, #1
49 ADD R2, R2, R3
50
51 LDRB R3, [R0, R1]
52 ADD R1, R1, #1
53 ADD R2, R2, R3
54
55 LDRB R3, [R0, R1]
56 ADD R1, R1, #1
57 ADD R2, R2, R3
58


C:\Users\sebas\wmu_classes\spring2026\intro_microprocessors\lab\lab3\task5\task5.uvprojx - µVision [Non-Commercial Use License]
File Edit View Project Flash Debug Peripherals Tools SVCS Window Help
Registers
Register Value
Core
R0 0x080003F4
R1 0x00000006
R2 0x00000015
R3 0x00000006
R4 0x00000000
R5 0x00000000
R6 0x00000000
R7 0x00000000
R8 0x00000000
R9 0x00000000
R10 0x00000000
R11 0x00000000
R12 0x00000000
R13 (SP) 0x20018000
R14 (LR) 0x080000C9
R15 (PC) 0x08000104
xPSR 0x01000000
Banked
System
Internal
Mode Thread
Privilege Privileged
Stack MSP
Disassembly
0x080000FE F1010101 ADD r1,r1,#0x01
57: ADD R2, R2, R3
58:
0x08000102 441A ADD r2,r2,r3
59: LDRB R3, [R0, R1]
0x08000104 5C43 LDRB r3,[r0,r1]
60: ADD R1, R1, #1
main.s
49 ADD R2, R2, R3
50
51 LDRB R3, [R0, R1]
52 ADD R1, R1, #1
53 ADD R2, R2, R3
54
55 LDRB R3, [R0, R1]
56 ADD R1, R1, #1
57 ADD R2, R2, R3
58
59 LDRB R3, [R0, R1]
60 ADD R1, R1, #1
61 ADD R2, R2, R3
62
63 LDRB R3, [R0, R1]

C:\Users\sebas\wmu_classes\spring2026\intro_microprocessors\lab\lab3\task5\task5.uvprojx - µVision [Non-Commercial Use License]

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Registers

Register	Value
Core	
R0	0x080003F4
R1	0x00000007
R2	0x0000001C
R3	0x00000007
R4	0x00000000
R5	0x00000000
R6	0x00000000
R7	0x00000000
R8	0x00000000
R9	0x00000000
R10	0x00000000
R11	0x00000000
R12	0x00000000
R13 (SP)	0x20018000
R14 (LR)	0x080000C9
R15 (PC)	0x0800010C
xPSR	0x01000000
Banked	
System	
Internal	
Mode	Thread
Privilege	Privileged
Stack	MSP
States	35
Sec	0.00000292
FPU	

Disassembly

```
0x0800010A 441A      ADD     r2,r2,r3
    63:                          LDRB R3, [R0, R1]
0x0800010C 5C43      LDRB    r3,[r0,r1]
    64:                          ADD R1, R1, #1
0x0800010E F1010101  ADD     r1,r1,#0x01
    65:                          ADD R2, R2, R3
    66:
```

main.s

```
49                  ADD R2, R2, R3
50
51                  LDRB R3, [R0, R1]
52                  ADD R1, R1, #1
53                  ADD R2, R2, R3
54
55                  LDRB R3, [R0, R1]
56                  ADD R1, R1, #1
57                  ADD R2, R2, R3
58
59                  LDRB R3, [R0, R1]
60                  ADD R1, R1, #1
61                  ADD R2, R2, R3
62
63                  LDRB R3, [R0, R1]
64                  ADD R1, R1, #1
65                  ADD R2, R2, R3
66
67                  LDRB R3, [R0, R1]
```

C:\Users\sebas\wmu_classes\spring2026\intro_microprocessors\lab\lab3\task5\task5.uvprojx - µVision [Non-Commercial Use License]

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Registers

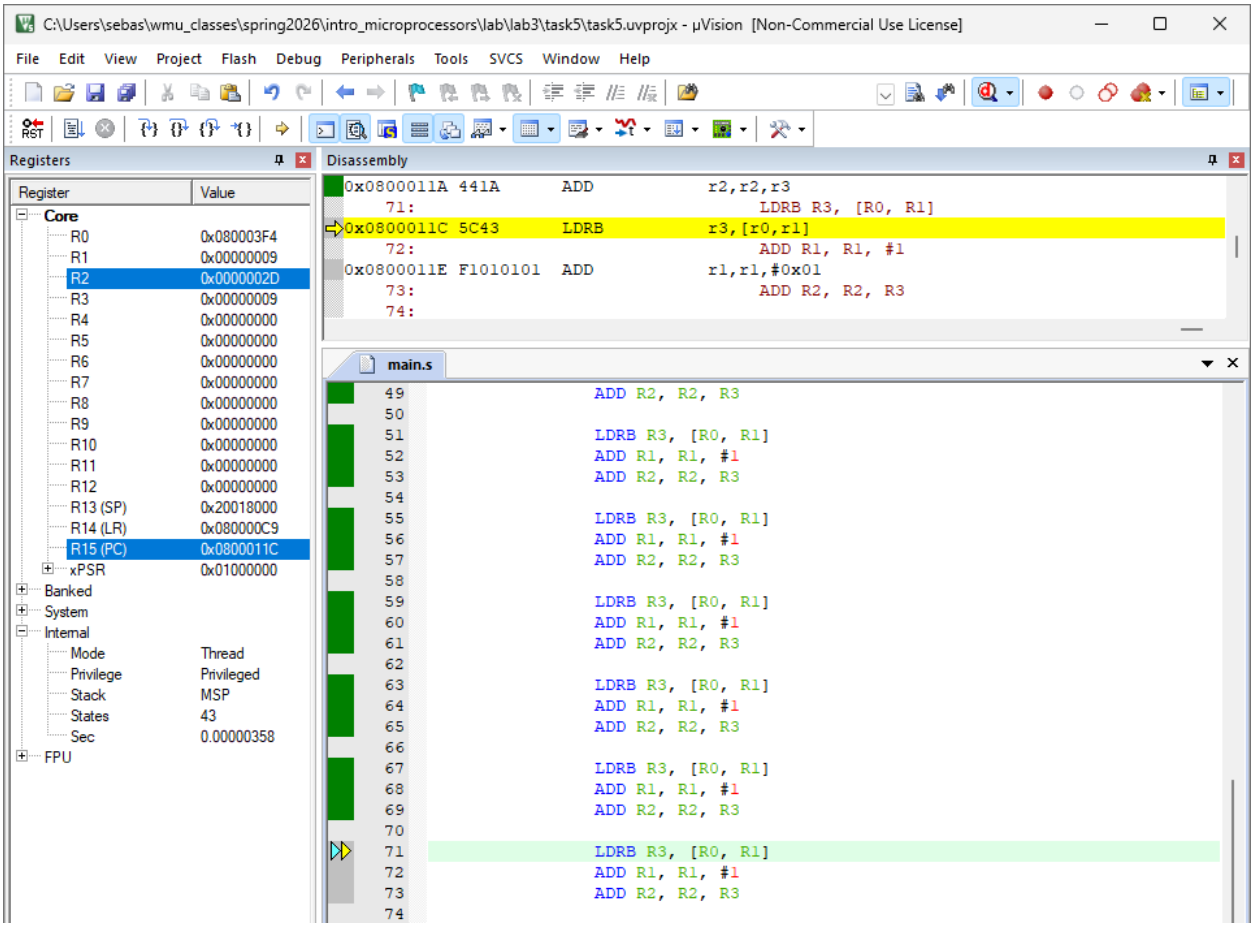
Register	Value
Core	
R0	0x080003F4
R1	0x00000008
R2	0x00000024
R3	0x00000008
R4	0x00000000
R5	0x00000000
R6	0x00000000
R7	0x00000000
R8	0x00000000
R9	0x00000000
R10	0x00000000
R11	0x00000000
R12	0x00000000
R13 (SP)	0x20018000
R14 (LR)	0x080000C9
R15 (PC)	0x08000114
xPSR	0x01000000
Banked	
System	
Internal	
Mode	Thread
Privilege	Privileged
Stack	MSP
States	39
Sec	0.00000325
FPU	

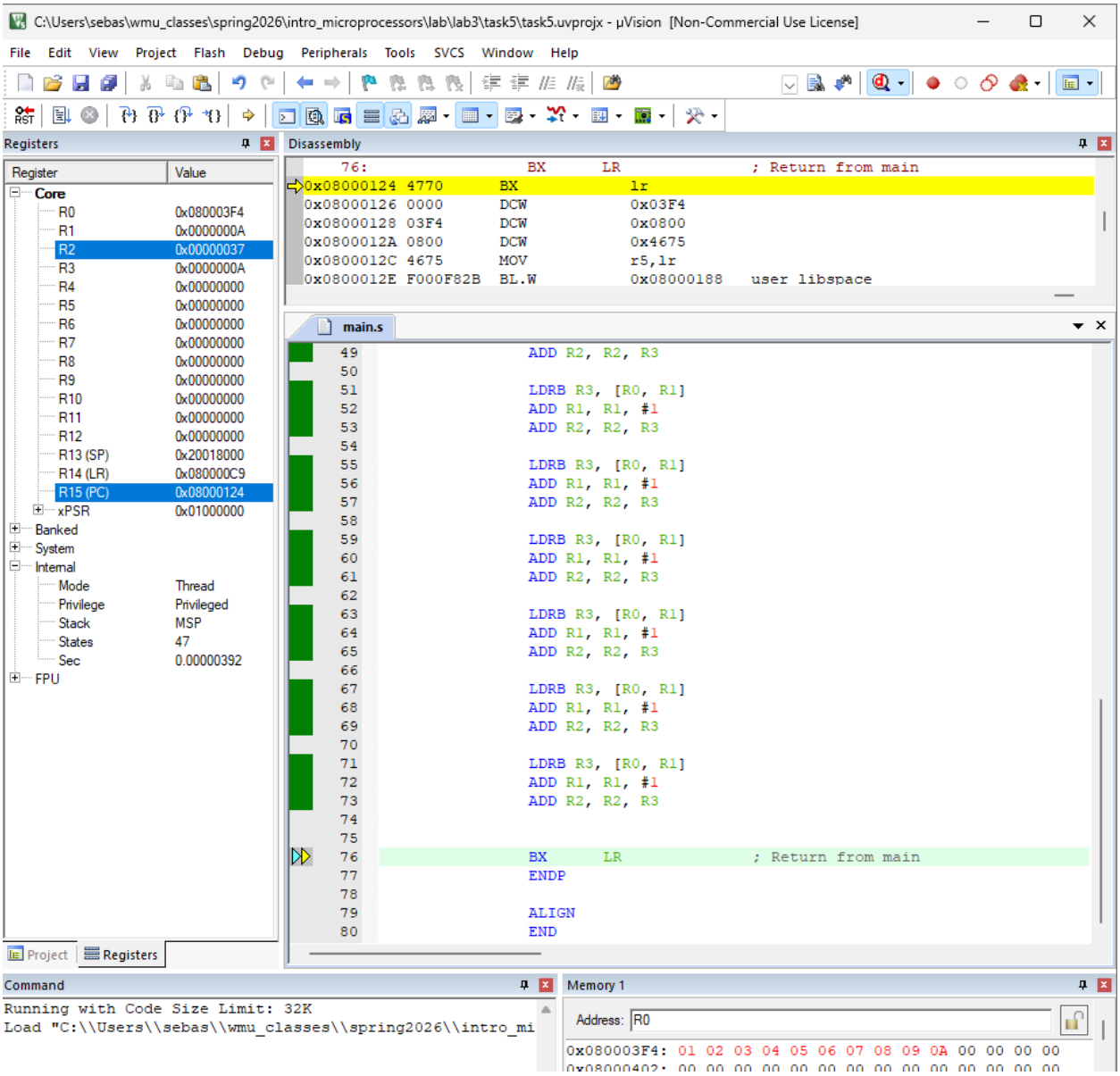
Disassembly

```
0x0800010E F1010101  ADD     r1,r1,#0x01
    65:                          ADD R2, R2, R3
    66:
0x08000112 441A      ADD     r2,r2,r3
    67:                          LDRB R3, [R0, R1]
0x08000114 5C43      LDRB    r3,[r0,r1]
    68:                          ADD R1, R1, #1
```

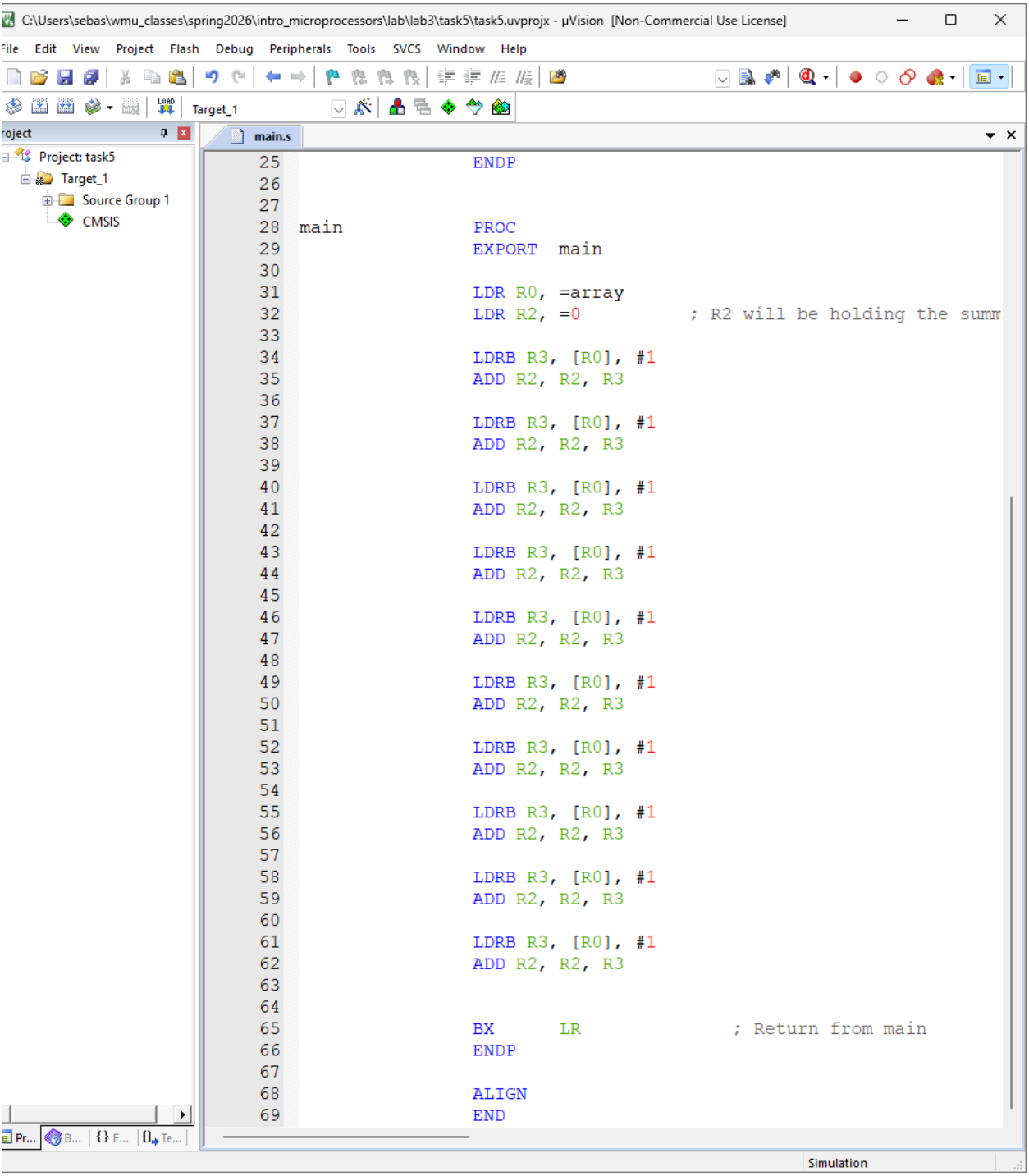
main.s

```
49                  ADD R2, R2, R3
50
51                  LDRB R3, [R0, R1]
52                  ADD R1, R1, #1
53                  ADD R2, R2, R3
54
55                  LDRB R3, [R0, R1]
56                  ADD R1, R1, #1
57                  ADD R2, R2, R3
58
59                  LDRB R3, [R0, R1]
60                  ADD R1, R1, #1
61                  ADD R2, R2, R3
62
63                  LDRB R3, [R0, R1]
64                  ADD R1, R1, #1
65                  ADD R2, R2, R3
66
67                  LDRB R3, [R0, R1]
68                  ADD R1, R1, #1
69                  ADD R2, R2, R3
70
71                  LDRB R3, [R0, R1]
```

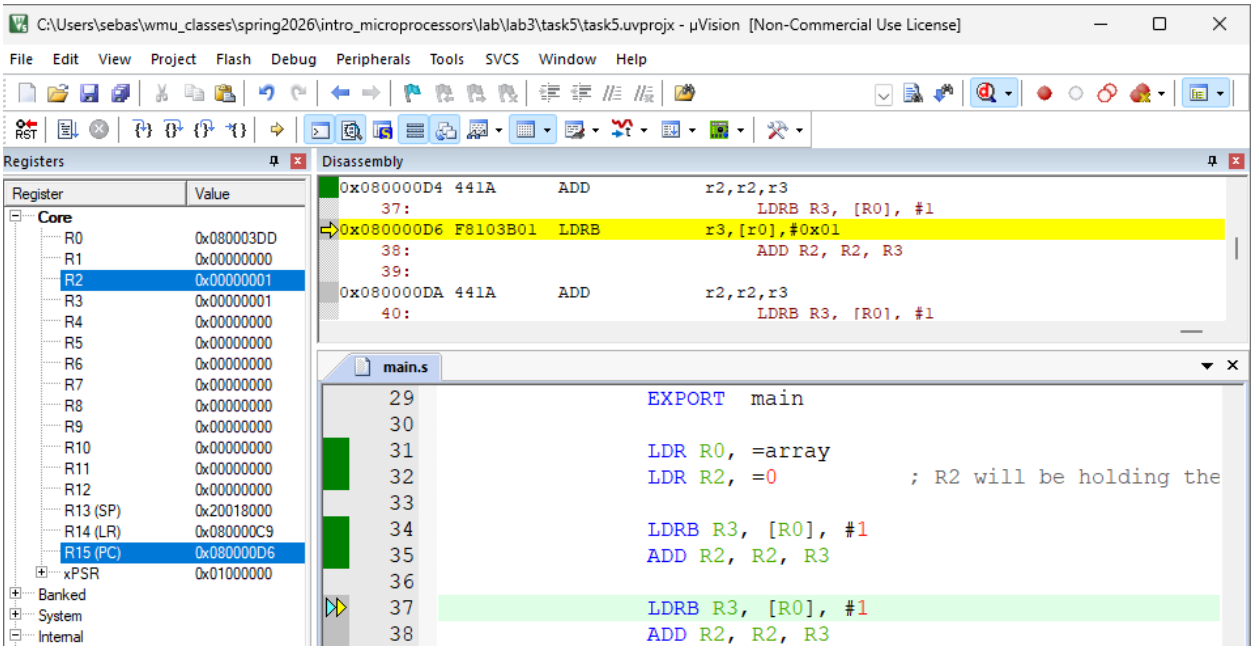

C:\Users\sebas\wmu_classes\spring2026\intro_microprocessors\lab\lab3\task5\task5.uvprojx - µVision [Non-Commercial Use License]
File Edit View Project Flash Debug Peripherals Tools SVCS Window Help
Registers
Register Value
Core
R0 0x080003F4
R1 0x00000009
R2 0x0000002D
R3 0x00000009
R4 0x00000000
R5 0x00000000
R6 0x00000000
R7 0x00000000
R8 0x00000000
R9 0x00000000
R10 0x00000000
R11 0x00000000
R12 0x00000000
R13 (SP) 0x20018000
R14 (LR) 0x080000C9
R15 (PC) 0x0800011C
xPSR 0x01000000
Banked
System
Internal
Mode Thread
Privilege Privileged
Stack MSP
States 43
Sec 0.00000358
FPU
Disassembly
0x0800011A 441A ADD r2,r2,r3
71: LDRB R3, [R0, R1]
0x0800011C 5C43 LDRB r3,[r0,r1]
72: ADD R1, R1, #1
0x0800011E F1010101 ADD r1,r1,#0x01
73: ADD R2, R2, R3
74:
main.s
49 ADD R2, R2, R3
50
51 LDRB R3, [R0, R1]
52 ADD R1, R1, #1
53 ADD R2, R2, R3
54
55 LDRB R3, [R0, R1]
56 ADD R1, R1, #1
57 ADD R2, R2, R3
58
59 LDRB R3, [R0, R1]
60 ADD R1, R1, #1
61 ADD R2, R2, R3
62
63 LDRB R3, [R0, R1]
64 ADD R1, R1, #1
65 ADD R2, R2, R3
66
67 LDRB R3, [R0, R1]
68 ADD R1, R1, #1
69 ADD R2, R2, R3
70
71 LDRB R3, [R0, R1]
72 ADD R1, R1, #1
73 ADD R2, R2, R3
74

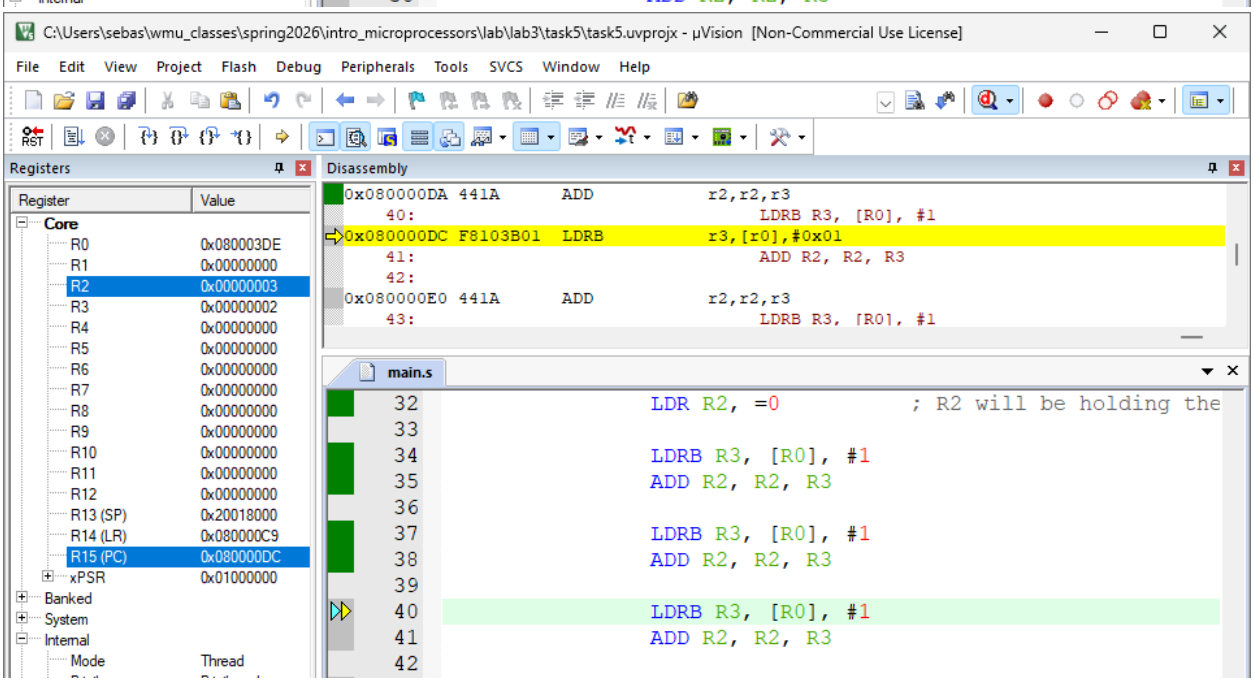

C:\Users\sebas\wmu_classes\spring2026\intro_microprocessors\lab\lab3\task5\task5.uvprojx - µVision [Non-Commercial Use License]
File Edit View Project Flash Debug Peripherals Tools SVCS Window Help
Registers
Register Value
Core
R0 0x080003F4
R1 0x0000000A
R2 0x00000037
R3 0x0000000A
R4 0x00000000
R5 0x00000000
R6 0x00000000
R7 0x00000000
R8 0x00000000
R9 0x00000000
R10 0x00000000
R11 0x00000000
R12 0x00000000
R13 (SP) 0x20018000
R14 (LR) 0x080000C9
R15 (PC) 0x08000124
xPSR 0x01000000
Banked
System
Internal
Mode Thread
Privilege Privileged
Stack MSP
States 47
Sec 0.00000392
FPU
Project Registers
Disassembly
76: BX LR ; Return from main
0x08000124 4770 BX lr
0x08000126 0000 DCW 0x03F4
0x08000128 03F4 DCW 0x0800
0x0800012A 0800 DCW 0x4675
0x0800012C 4675 MOV r5,lr
0x0800012E F000F82B BL.W 0x08000188 user_libspace
main.s
49 ADD R2, R2, R3
50
51 LDRB R3, [R0, R1]
52 ADD R1, R1, #1
53 ADD R2, R2, R3
54
55 LDRB R3, [R0, R1]
56 ADD R1, R1, #1
57 ADD R2, R2, R3
58
59 LDRB R3, [R0, R1]
60 ADD R1, R1, #1
61 ADD R2, R2, R3
62
63 LDRB R3, [R0, R1]
64 ADD R1, R1, #1
65 ADD R2, R2, R3
66
67 LDRB R3, [R0, R1]
68 ADD R1, R1, #1
69 ADD R2, R2, R3
70
71 LDRB R3, [R0, R1]
72 ADD R1, R1, #1
73 ADD R2, R2, R3
74
75
76 BX LR ; Return from main
77 ENDP
78
79 ALIGN
80 END
Command
Running with Code Size Limit: 32K
Load "C:\\Users\\sebas\\wmu_classes\\spring2026\\intro_mi
Memory 1
Address: R0
0x080003F4: 01 02 03 04 05 06 07 08 09 0A 00 00 00 00

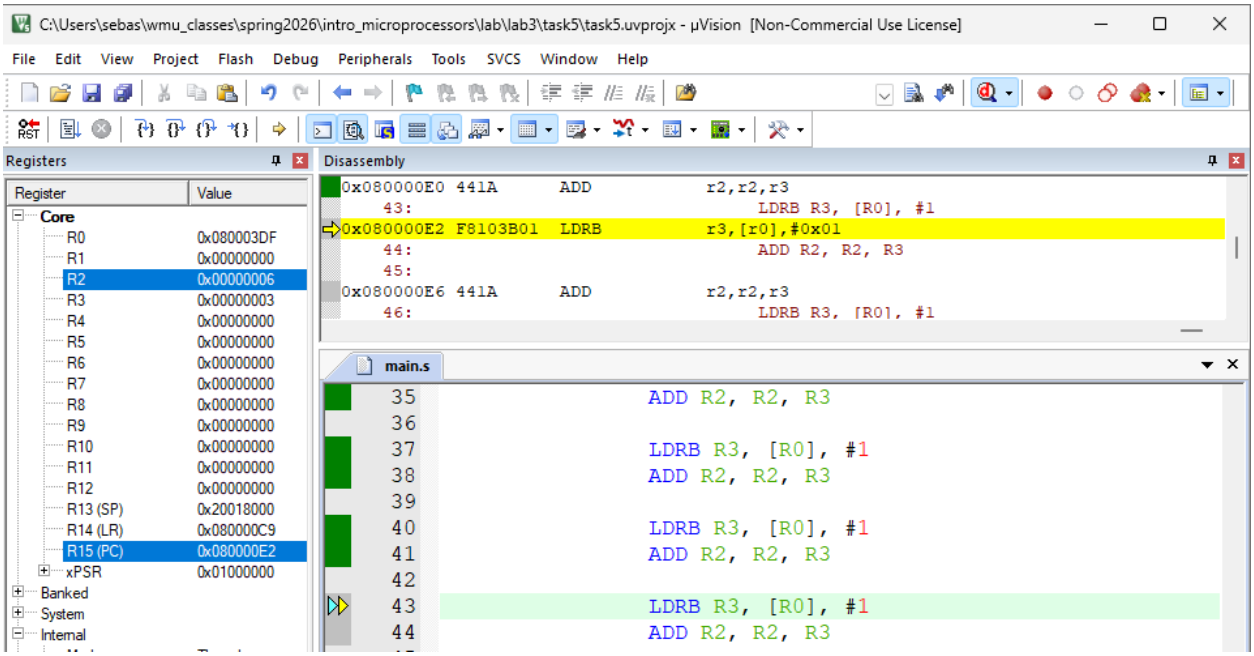
For post indexing:

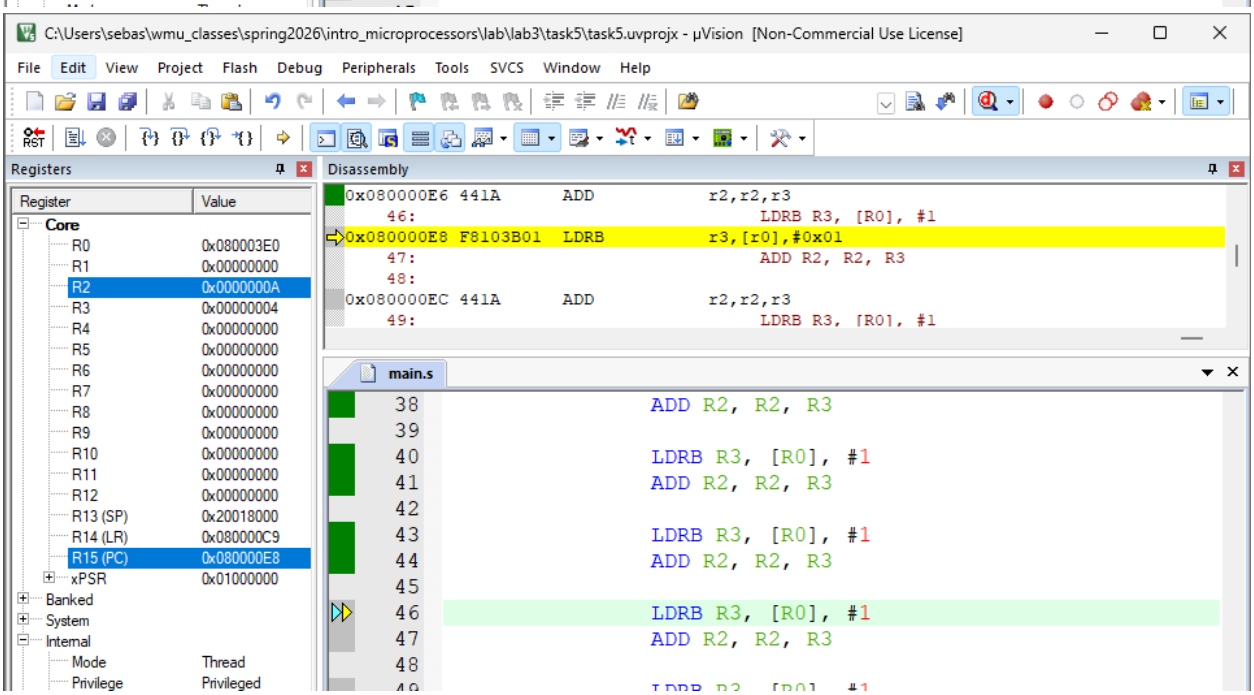


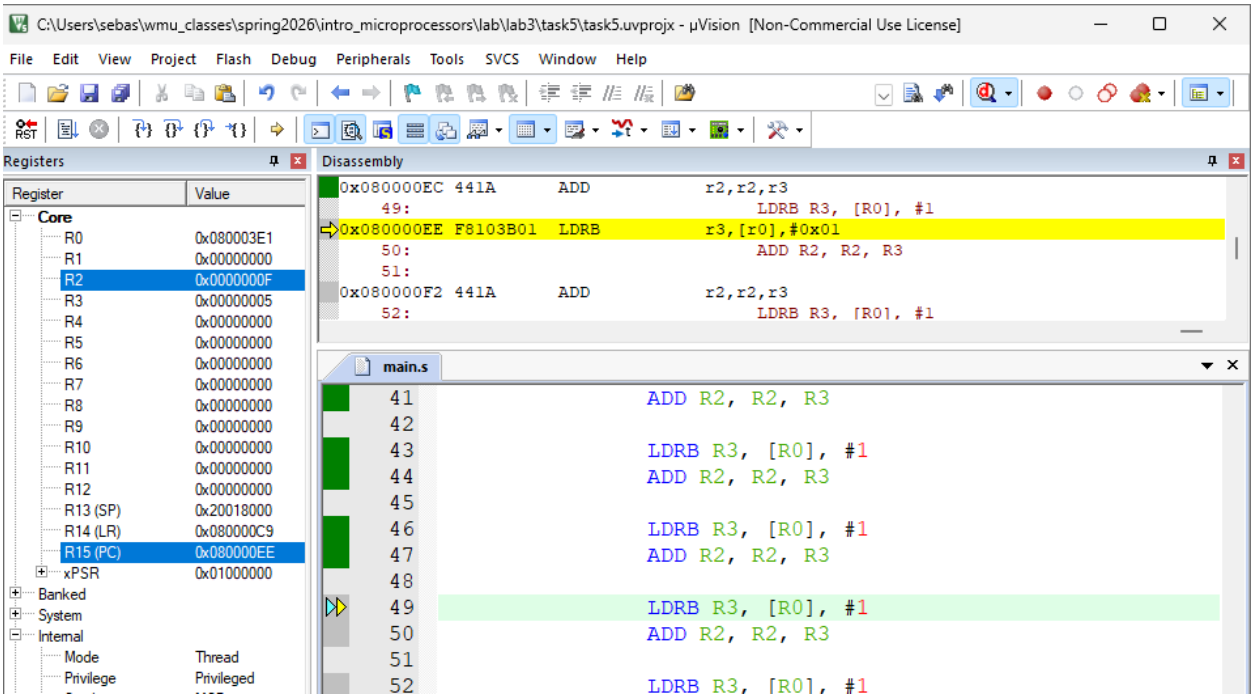

```
                ENDP


main            PROC
                EXPORT  main

                LDR R0, =array
                LDR R2, =0          ; R2 will be holding the summ

                LDRB R3, [R0], #1
                ADD R2, R2, R3

                LDRB R3, [R0], #1
                ADD R2, R2, R3

                LDRB R3, [R0], #1
                ADD R2, R2, R3

                LDRB R3, [R0], #1
                ADD R2, R2, R3

                LDRB R3, [R0], #1
                ADD R2, R2, R3

                LDRB R3, [R0], #1
                ADD R2, R2, R3

                LDRB R3, [R0], #1
                ADD R2, R2, R3

                LDRB R3, [R0], #1
                ADD R2, R2, R3

                LDRB R3, [R0], #1
                ADD R2, R2, R3


                BX      LR              ; Return from main
                ENDP

                ALIGN
                END
```

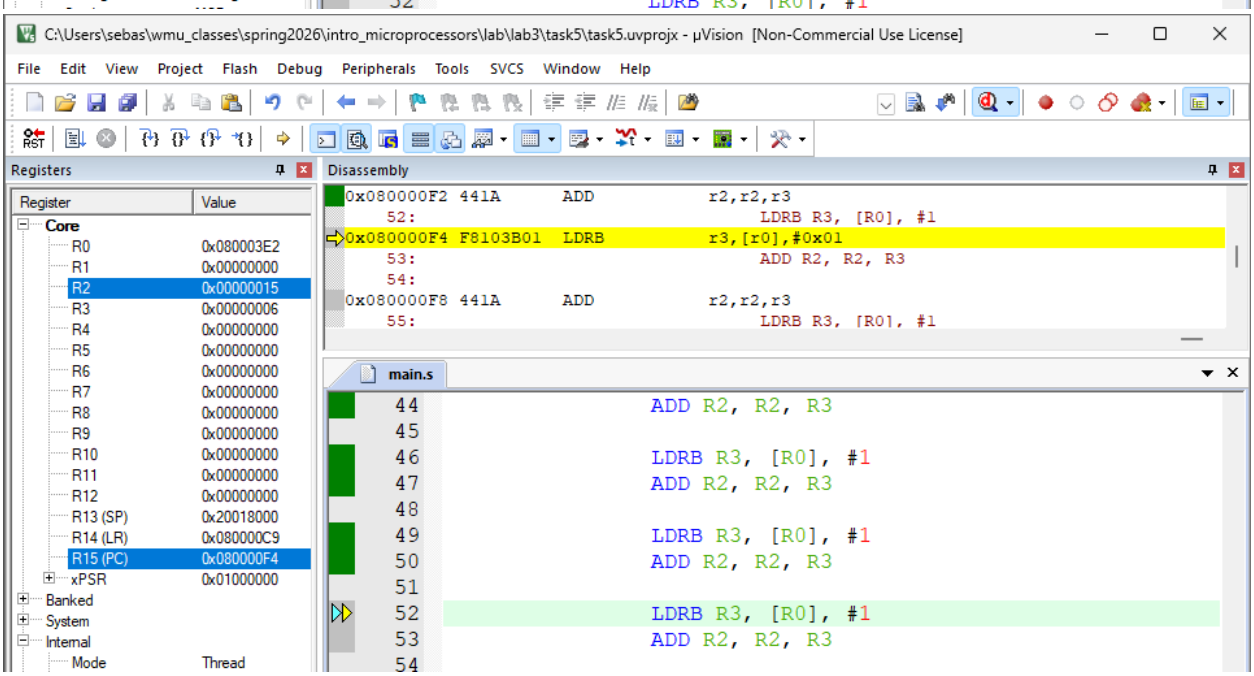

C:\Users\sebas\wmu_classes\spring2026\intro_microprocessors\lab\lab3\task5\task5.uvprojx - µVision [Non-Commercial Use License]
File Edit View Project Flash Debug Peripherals Tools SVCS Window Help
Registers
Register Value
Core
R0 0x080003DD
R1 0x00000000
R2 0x00000001
R3 0x00000001
R4 0x00000000
R5 0x00000000
R6 0x00000000
R7 0x00000000
R8 0x00000000
R9 0x00000000
R10 0x00000000
R11 0x00000000
R12 0x00000000
R13 (SP) 0x20018000
R14 (LR) 0x080000C9
R15 (PC) 0x080000D6
xPSR 0x01000000
Banked
System
Internal
Disassembly
0x080000D4 441A ADD r2,r2,r3
37: LDRB R3, [R0], #1
0x080000D6 F8103B01 LDRB r3,[r0],#0x01
38: ADD R2, R2, R3
39:
0x080000DA 441A ADD r2,r2,r3
40: LDRB R3, [R0], #1
main.s
29 EXPORT main
30
31 LDR R0, =array
32 LDR R2, =0 ; R2 will be holding the
33
34 LDRB R3, [R0], #1
35 ADD R2, R2, R3
36
37 LDRB R3, [R0], #1
38 ADD R2, R2, R3

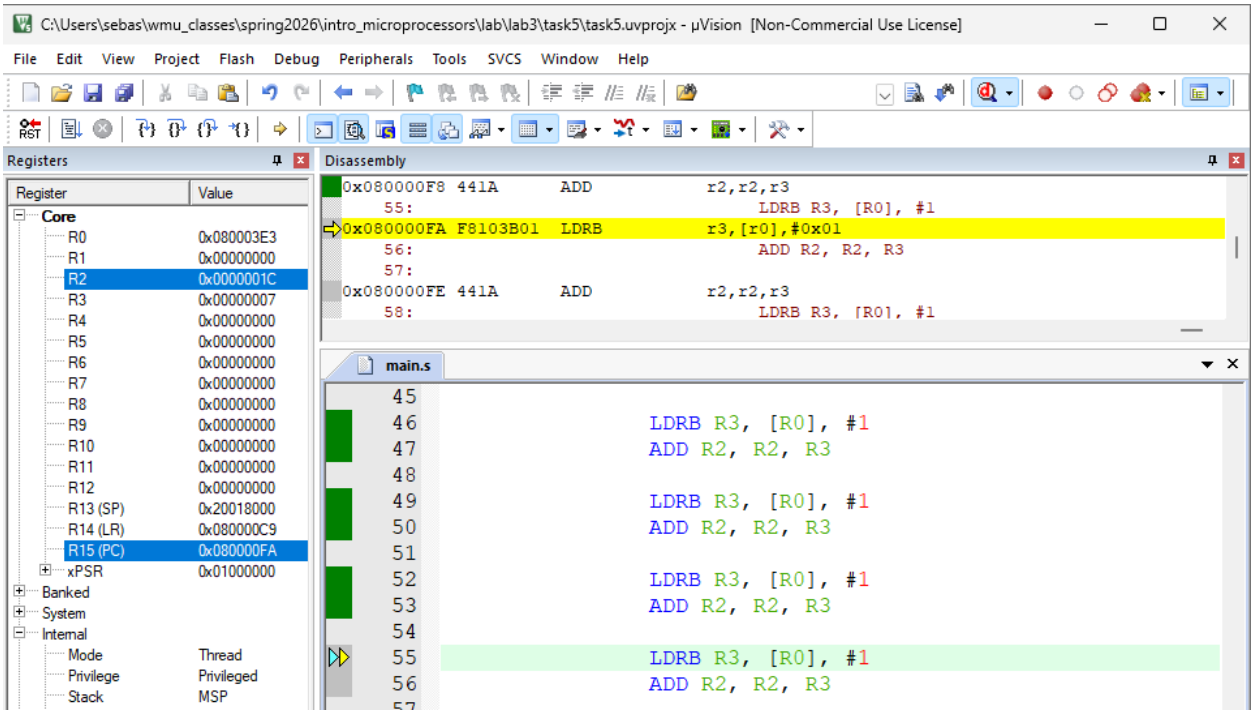

C:\Users\sebas\wmu_classes\spring2026\intro_microprocessors\lab\lab3\task5\task5.uvprojx - µVision [Non-Commercial Use License]
File Edit View Project Flash Debug Peripherals Tools SVCS Window Help
Registers
Register Value
Core
R0 0x080003DE
R1 0x00000000
R2 0x00000003
R3 0x00000002
R4 0x00000000
R5 0x00000000
R6 0x00000000
R7 0x00000000
R8 0x00000000
R9 0x00000000
R10 0x00000000
R11 0x00000000
R12 0x00000000
R13 (SP) 0x20018000
R14 (LR) 0x080000C9
R15 (PC) 0x080000DC
xPSR 0x01000000
Banked
System
Internal
Mode Thread
Disassembly
0x080000DA 441A ADD r2,r2,r3
40: LDRB R3, [R0], #1
0x080000DC F8103B01 LDRB r3,[r0],#0x01
41: ADD R2, R2, R3
42:
0x080000E0 441A ADD r2,r2,r3
43: LDRB R3, [R0], #1
main.s
32 LDR R2, =0 ; R2 will be holding the
33
34 LDRB R3, [R0], #1
35 ADD R2, R2, R3
36
37 LDRB R3, [R0], #1
38 ADD R2, R2, R3
39
40 LDRB R3, [R0], #1
41 ADD R2, R2, R3
42

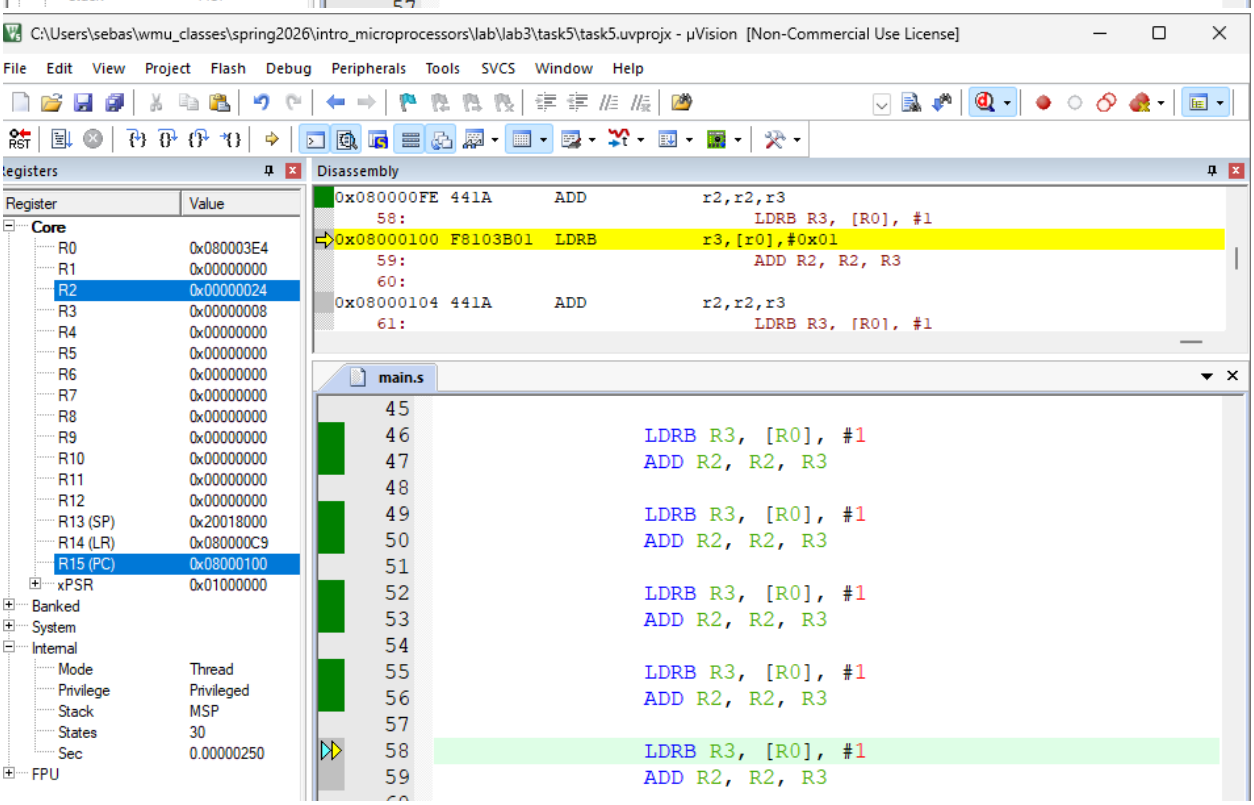

C:\Users\sebas\wmu_classes\spring2026\intro_microprocessors\lab\lab3\task5\task5.uvprojx - µVision [Non-Commercial Use License]
File Edit View Project Flash Debug Peripherals Tools SVCS Window Help
Registers
Register Value
Core
R0 0x080003DF
R1 0x00000000
R2 0x00000006
R3 0x00000003
R4 0x00000000
R5 0x00000000
R6 0x00000000
R7 0x00000000
R8 0x00000000
R9 0x00000000
R10 0x00000000
R11 0x00000000
R12 0x00000000
R13 (SP) 0x20018000
R14 (LR) 0x080000C9
R15 (PC) 0x080000E2
xPSR 0x01000000
Banked
System
Internal
Disassembly
0x080000E0 441A ADD r2,r2,r3
43: LDRB R3, [R0], #1
0x080000E2 F8103B01 LDRB r3,[r0],#0x01
44: ADD R2, R2, R3
45:
0x080000E6 441A ADD r2,r2,r3
46: LDRB R3, [R0], #1
main.s
35 ADD R2, R2, R3
36
37 LDRB R3, [R0], #1
38 ADD R2, R2, R3
39
40 LDRB R3, [R0], #1
41 ADD R2, R2, R3
42
43 LDRB R3, [R0], #1
44 ADD R2, R2, R3

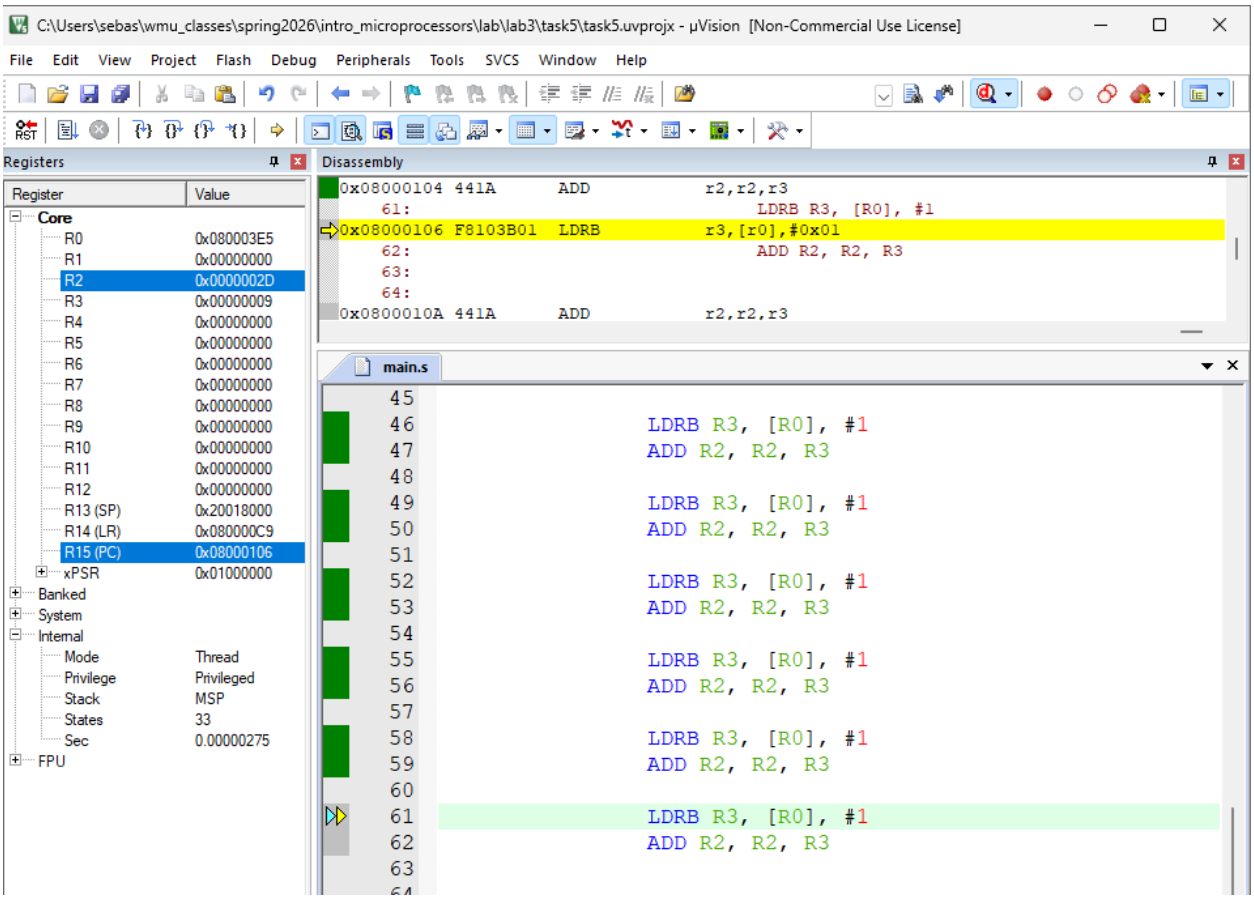

C:\Users\sebas\wmu_classes\spring2026\intro_microprocessors\lab\lab3\task5\task5.uvprojx - µVision [Non-Commercial Use License]
File Edit View Project Flash Debug Peripherals Tools SVCS Window Help
Registers
Register Value
Core
R0 0x080003E0
R1 0x00000000
R2 0x0000000A
R3 0x00000004
R4 0x00000000
R5 0x00000000
R6 0x00000000
R7 0x00000000
R8 0x00000000
R9 0x00000000
R10 0x00000000
R11 0x00000000
R12 0x00000000
R13 (SP) 0x20018000
R14 (LR) 0x080000C9
R15 (PC) 0x080000E8
xPSR 0x01000000
Banked
System
Internal
Mode Thread
Privilege Privileged
Disassembly
0x080000E6 441A ADD r2,r2,r3
46: LDRB R3, [R0], #1
0x080000E8 F8103B01 LDRB r3,[r0],#0x01
47: ADD R2, R2, R3
48:
0x080000EC 441A ADD r2,r2,r3
49: LDRB R3, [R0], #1
main.s
38 ADD R2, R2, R3
39
40 LDRB R3, [R0], #1
41 ADD R2, R2, R3
42
43 LDRB R3, [R0], #1
44 ADD R2, R2, R3
45
46 LDRB R3, [R0], #1
47 ADD R2, R2, R3
48

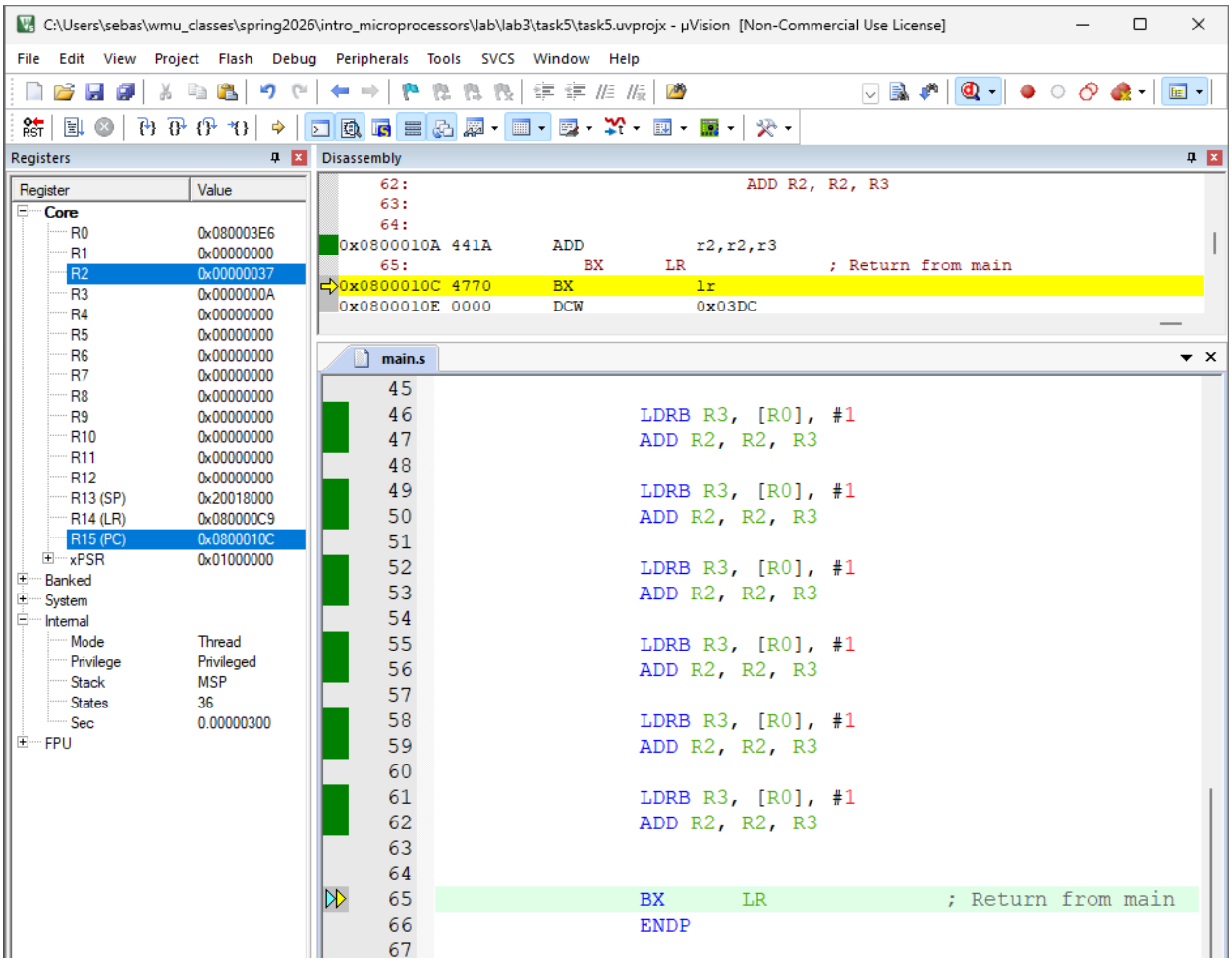

C:\Users\sebas\wmu_classes\spring2026\intro_microprocessors\lab\lab3\task5\task5.uvprojx - µVision [Non-Commercial Use License]
File Edit View Project Flash Debug Peripherals Tools SVCS Window Help
Registers
Register Value
Core
R0 0x080003E1
R1 0x00000000
R2 0x0000000F
R3 0x00000005
R4 0x00000000
R5 0x00000000
R6 0x00000000
R7 0x00000000
R8 0x00000000
R9 0x00000000
R10 0x00000000
R11 0x00000000
R12 0x00000000
R13 (SP) 0x20018000
R14 (LR) 0x080000C9
R15 (PC) 0x080000EE
xPSR 0x01000000
Banked
System
Internal
Mode Thread
Privilege Privileged
Disassembly
0x080000EC 441A ADD r2,r2,r3
49: LDRB R3, [R0], #1
0x080000EE F8103B01 LDRB r3,[r0],#0x01
50: ADD R2, R2, R3
51:
0x080000F2 441A ADD r2,r2,r3
52: LDRB R3, [R0], #1
main.s
41 ADD R2, R2, R3
42
43 LDRB R3, [R0], #1
44 ADD R2, R2, R3
45
46 LDRB R3, [R0], #1
47 ADD R2, R2, R3
48
49 LDRB R3, [R0], #1
50 ADD R2, R2, R3
51
52 LDRB R3, [R0], #1


C:\Users\sebas\wmu_classes\spring2026\intro_microprocessors\lab\lab3\task5\task5.uvprojx - µVision [Non-Commercial Use License]
File Edit View Project Flash Debug Peripherals Tools SVCS Window Help
Registers
Register Value
Core
R0 0x080003E2
R1 0x00000000
R2 0x00000015
R3 0x00000006
R4 0x00000000
R5 0x00000000
R6 0x00000000
R7 0x00000000
R8 0x00000000
R9 0x00000000
R10 0x00000000
R11 0x00000000
R12 0x00000000
R13 (SP) 0x20018000
R14 (LR) 0x080000C9
R15 (PC) 0x080000F4
xPSR 0x01000000
Banked
System
Internal
Mode Thread
Disassembly
0x080000F2 441A ADD r2,r2,r3
52: LDRB R3, [R0], #1
0x080000F4 F8103B01 LDRB r3,[r0],#0x01
53: ADD R2, R2, R3
54:
0x080000F8 441A ADD r2,r2,r3
55: LDRB R3, [R0], #1
main.s
44 ADD R2, R2, R3
45
46 LDRB R3, [R0], #1
47 ADD R2, R2, R3
48
49 LDRB R3, [R0], #1
50 ADD R2, R2, R3
51
52 LDRB R3, [R0], #1
53 ADD R2, R2, R3
54


C:\Users\sebas\wmu_classes\spring2026\intro_microprocessors\lab\lab3\task5\task5.uvprojx - µVision [Non-Commercial Use License]
File Edit View Project Flash Debug Peripherals Tools SVCS Window Help
Registers
Register Value
Core
R0 0x080003E3
R1 0x00000000
R2 0x0000001C
R3 0x00000007
R4 0x00000000
R5 0x00000000
R6 0x00000000
R7 0x00000000
R8 0x00000000
R9 0x00000000
R10 0x00000000
R11 0x00000000
R12 0x00000000
R13 (SP) 0x20018000
R14 (LR) 0x080000C9
R15 (PC) 0x080000FA
xPSR 0x01000000
Banked
System
Internal
Mode Thread
Privilege Privileged
Stack MSP
Disassembly
0x080000F8 441A ADD r2,r2,r3
55: LDRB R3, [R0], #1
0x080000FA F8103B01 LDRB r3,[r0],#0x01
56: ADD R2, R2, R3
57:
0x080000FE 441A ADD r2,r2,r3
58: LDRB R3, [R0], #1
main.s
45
46 LDRB R3, [R0], #1
47 ADD R2, R2, R3
48
49 LDRB R3, [R0], #1
50 ADD R2, R2, R3
51
52 LDRB R3, [R0], #1
53 ADD R2, R2, R3
54
55 LDRB R3, [R0], #1
56 ADD R2, R2, R3


C:\Users\sebas\wmu_classes\spring2026\intro_microprocessors\lab\lab3\task5\task5.uvprojx - µVision [Non-Commercial Use License]
File Edit View Project Flash Debug Peripherals Tools SVCS Window Help
Registers
Register Value
Core
R0 0x080003E4
R1 0x00000000
R2 0x00000024
R3 0x00000008
R4 0x00000000
R5 0x00000000
R6 0x00000000
R7 0x00000000
R8 0x00000000
R9 0x00000000
R10 0x00000000
R11 0x00000000
R12 0x00000000
R13 (SP) 0x20018000
R14 (LR) 0x080000C9
R15 (PC) 0x08000100
xPSR 0x01000000
Banked
System
Internal
Mode Thread
Privilege Privileged
Stack MSP
States 30
Sec 0.00000250
FPU
Disassembly
0x080000FE 441A ADD r2,r2,r3
58: LDRB R3, [R0], #1
0x08000100 F8103B01 LDRB r3,[r0],#0x01
59: ADD R2, R2, R3
60:
0x08000104 441A ADD r2,r2,r3
61: LDRB R3, [R0], #1
main.s
45
46 LDRB R3, [R0], #1
47 ADD R2, R2, R3
48
49 LDRB R3, [R0], #1
50 ADD R2, R2, R3
51
52 LDRB R3, [R0], #1
53 ADD R2, R2, R3
54
55 LDRB R3, [R0], #1
56 ADD R2, R2, R3
57
58 LDRB R3, [R0], #1
59 ADD R2, R2, R3


C:\Users\sebas\wmu_classes\spring2026\intro_microprocessors\lab\lab3\task5\task5.uvprojx - µVision [Non-Commercial Use License]
File Edit View Project Flash Debug Peripherals Tools SVCS Window Help
Registers
Register Value
Core
R0 0x080003E5
R1 0x00000000
R2 0x0000002D
R3 0x00000009
R4 0x00000000
R5 0x00000000
R6 0x00000000
R7 0x00000000
R8 0x00000000
R9 0x00000000
R10 0x00000000
R11 0x00000000
R12 0x00000000
R13 (SP) 0x20018000
R14 (LR) 0x080000C9
R15 (PC) 0x08000106
xPSR 0x01000000
Banked
System
Internal
Mode Thread
Privilege Privileged
Stack MSP
States 33
Sec 0.00000275
FPU
Disassembly
0x08000104 441A ADD r2,r2,r3
61: LDRB R3, [R0], #1
0x08000106 F8103B01 LDRB r3,[r0],#0x01
62: ADD R2, R2, R3
63:
64:
0x0800010A 441A ADD r2,r2,r3
main.s
45
46 LDRB R3, [R0], #1
47 ADD R2, R2, R3
48
49 LDRB R3, [R0], #1
50 ADD R2, R2, R3
51
52 LDRB R3, [R0], #1
53 ADD R2, R2, R3
54
55 LDRB R3, [R0], #1
56 ADD R2, R2, R3
57
58 LDRB R3, [R0], #1
59 ADD R2, R2, R3
60
61 LDRB R3, [R0], #1
62 ADD R2, R2, R3
63



Discussion

With the register offset, I had to keep a counter register to traverse through the array declared with DCB and use each value individually using the position marked by my register offset,

For the Post-indexed addressing mode, there are two actions within a single instruction, first get the byte found in the address at R0 and loads it to R3, then add the offset to the register R0 used as base, then R0 points to the next byte in the array, ready for the next instruction